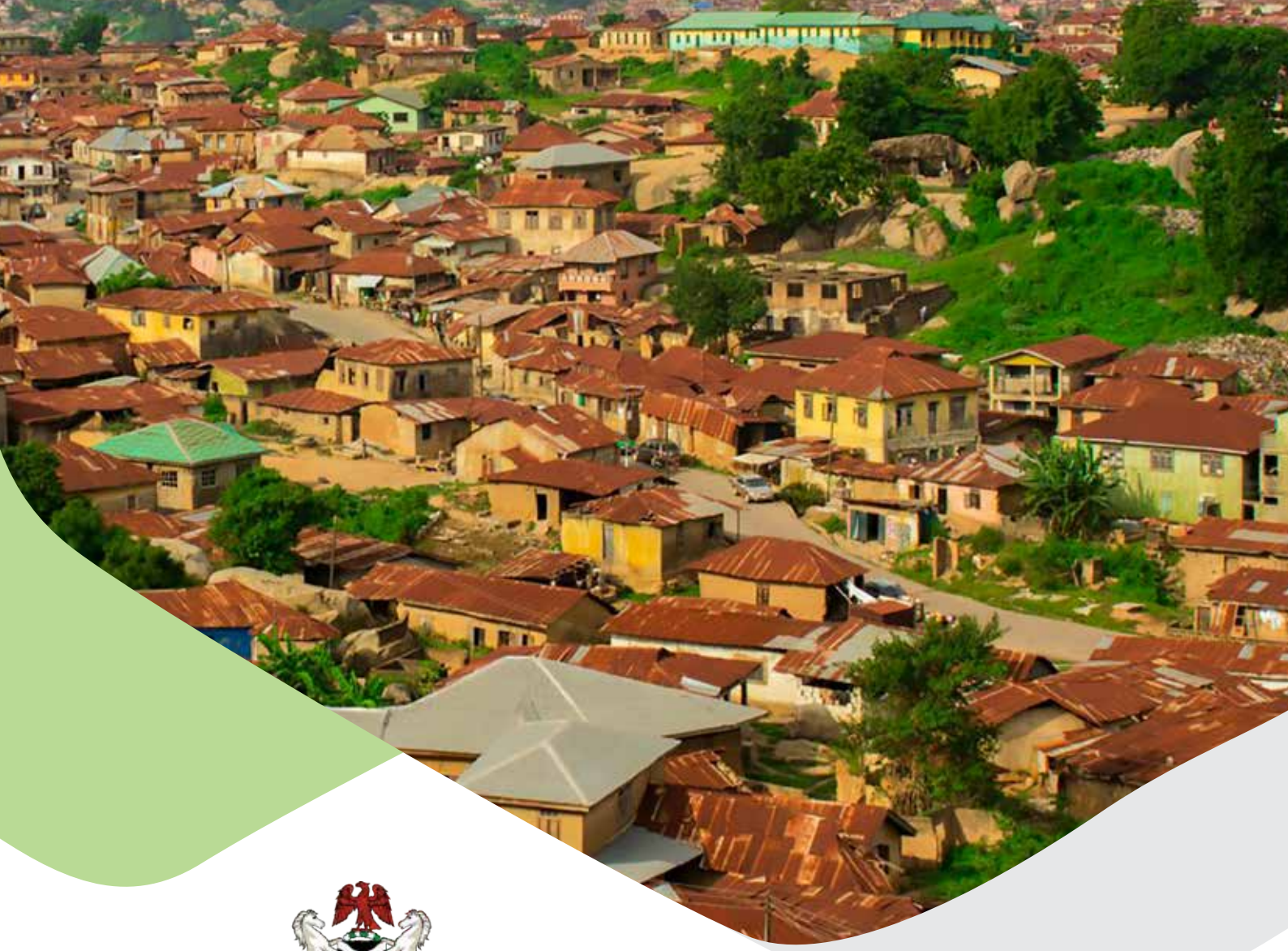


FEDERAL MINISTRY
OF ENVIRONMENT

NIGERIA'S NATIONAL ADAPTATION PLAN (NAP)

JULY, 2026



Nigeria's National Adaptation Plan (NAP)

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July, 2026

Foreword

Nigeria progresses towards a low-carbon and climate-resilient economy, adopting adaptation strategies to address the numerous climate change impacts that plague us as a Nation. Reaffirming the country's resolve to act decisively in tackling climate change.

The Nation have made remarkable efforts toward climate adaptation, this includes the preparation of its National Adaptation Strategy and Plan of Action on Climate Change in Nigeria (NASPA-CCN) in 2011, which identified 13 thematic areas that required urgent adaptation. A National Adaptation Framework in 2020 to provide guidelines for adaptation planning and governance. In 2021, it also developed its Adaptation Communication, which was submitted to the United Nations Framework Convention on Climate Change (UNFCCC).

It is against this backdrop that Nigeria's National Adaptation Plan (NAP), supported by the Green Climate Fund and United Nations Environment Programme (UNEP), builds on the National Adaptation Strategy and Plan of Action on Climate Change for Nigeria (NASPA-CCN) developed in 2011. It identified 13 thematic areas requiring adaptation planning. It is formulated in line with the Cancun Adaptation Framework, which enables Parties to formulate and implement NAPs as a means of identifying medium to long-term adaptation needs and developing and implementing strategies and programmes to address those needs. The objectives of the NAP are;

- To reduce vulnerability to the impacts of climate change, by building adaptive capacity and resilience;

- To facilitate the integration of climate change adaptation, coherently, into relevant new and existing policies, programmes and activities, in particular development planning processes and strategies, within all relevant sectors and at different levels, as appropriate

A major activity of the NAP Process is the conduct of Climate Risks and Vulnerability Assessment (CRVA), which identifies communities' vulnerabilities and suggests possible adaptation strategies and options to address those challenges. These identified options align with Government priorities and development plans in line

with global best practices. Nigeria's NAP is expected to provide the blueprint for adaptation planning and governance towards a climate-resilient society for sustainability.

My sincere gratitude goes to the Green Climate Fund (GCF) for extending this Support to the Government of Nigeria, and to the United Nations Environment Programme (UNEP) for their technical contributions towards the delivery of this Support. I appreciate the Federal Ministry of Environment for facilitating the development of this important National Document.



Balarabe Abbas Lawal
Honourable Minister,
Federal Ministry of Environment

Preface

Nigeria has been actively engaged in international climate negotiations since it became a Party to the United Nations Framework Convention on Climate Change (UNFCCC) in 1994, the Kyoto Protocol in 2004, and the Paris Agreement in 2017. As one of the developing countries adversely affected by the impacts of climate change, it has made giant strides in the area of Adaptation Planning and Governance.

Following the adopton of the Cancun Adaptation Framework (CAF), which established the NAP Process at COP 15 in Cancun, Mexico. The CAF enables Parties to formulate and implement NAPs as a means of identifying medium to long-term adaptation needs, developing and implementing strategies and programmes to address those needs. Nigeria developed the National Adaptation Strategy and Plan of Action for Climate Change in Nigeria (NASPA-CCN) in 2011, the National Adaptation Plan Framework in 2020, and Nigeria's Adaptation Communication (ADCOM) in 2021. Nigeria accessed the Green Climate Fund Readiness Support to advance its National Adaptation Plan. These documents were developed to enhance climate resilience and build Nigeria's capacity to adapt to climate change.

The Climate Risk Vulnerability Assessment (CRVA) conducted during the NAP Formulation identified vulnerabilities peculiar to each ecological zone of the Federation, appropriate adaptation strategies, and options based on priority sectors that align with the United Arab Emirates (UAE) Framework and Nigeria's Nationally Determined Contributions (NDCs) sectors. These adaptation strategies were costed, with a positive Benefit-Cost-Ratio and an Adaptation Finance Framework developed.

While formulating Nigeria's NAP, further support was obtained to integrate gender issues as well as conflict sensitivity and peace-building into the NAP process. These integrations were done in line with identified gaps and to further enrich the NAP Document.

As Nigeria commences the implementation of its NAP, It is estimated that over N18 trillion the country seeks the support of development partners, both local and international, and other relevant stakeholders, to implement its NAP successfully in line with the objectives of the UNFCCC.



Salihu Aminu Usman
Permanent Secretary,
Federal Ministry of Environment

Acknowledgement

The Federal Ministry of Environment extends its deepest gratitude to His Excellency, Bola Ahmed Tinubu (GCFR), President and Commander-in-Chief of the Armed Forces of the Federal Republic of Nigeria, for his leadership and constant support for Nigeria's climate action to enhance sustainability. His commitment to sustainable environmental stewardship and international cooperation indicates Nigeria's commitment to playing a prominent role in the international fight against climate change.

We would also like to appreciate the Honourable Minister of Environment, Balarabe Abbas Lawal, and the Permanent Secretary, Federal Ministry of Environment, Dr. Salihu Aminu Usman, for strategic leadership and guidance to the successful formulation of the National Adaptation Plan (NAP) Document. Further appreciation to the Director, Department of Climate Change, Dr. Iniobong Abiola-Awe, for technical leadership and commitment throughout the entire process, and the entire Management and Staff of the Department for their technical support and dedication to a successful NAP formulation for Nigeria. We also acknowledge the tireless work of the NAP Secretariat, led by Mr. Kayode Aboyeji, to ensure the successful completion of this project.

Special thanks to the Green Climate Fund for extending this support to the Government of Nigeria and the Implementing Agency, the United Nations Environment Programme (UNEP) team led by the Country's Task Manager, Mr. Gift Gewona, for the unwavering support provided throughout the entire process. Furthermore, our appreciation goes to the NAP Global Network, International Organisation for Migration (IOM), Food and Agriculture Organisation (FAO) and Global Centre for Climate Mobility (GCCM). International and local consultants for technical support to the success of Nigeria's National Adaptation Plan.

Special appreciation to the federal ministries, departments, and agencies, States' ministries of environment, NGOs & CSOs, academia/researchers, people with disabilities, development partners, media, and all stakeholders whose valuable inputs and constructive feedback have helped shape and enhance the information in this document. Your numerous contributions at the various stages of this project have made the document very rich and robust.

Finally, we salute all Nigerians, particularly those in vulnerable and climate-impacted communities, whose resilience, life experiences, and aspirations are the true motivation behind this document.

God bless the Federal Republic of Nigeria.

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Executive Summary

Nigeria contributes very minimally to the global emissions that are culprits of the current climate change discourse, but it is highly vulnerable to the impact of climate change. The country ranked as 160 out of 181 countries in the 2020 ND-GAIN Index, which summarises a country's vulnerability to climate change and other global challenges (the more vulnerable a country is, the lower the score), in combination with its readiness to improve resilience (the more ready a country is to improve its resilience, the higher the score)[1]. As a result of this vulnerability, climate change poses profound and pressing challenges to Nigeria's development. In recent decades, rising temperatures, erratic rainfall patterns, severe droughts, desertification in the northern regions, and devastating floods and coastal erosion in the south continue to be manifestations of climate change in the country. These environmental disruptions not only undermine socio-economic development and food security but also exacerbate poverty, conflict, migration, and public health risks, particularly among Nigeria's most vulnerable populations.

In response to these escalating threats, Nigeria has demonstrated commendable ambition and commitment at both global and national levels. This includes its ratification of the Paris Agreement (United Nations Framework Convention on Climate Change [UNFCCC], 2015) and the submission of revised Nationally Determined Contributions (NDCs) in 2021 and 2025, which outline specific targets for greenhouse gas emission reductions and adaptation efforts. Crucially, these international commitments are underpinned by the passage of the Climate Change Act, 2021, which provides a comprehensive legal and institutional framework for national climate governance. This Act notably established the National Council on Climate Change (NCCC) as the apex body mandated to coordinate and monitor climate actions across all sectors and tiers of government.

Nigeria recognizes the imperative for effective adaptation to climate change to reduce its impact on the people and make the society more resilient. To this end, the country has implemented a number of adaptation projects. A major one is the $500 million World Bank assisted Nigeria Erosion and Watershed Management Project (NEWMAP), which was designed in collaboration with the Federal Ministry of Environment to tackle the menace of gully erosion in south-east Nigeria and

[1] Climate Risk Profile: Nigeria (2021): World Bank Group

other forms of land degradation in northern Nigeria, The successor to NEWMAP is the six-year (2022 -2028) $700 million Agro-Climatic Resilience in Semi-Arid Landscapes (ACReSAL) Project. It aims at addressing the challenges of large-scale watershed degradation in northern Nigeria, improving community climate resilience, strengthening institutional capacity and enabling institutional and policy foundation for multisectoral integrated landscape management and climate resilience and, lastly, to install a financing mechanism available to borrowers in investment project financing operations and, thus, to enable quick deployment of uncommitted funds to respond to eligible crises or emergencies.

There was also the $4.9 million Building Nigeria's Response to Climate Change (BNRCC) project (2007 - 2012), which researched vulnerability (current and future), and awareness and pilot projects, mainstreaming gender throughout the project lifecycle. and resulted in the development of the National Adaptation Strategy and Plan of Action on Climate Change for Nigeria (NASPA-CCN). The pilot projects of the BNRCC had the goals of utilising improved varieties to improve food security, providing fuel-efficient wood stoves, proposing aquaculture as an alternative source of income and weaning communities off forest products, improving access to water sources and rehabilitating ecosystems by planting trees. There is also the Great Green Wall Initiative, which has built about 1,359 km of shelterbelt, planted over 7.6 million seedlings, created nearly 3,000 hectares of reforested land, and trained over 1,200 people in land management and food security. The project also aims to create jobs, reduce tree cutting with alternative energy sources like solar-powered pumps and stoves, and improve agricultural land use practices. In addition, the Africa Development Bank (AfDB) has launched the $25 billion Africa Adaptation Acceleration Program (AAAP) in collaboration with the Global Centre on Adaptation (GCA) to unlock financing from African governments, investors, foundations, resilience bonds and debt for climate adaptation swaps.

Significant improvement has also been recorded on adaptation awareness, policies and implementation of adaptation actions in Nigeria as shown by NASPA-CCN 2011; the Nigeria's National Adaptation Plan Framework 2020; the National Agricultural Resilience Framework (NARF) 2016 and the National Agricultural Technology and Innovation Policy 2022 - 2027 that replaced the Agricultural Promotion Policy (APP) 2016. In 2021, the Climate Change Act was enacted, demonstrating Nigeria's commitment to meet its obligations to combat climate change with a comprehensive regulatory framework.

Despite these significant milestones, Nigeria's national response to climate change has historically been characterised by fragmentation, a lack of coordination, and institutional weaknesses. A critical challenge has been the relatively uncoordinated adaptation initiatives and proliferation of multiple sector-specific policies, strategies and plans. To enhance national adaptation response to the impact

of climate change, Nigeria obtained a Readiness and Preparatory Support from the Green Climate Fund (GCF) to advance its National Adaptation Plan process through a project titled "Strengthening Nigeria's Capacity to Advance the National Adaptation Plan (NAP) Process". The main objective is to strengthen the capacity of Nigeria's government at all levels to implement a NAP process, including planning and budgeting for adaptation towards building adaptive capacity and resilience in Nigeria while also facilitating the integration of climate change into new and existing policies, plans and strategies.

The Nigerian government's request for GCF support is anchored on the decision of the 2010 Sixteenth Conference of the Parties (COP16) of the United Nations Framework Convention on Climate Change (UNFCCC), which emphasised enhanced action through the Cancun Framework to reinforce the global drive towards climate adaptation. This decision established the National Adaptation Plan (NAP) process to identify and address countries' medium- and long-term adaptation needs, facilitate effective adaptation planning and integrate climate change adaptation within national development processes. The 2015 Paris Agreement further reinforces adaptation planning as vital to socioeconomic development and integral to the global response to climate change. It calls on all countries to initiate the preparation and implementation of NAPs to further reinforce climate adaptation actions, to which Nigeria complied.

The implementation of the NAP Process Project included extensive modelling assessments on the country's vulnerability, the economic appraisal of priority adaptation measures in various sectors, gender analysis, conflict sensitivity and peace-building analysis, and climate change mainstreaming processes into development as the building blocks. The NAP aligns with some national and sectoral policies, including the NASPA-CCN, NAPF, NCCPRS, the ADCOM; Gender and Climate Change Action Plan (NAPGCC), ERGP, Climate Change Act 2021, NDC 3.0, NDP 2021-2050. These policies are the sources from which Nigeria's NAP draws its mandate, in addition to the backing from Decision 1/CP.16 of the UNFCCC CAF agreement.

Nigeria's NAP is envisioned to strengthen the effectiveness of adaptation governance and coordination; build national capacity to undertake NAP planning, analyse climate information and prioritise adaptation options. The governance structure of NAP is as outlined in Nigeria's Adaptation Communication of 2021, which provides for a steering committee that gives leadership and guidance to the NAP process with membership drawn from the Inter-ministerial Committee on Climate Change (ICCC).

This NAP process and priority strategies took cognisance of Nigeria's broader development priorities as contained in the NDP 2021-2050, the Climate Change

Act (2021), and international commitments such as the Paris Agreement, the SDGs and the Sendai Framework for Disaster Risk Reduction. Relevant data on national context across all the ecological regions, which reflected gender-responsiveness and followed best global standards, as captured in the Climate modeling and climate vulnerability assessment, were used in the development of the NAP.

Following a detailed analysis of the climate risk analysis (CRA), the adaptation priorities were selected based on strategic and weighted sectoral criteria and measures identified from relevant data, economic and finance indices and considerations. The CRA covered Nigeria's five major agro-ecological zones as well as urban areas using nine representative States. It used impact story lines (ISL) and down-scaled or area-based climate change projections to explore a range of potential future conditions based on different trajectories of greenhouse gas (GHG) emissions and socio-economic developments, based on different scenarios generated by the Shared Socioeconomic Pathways (SSPs) framework, that integrates climate projections with societal trends, and the Representative Concentration Pathways (RCP) to inform priority strategies for mitigation and adaptation, using a 30-year period (1981-2010) as the reference period, while 2011 - 2040, 2041 - 2070 and 2071 -2100 are near-term, mid-term, and long-term respectively.

Nigeria's NAP then integrated the findings of a rigorously produced cost-benefit analysis (CBA) of Nigeria's priority adaptation strategies. The CBA adopted RCP 4.5 (aligned with SSP2-4.5) and RCP 8.5 (aligned with SSP5-8.5), which were also used in the CRA. It worked with five adaptation strategies in the agriculture, water resources and energy sectors, which were identified through the CRA impact-chain process as highly vulnerable to the impacts of climate change. In addition, the findings and recommendations of the Conflict sensitivity and peace-building assessment were integrated into the CRA approach, giving NAP the capacity to respond to socioeconomic conflict issues in Nigeria. Similarly, Nigeria's NAP aligns and strongly complements the Sendai Framework 2015-2030 and Sustainable Development Goals (SDGs).

Findings from the CRA analysis indicate that mean annual temperatures have increased significantly across Nigeria's coastal, rainforest, Sahel, Sudan savannah and Guinea savannah agroecological zones in the period 1901-2022. There was an average of 16 heatwave days per year experienced during the reference period (1981-2010), with the highest number of heatwave days occurring in the Sahel (CRA 2025). Similarly, significant variability in the amount, distribution and pattern of rainfall occurred over the period across the different agroecological zones, resulting in floods and droughts indifferent parts of the country at different times. In 2022, for example, rain-induced floods destroyed ₦700 billion worth of agricultural investments and affected 9 million people across 29 States, many of



Houses are partially submerged following a dam collapse in Maiduguri, Nigeria, Tuesday, Sept 10, 2024. (AP Photos/ Musa Ajit Borno)

them farming households (CRA, 2025), particularly in the tropical rainforest and mangrove/swamp forest ecosystems. On the other hand, droughts that occurred affected the Sahel, Sudan savannah and Guinea savannah areas at different times and with different intensities in the reference period of the CRA analysis. Sea-level rise of 0.462m resulted in the flooding of many communities along the low-lying coastline with the destruction of human settlements and infrastructure, and the erosion of mangroves.

Future projections indicate that changes in the mean annual temperatures will occur across all the agroecological zones of Nigeria, with significant warming of between +2.5 to +3.2 °C projected to occur by 2050 under both RCP 4.5 and RCP 8.5 scenarios (CRA 2025). Similarly, under both scenarios, there will be more frequent heavy rainfall events that will increase the frequency of floods. A slight increase in the number of extremely heavy rainy days from near-term to long-term may be observed under both SSPs, which spreads across all ecological zones; with the Sahel and Guinea Savannah likely to experience the biggest increase of such in the future (CRA 2025). Future projections show an average increase of heavy rainy days from near-term to long-term across Nigeria, especially in the northern ecological zones of the Sahel and Sudan Savannah (CRA 2025). The longest period without precipitation will gradually decrease in the country under SSP2-4.5 and SSP5-8.5 on average and this decrease will be most marked in the Sahel and Sudan Savannah. Under the RCP 8.5 scenario of the CRA Analysis, however, the frequency of severe/extreme drought will be more intense towards the end

of the 21st century, adversely affecting water availability. With the current global estimate of sea level rise of 0.21m that is projected to increase to 1.32 m by the year 2100 and 5.61m by 2300 under the RCP 8.5 (World Economic Forum, 2024; Horton et al., 2020), the inundated area of 3,400 km^2 in the coastal regions of Nigeria will increase to 18,400 km^2 under the projected 1m scenario (Federal Republic of Nigeria, 2023). A one metre rise in sea level would inundate about three-quarters of the Niger Delta area, leading to a huge loss in Nigeria's GDP as the oil and gas and fisheries sectors will be severely affected (Oyegun 2018; Danladi, 2017). Given these projected trends, Nigeria may become a hotspot of food insecurity in the future, with accompanying significant economic losses, damage to arable lands and infrastructure, loss of lives and livelihoods

In addition to the CRA. Nigeria's NAP also undertook an economic appraisal to ascertain the profitability or otherwise of adaptation investments. The appraisal prioritised five adaptation strategies in three sectors (agriculture, water resources, and energy) based on ecological, social and economic indices. Using net present value (NPV) and benefit-cost ratio (BCR), all five strategies offered highly favourable economic outcomes (CBA, 2025). When ranked based on relative performance and attractiveness using BCR, the cross-cutting agriculture measures stand out, giving a BCR of 5.80-5.89, followed by flood protection systems with a BCR of 5.51-5.74, showing a delivery of nearly USD$6 in benefits for every dollar invested and making the 2 strategies "quick win" measures that can deliver resilience dividends at low cost. When ranked by absolute net benefits (NPV), the energy sector adaptation strategy stands out in magnitude higher than all other strategies combined with USD 7.8-8.5 billion followed by water supply augmentation with USD 947 million-1.0 billion and heat impacts management in agriculture with USD 861-931 million. With an average BCRs of 2.32 in 2050 and 2.41 in 2080, and NPVs of ₦15.3-16.6 trillion (USD 10.4-11.3 billion), for every ₦1 (or dollar) investment in any of the adaptation strategies across all three sectors, Nigeria gains at least ₦2.3 (or dollar) in avoided damages, productivity gains, and resilience co-benefits; thus confirming the economic viability of the combined climate adaptation portfolio.

Nigeria's diverse ecological zones create opportunities for multiple livelihoods, agricultural practices, and commodities, all of which are affected by climate change and shocks. Largely rainfed agriculture, which contributed to 21.07% of the GDP in the first quarter of 2024 and employs over 70% of Nigerians is already suffering a myriad of problems, ranging from poor soil fertility, land fragmentation and inefficient tenure system, weak agricultural extension system, poor adoption of improved technology, substantial post-harvest losses, and lack of access to inputs and finance and poor crop insurance.

In general, NAP analysis indicates clearly that many sectors of Nigeria's development will be impacted by changes in the climatic conditions, as depicted in the following:

- **Agriculture and Forestry:** Extreme weather events caused by climate change will exacerbate existing challenges in both crop and livestock sub-sectors, leading to food insecurity. The country's fishery sector will face decreased fish stocks, higher prices of fish products, and loss of livelihoods for many fishing-dependent communities because of future climate risks from rising water temperatures, altered rainfall patterns, and sea-level rise. The forestry sub-sector, already challenged by deforestation, illegal logging, and land conversion, will witness accelerated degradation and suffer further biodiversity loss, among others, because of climate impacts.

- **Water resources:** Water resources in Nigeria will face serious risks from climate change stemming from a combination of rising temperatures, changing rainfall patterns, increased frequency of droughts and floods, sea-level rise and other climate-induced phenomena. These cascading impacts will not only limit access to safe and clean water but also increase the burden on public health systems and reduce agricultural productivity. The inherent vulnerabilities of Nigeria's water sector, including poor water resources management and over-reliance on surface water resources, will be worsened by climate hazards.

- **Health:** Nigeria's H-NAP indicates that Nigeria will experience a 21% increase in disease burden due to climate change by 2030, with significant increases in water-borne diseases, non-communicable diseases and mental health disorders (Nigeria Climate Change and Health, 2024). Climate hazards will worsen the health sector's existing challenges, such as poor access to healthcare, unreliable energy supply that undermines the effectiveness of cooling systems, poor sanitation, limited disease surveillance and prevention.

- **Energy:** Nigeria's energy sector, including oil and gas, wood biomass, renewable infrastructure, and power infrastructure, is not spared by the intensifying climate hazards. The oil and gas operations, which are mostly located in the Niger Delta will witness frequent disruptions due to rain-induced flood and sea-level rise. In addition, 15% of the country's electricity supply is generated from hydropower, which is vulnerable to drought, high temperatures and floods. The energy sector will face frequent disruptions and the collapse of infrastructure, causing massive loss of revenue due to increasing vulnerability to immediate and long-term climate risks.

- **Transport:** In recent years, Nigeria made significant investments to improve the transport sector, including air, rail, water and road transport which is the dominant means of moving goods and services across the country. These investments are highly threatened by rising temperatures, erratic rainfall patterns, sea-level rise, and more frequent extreme weather events, causing



How erosion crisis threatens homes and disrupts livelihoods in Nigeria (https://climateaction.africa/)

the collapse of roads, bridges, and railways, leading to disruptions in the transport sector with negative ripple effects on the economy. Rural areas are particularly vulnerable, as disruptions often leave these areas isolated and hinder access to essential services.

On an agroecological basis, the following impacts are expected:

- **Swamp Forest:** The swamp forest region found along the coastal areas is particularly vulnerable to flooding and sea-level rise because of hydrological and ecological characteristics, causing displacement of fishing communities, loss of mangrove biodiversity, and worsening coastal erosion.

- **Tropical Rainforest:** The tropical rainforest agroecological zone, which provides immense ecosystem services, is already facing deforestation and over-exploitation due to human activities. The degradation will be exacerbated by rain-induced floods, heat waves, and changing rainfall patterns, causing soil erosion, loss of biodiversity and ecosystem services etc.

- **Guinea Savannah:** The Guinea Savannah covers nearly half of Nigeria's land area with generally fertile soil that favours agricultural activities. Erratic rainfall patterns, heat stress, and flooding will impair production of staple crops and livestock, exacerbate food insecurity and lead to financial losses and escalating farmers and herder clashes.

- **Sudan Savannah:** The Sudan Savannah region is particularly vulnerable to desertification, heat waves, erratic rainfall patterns and increased frequency of extreme weather events which are likely to cause frequent crop and livestock losses, more scarcity of water and pastureland, heighten food insecurity with increases in food prices and cases of malnutrition.

- **Sahel:** The Sahel agroecological zone occurring in the northernmost states of Nigeria will experience increasing risks from extreme weather events, drought, high variability of rainfall, high temperatures causing drying up of water bodies, death of livestock, crop failures, forced migration of residents, desert encroachment, and the spread of waterborne diseases.

The NAP came up with several adaptation strategies that will build the country's resilience. Although only 3 sectors were prioritised in this NAP, the Government of Nigeria recognises that climate change affects every sector of the economy and all the sectors are interconnected, thus the crucial importance of addressing other sectors as well. The following are some key adaptation strategies identified for agriculture, water resources and energy sectors:

- **Agriculture**: In the agriculture sector, the prioritised adaptation strategies to fast-track increased agricultural productivity and build resilience of the economy are: Flood management Agricultural Adaptation Strategy; Heat impacts management Agricultural Adaptation Strategy; and Cross-cutting Agriculture Adaptation Strategy.

- **Water resources:** In the water resources sector, the Water Resource Management Adaptation Strategy is the recommended strategy, which consists of Water supply augmentation and Flood Protection and Demand Management. Water Augmentation strategy has the benefit of expanding access to water; saving time and reducing the burden on women and children; strengthening public health through improved hygiene and sanitation; strengthening drought resilience etc.

- **Energy:** In the energy sector, the Adaptation Strategy includes Grid Hardening and Modernisation, Expanding sustainable energy sources through Renewable Energy Expansion, and decentralising transmission. It also includes strengthening existing infrastructure using Energy Storage & Backup Systems, etc. Nature-based solutions will be implemented as an effective adaptation measure with significant environmental co-benefits, given Nigeria's rich diversity of ecosystem services. In addition, gender-responsive, conflict-sensitive, and peace-building adaptation strategies will be adopted to avoid worsening poverty, economic instability leading to social unrest, and social and cultural disruptions and conflicts. The implementation

of Nigeria's NAP shall adopt strategies to ensure that gender issues play a crucial role in achieving the success of climate change adaptation in Nigeria.

The government of Nigeria will demonstrate sufficient goodwill by mobilising the substantial financial resources required to implement the identified strategies, taking into account the population, landmass and the country's ecological diversity. Potential financial sources to fund the NAP include government (Federal and State) budgets, bilateral and multilateral partners, private investment, Civil Society Organisations, and community contributions. A favourable financing outlook for Nigeria's NAP is supported by its alignment with the goals of the Green Climate Fund (GCF), the Adaptation Fund (AF), and the Special Climate Change Fund (SCCF) of the UNFCCC, which support developing countries like Nigeria in responding to climate change through mitigation and adaptation actions. It is envisaged that the private sector will play a key role in the realisation of NAP goals, particularly in financing investments in adaptation solutions in key priority sectors.

An assessment of previous investment patterns showed that only 11% of the estimated USD 17.7 billion needed annually to meet the conditional NDC targets to reduce emissions by 47% was mobilised. The tracked USD 1.9 billion of climate finance flowing to and within Nigeria is meagre relative to the size of the country's economy and the opportunities for low-carbon development. Nigeria's finance largely came from the public sector (USD 1.5 billion, 77%), while the private sector lagged (USD 0.4 billion, 23%). Evidently, Nigeria has accessed a very meagre amount of international public climate finance to date and continues to face significant challenges in accessing international funds because of the complexity of the application and approval process and the need for mediated access through an MDB. Nigeria needs stronger mobilisation of both domestic resources and international climate finance to meet the expected NDC adaptation targets. The country has a profitable investment environment for adaptation opportunities, as all six adaptation strategies generate positive returns, with Benefit-Cost Ratios (BCRs) above 2 and NPVs strongly positive for both 2050 and 2080.

The Cost–benefit analysis of the adaptation strategies demonstrates strong economic viability of adaptation investments: Flood management in agriculture has Low-cost, high-return with BCR ≈ 4.3; NPV ≈ USD 150 m; Heat-impact management in agriculture is Capital-intensive but transformative with BCR ≈ 2.4; NPV ≈ USD 900m; Cross-cutting agriculture and insurance systems is highly efficient and attractive with BCR ≈ 5.8; NPV ≈ USD 400m; Water supply & flood protection offers Social dividends and quick wins with BCR 3–5.7; NPV ≈ USD 1.2b; Energy-sector adaptation offers the Largest absolute return with NPV ≈ USD 8b. Overall, ₦18.3 trillion (≈ USD 12.4 b) is required by 2050, but each Naira invested yields more than double in returned benefits, confirming high national and social value.

Nigeria will use a diversified, layered financing architecture to implement the NAP, with funding from public finance (25–35 %); international climate finance (30–40 %); and private and blended finance (20–30 %).

Part of the country's initiatives for implementing the NAP include strengthening the institutional and policy arrangements by establishing a National Climate Adaptation Finance Facility (NCAFF) under the Ministry of Finance and the Ministry of Environment to coordinate funding streams and track adaptation outcomes; pursuing Nigeria's accreditation capacity to access GCF and Adaptation Fund finances directly; and introducing climate budget tagging and integrating adaptation into medium-term expenditure frameworks, etc.

Nigeria will ensure the realisation of the goals of the NAP by tracking performance using a Results-Based Monitoring (RBM) tool for monitoring and evaluation across all five strategies. NAP implementation will be monitored annually and reviewed biennially, with stakeholders conducting annual and biennial review workshops. The National Adaptation Plan M&E system will be monitored by project-executing entities and supervised by the Federal Ministry of Environment (the national implementing entities (NIEs)). To support effective implementation of the NAP monitoring and evaluation framework, the plan includes a process of continuous training for relevant stakeholders on the various facets.

Nigeria's NAP will be revised and updated every five-year cycle, taking into account the uncertainty of climate change and the country's diversity. The review and updates will take into account progress achieved and previous adaptation barriers to address them, as well as social and environmental trends, new adaptation issues and policies, and national and international government pledges. The Federal Ministry of Environment, as the national implementing entity (NIE), shall trigger the revision process one year before each five-year cycle to enable the integration of lessons into the national development planning process.

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Acronyms

AF	-	Adaptation Fund
AAP	-	Africa Adaptation Programme
ADCOM	-	Adaptation Communication
BOF	-	Budget Office of the Federation
BIRs	-	budget implementation reports
CAF	-	Cancun Adaptation Framework
CBA	-	community-based adaptation
COP	-	Conference of the Parties
CSA	-	Climate-smart agriculture
CSO	-	Civil Society organization
DCC	-	Department of Climate Change
EBA	-	Ecosystem-based adaptation
ECOWAS	-	Economic Community of West African States
ERGP	-	Economic Recovery and Growth Plan
FME	-	Federal Ministry of Environment
FMWA	-	Federal Ministry of Women Affairs
FRC	-	Fiscal Responsibility Commission
GCF	-	Green Climate Fund
IDP	-	Internally displaced persons
ICCC	-	Inter-Ministerial Committee on Climate Change
KPI	-	Key performance indicators
MBNP	-	Ministry of Budgets and National Planning
MDAs	-	Ministries, Departments, and Agencies
M&E	-	Monitoring and Evaluation
MTSS	-	Medium-Term Sector Strategy
NACA	-	National Agency for the Control of AIDS
NAP	-	National Adaptation Plan
NARF	-	National Agricultural Resilience Framework
NASC	-	National Agricultural Seeds Council
NASPA-CCN	-	National Adaptation Strategy and Plan of Action on Climate

		Change for Nigeria
NASS	-	National Assembly of Nigeria
NCCC	-	National Council on Climate Change
NCF	-	Nigerian Conservation Foundation
NCDP	-	National Council for Development Planning
NDC	-	Nationally Determined Contribution
NES	-	Nigerian Environmental Society
NGO	-	Non-Govenmental Organization
NCCP-RS	-	National Climate Change Policy Response and Strategy
RCP	-	Representative Concentration Pathway
REDD	-	Reducing Emissions from Deforestation and Forest Degradation
UNCBD	-	United Nations Convention on Biodiversity
UNCDD	-	United Nations Convention to Combat Desertification
UNDP	-	United Nations Development Programme
UNFCCC	-	United Nations Framework Convention on Climate Change
UNGA	-	United Nations General Assembly
USAID	-	United States Agency for International Development
SDGs	-	Sustainable Development Goals
SSP	-	Shared Socioeconomic pathways



Chapter One

Background

1.1 Introduction

The United Nations Framework Convention on Climate Change (UNFCCC) supports country Parties in strengthening their adaptive capacities and building resilience as effective means of reducing vulnerabilities, particularly in developing countries. From the COP meetings held at Cancun, Marrakesh, Paris, to Baku (2024), the national adaptation process (NAP) has evolved through adaptation planning, with strengthened implementation in critical sectors such as agriculture, forestry, water resources, housing, transport, health and waste management in Nigeria. The decision of the 2010 Sixteenth Conference of the Parties (COP16) of the UNFCCC emphasised enhanced action through the Cancun Framework to reinforce the global drive towards climate adaptation. This decision established the NAP process to identify and address countries' medium- and long-term adaptation needs, facilitate effective adaptation planning and integrate climate change adaptation within national development processes. The 2015 Paris Agreement further reinforces adaptation planning as vital to socioeconomic development and integral to the global response to climate change. It calls on all countries to initiate the preparation and implementation of NAPs to further reinforce climate adaptation actions, to which Nigeria complied. Thus, global climate governance through the Paris Agreement establishes climate change adaptation as a key approach to addressing climate change. Article 7(9) of the Agreement requires all parties to engage in adaptation planning and implementation, which includes the development of a national adaptation plan.

To enhance its national adaptation response to the impacts of climate change, Nigeria obtained *Readiness and Preparatory Support* from the Green Climate Fund (GCF) to advance its National Adaptation Plan process through a project titled "Strengthening Nigeria's Capacity to Advance the National Adaptation Plan (NAP) Process". The main objective is to strengthen the capacity of Nigeria's government at all levels to implement a NAP process, including planning and budgeting for adaptation to build adaptive capacity and resilience in Nigeria, while also facilitating the integration of climate change into new and existing policies, plans and strategies.

The elaboration of Nigeria's NAP employed a desk review of relevant scientific literature and historical climate data to gain a broad understanding of the context and to identify key trends, vulnerabilities, and where knowledge gaps exist. Field missions and multiple layers of inclusive stakeholder consultations, involving men, women, and other vulnerable groups across the public and private sectors, communities, civil society, indigenous groups, and policymakers, were undertaken. The field missions gathered site-specific data and observed local conditions, allowing a nuanced understanding of vulnerabilities across the country's agroecological zones. The methodology included interviews with identified key stakeholders on diverse issues relating to the development of a functional national adaptation plan of action for the country. In addition, the NAP process benefited from commissioned empirical reports from Nigeria's NAP Readiness and Preparatory Project, such as the Climate Modelling and Vulnerability Assessments, Economic Appraisal, Conflict Sensitivity and Peace-building, Gender Analysis, Finance Strategy, and Monitoring and Evaluation frameworks. These studies were community-driven, emphasising adaptation cases and reflecting the diversity of national ecosystems and sectors, providing evidence-based and gender-sensitive data for the elaboration of the NAP. The findings, observations, and propositions from all the exercises were actively documented and integrated into Nigeria's NAP document and validated in a national stakeholder workshop..

1.2 National Circumstances

1.2.1 Physical characteristics

Nigeria lies roughly between latitudes 4°N and 14°N and longitudes 3°E and 15°E. It is bordered by Niger to the north, Chad to the northeast, Cameroon to the east, and Benin to the west. Nigeria's southern border lies on the Gulf of Guinea, with a coastline of around 853 kilometres. Nigeria has an area of approximately 923,768 km^2 and is a Federal Republic consisting of 36 States and the Federal Capital Territory (FCT) (Fig. 1.1a & 1.1b). The capital of Nigeria is Abuja, in the North-Central part of the country.



Figure 1.1: Location and Geopolitical Zones of Nigeria
Source: (Federal Republic of Nigeria, 2020)

The country's geography is characterised by high plateaus and lowlands. The high plateaus (North Central Plateau, Eastern and Northeastern Highlands, and the Western Uplands) range between 300 meters and 900 meters above sea level, while the lowlands (Sokoto Plains, the Niger-Benue Trough, the Chad Basin, the interior coastal lowlands of western Nigeria, along with the lowlands and scarp lands of southeastern Nigeria and coastlands) are generally less than 300 meters above sea level. The highest elevation in Nigeria (Chappal Waddi) reaches 2,419 m and is located near the Cameroon border. The main watercourses are the Niger River and its tributary and the Benue River with both rivers converging before discharging into the Atlantic Ocean. The rivers are vital for transportation, irrigation and fishing (CCKP, 2021a). Figure 1.2a depicts a map of Nigeria's main river systems. In contrast, Figure 1.2b shows the topographic map of Nigeria and rivers of Nigeria broken down by their major hydrological basin and scaled by their discharge.

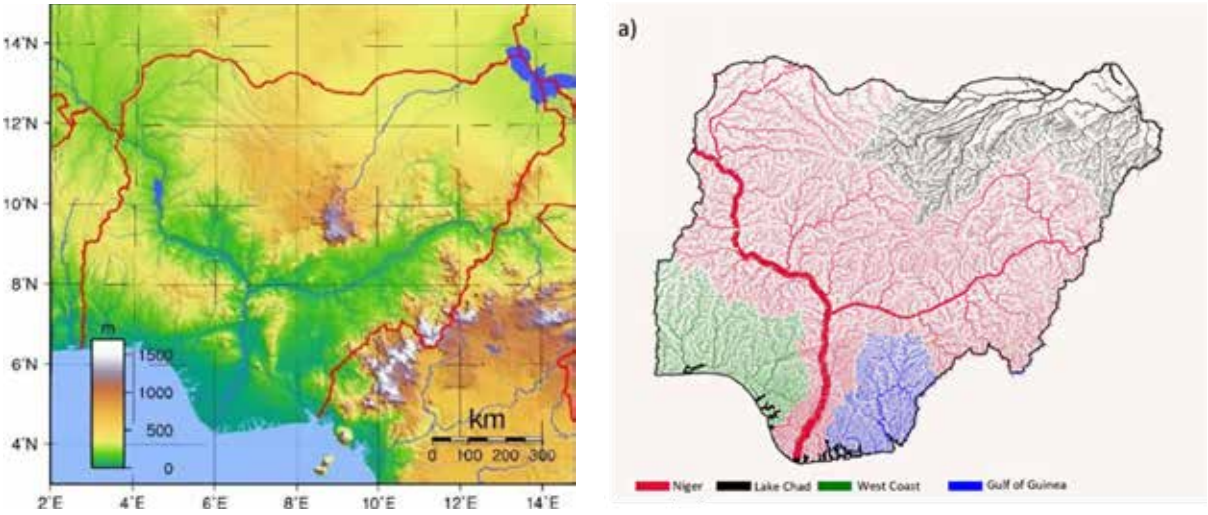


Figure 1.2 a & b Source: left: (Sadalmelik, 2007) right:(PythonMaps, 2023)

Nigeria comprises four major climate categories according to the Köppen-Geiger climatic classification (1961), namely Tropical monsoon (Am) found in the southern part of Nigeria, characterised by high rainfall and temperatures with a distinct dry season; Tropical savanna, dominating the central regions, with a distinct dry season followed by a long wet season; warm deserts (BWh) occurring in the northern part of the country, characterised by very low rainfall and high temperatures; and warm semi-arid (Bsh) climates in the northern part of Nigeria, which form the transition zone from the desert climate towards the savannah (https://climateknowledgeportal.worldbank.org/country/nigeria).

The two major seasons in Nigeria are the dry season (typically from November to March) and the rainy season (typically from April to October) (Ibeabuchi & Itohan-osa, 2023). Rainfall occurs throughout the year, and the mean annual precipitation in Nigeria is 1,162.8 mm. States in the southern regions experience heavier rainfall during the rainy season, with annual rainfall usually above 2,000 mm and reaching 4,000 mm around the Niger Delta (FMEnv, 2020). Northern regions show high annual variation in their rainfall regime, with mean annual precipitation ranging from 500 mm to 750 mm, occurring from June to September. The central regions are governed by a well-defined single rainy season (April to October) and dry season (November to March). The dry season is influenced by the Harmattan wind from the Sahara. Nigeria's mean annual temperature ranges from 17°C to 37°C in the south and from 12°C to 45°C in the north. The highest temperatures in Nigeria occur during the dry season and vary little from the coast to inland areas of the country.

Nigeria's vegetation varies significantly from north to south due to its diverse climatic conditions. The country can be categorised into seven agroecological zones based on climatic, soil, and topographical characteristics, as well as agricultural potential

(Figure 1.3). In the far north, the Sahel Savannah is characterised by sparse vegetation of drought-resistant shrubs and grasses, forming a semi-arid zone that transitions into the Sahara Desert. South of the Sahel, the Sudan Savannah features a mix of grasses and acacia trees, providing a more suitable environment for agriculture and livestock grazing. Further south, the Guinea Savannah has denser grasses and scattered trees, supporting both agriculture and forestry. In the southern part of Nigeria, the rainforest zone is rich in biodiversity, with dense forests and tall trees such as mahogany and iroko. The forest types include saltwater swamps (Mangrove), freshwater swamps, and the high forest. Forest ecosystems are vital sources of livelihoods for millions of people, providing resources such as food, fuel, and medicinal plants. Despite this importance, Nigeria's forests are rapidly declining at a rate of 110,000 ha per annum (0.8%), primarily driven by agricultural expansion, industrial activities, human settlement, infrastructure development, extraction of forest products, production of fuelwood and charcoal, and livestock grazing (Federal Republic of Nigeria, 2020).

The region has longer rainy seasons and is therefore suitable for growing plantation crops such as cocoa and staple foods such as yams, manioc and maize. The coastal region and the Niger Delta are rich in biodiversity with mangrove swamps suitable for fishing and aquaculture (Central Intelligence Agency, 2024).

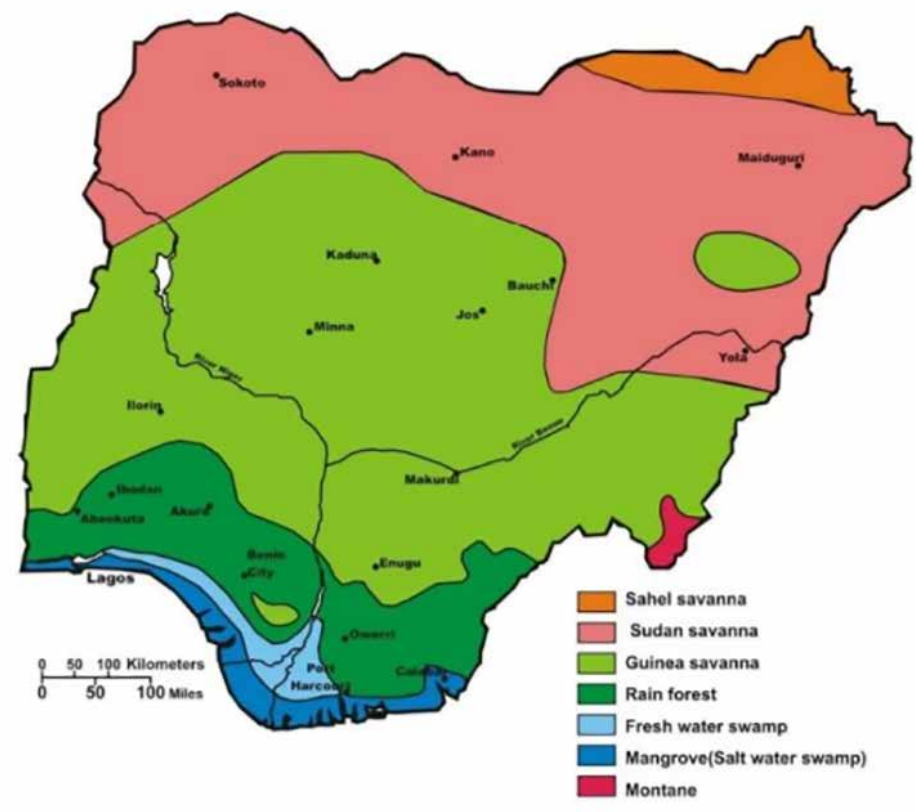


Figure 1.3: Agro-Ecological Zones of Nigeria
Source: (FMEnv, 2020)

1.2.2 Socio-economic

Population

Nigeria is the most populous country in Africa, with approximately 223 million people and an average growth rate of 2.4% in recent years (WB, 2024c). Nigeria's population is projected to reach 263 million by 2035 and over 300 million by 2050 (FMEnv EA, 2025), while the urban population is expected to rise from 50% to 70% by 2050. The demographic profile is heavily skewed toward youth, characterized by a median age of 16.9 years. (Statista, 2024b). Nigeria's population density is high, averaging around 218 people per km2, with significant variation between urban and rural areas (Figure 1.4). The north-east and north-central regions have low population densities, while most of the south and the north-west are very densely populated. Nigeria is known for its cultural and linguistic diversity, with over 250 ethnic groups and 50 languages spoken. The most prominent ethnic groups are Hausa, Ibo, and Yoruba, which together constitute over 40% of the population. Other large ethnic groups include Tiv, Ibibio, Ijaw, Kanuri, Nupe, Gwari, Igala, Jukun, Idoma, Fulani, Edo, Urhobo, and Ijaw. The gender distribution of Nigeria's population is 51% male and 49% female. The two main religions practised in Nigeria are Islam and Christianity, with significant populations practising various indigenous African religions (Ministry of Foreign Affairs, 2024).

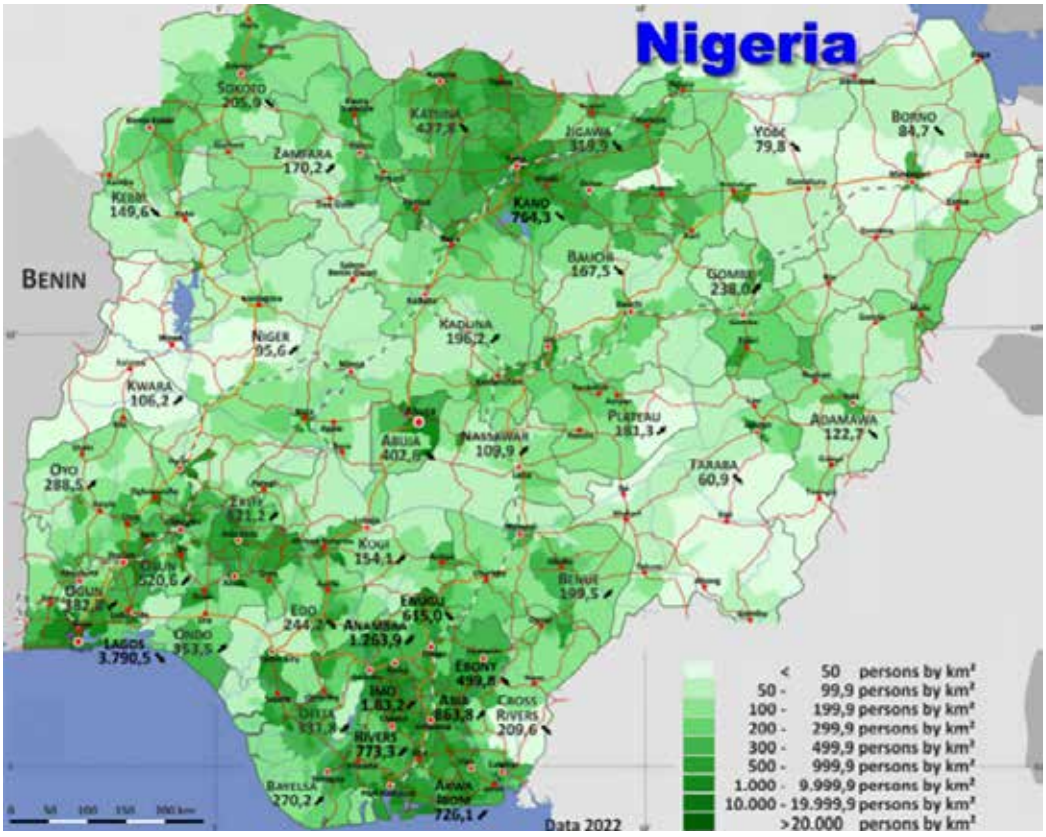


Figure 1.4: Population Density in Nigeria (2022)
Source: (Geo-Ref.net, 2023)

Economy

Nigeria is categorised as a middle-income country and one of the largest economies in Africa (WB, 2024d). The country has a Gross Domestic Product (GDP) of $432.29 billion (2020) and a current annual growth rate of 2.2% in 2019, 01.8% in 2020 and 3-4% per annum in 2024 (FMEnv EA, 2025), while GDP per capita is relatively low, standing at around USD 1,621 (WB, 2024d).

Nigeria is a driving force on the continent through its strategic and financial leadership in the Economic Community of West African States (ECOWAS), which has had its headquarters in Abuja since 1975. Due to the COVID-19 pandemic in 2020, Nigeria faced its most severe recession in decades (estimated GDP contraction of –4%), with negative impacts on overall economic and social development that are anticipated to persist for years (Berger et al., 2021).

Nigeria's economy is highly diversified and can be segmented into agriculture, oil and gas, manufacturing and industry, solid minerals and mining, service sectors, power and energy, infrastructure and transport, financial systems and the environment. Oil and gas are the dominant foreign exchange earners. Nigeria is Africa's biggest crude oil exporter, and the oil and gas (O&G) sector is the cornerstone of Nigeria's economy, accounting for about 10% of GDP, 90% of export revenues and over half of the government's annual revenues. Due to the high dependence of Nigeria's economy on its O&G industry, changes in the international oil market price consequently have a significant impact on Nigeria's overall revenue. Agriculture remains a vital part of the economy, contributing roughly 21% to GDP and employing about 70% of the workforce, primarily in subsistence farming. Nigeria has 70.8 million hectares of agricultural land, including about 37 million hectares of arable land, 6 million hectares of permanent crops and 25 million hectares of permanent meadows and pastures. The staple crops include maize, cassava, guinea corn, yam, beans, millet and rice, and cash crops include cocoa, palm oil, cotton, groundnut and rubber. Nigeria is a major exporter of sorghum, palm oil and cocoa beans.

Livestock, such as poultry, goats, sheep and cattle, are mainly reared by farming families. The north is suitable for cattle rearing, while the south is suited to fish farming (FAO, 2024; Statista, 2024c). Nigeria's agriculture is largely rainfed and dominated by non-homogeneous smallholder farmers, which increases the sector's sensitivity to climate change. The sector has been affected by several shocks, including regular flooding, desertification of arable and grazing lands, extremist insurgency, devaluation of the national currency and conflicts between herders and farmers. Consequently, productivity has declined and food imports have risen due to population growth. For example, between 2016 and 2019, Nigeria's cumulative agricultural imports were four times higher than its agricultural exports. Similarly, livestock production fails to meet domestic demand (FAO, 2024). S

The services sector, including telecommunications, banking, and trade, contributes approximately 52% to GDP. Nigeria boasts a rapidly growing telecommunications sector, with mobile phone penetration exceeding 90%. The manufacturing and industry sector, consisting of food and beverages, cement, textiles, chemicals, steel, consumer goods, and automobile assembly, contributes 9% to the national GDP (Statista, 2024c). Power and energy systems include electricity generation (gas, hydro, renewables such as solar and wind), transmission, and distribution. The energy sector faces significant challenges, with frequent outages and limited coverage in rural regions. Efforts are underway to improve electricity generation and distribution through reforms and investments in renewable energy sources. The solid mineral and mining sector includes gold, limestone, coal, iron ore, bitumen, lead/zinc, and gemstones, which contribute less than 1% to GDP but have great future potential.

Urbanization

Nigeria is rapidly becoming urbanised. In 2023, over 50% of the population was estimated to live in urban areas, and this is projected to increase to 60% and 70% by 2030 and 2050, respectively (WB 2021). Consequently, pressure on infrastructure has become a challenge with rapid urban expansion, as economic activities concentrate primarily in commercial cities like Lagos, Kano, Onitsha, Kaduna, Port Harcourt, etc. Great potential abounds with the expansion of urban centres, through rising average incomes, standards of living, and productivity. On the side of climate change, cities produce more than 60% of global GHG emissions from power generation, transport, and industry. Also, urbanisation attracts higher risks of the impacts of climate hazards such as severe storms and floods, sea-level rise, increased rainfall, severe and frequent storms and floods, and prolonged intense heat. The flood events of 2012, 2018 and 2022 clearly demonstrated the grave danger faced by urban centres such as Lagos, Makurdi, Kaduna, etc. Nigeria evidently needs to institute effective urban management and resilience plans to minimise the risks and impacts of climate change on its growing urban population, using a low-carbon development model in medium- and long-term plans.

Water

Nigeria is considered an economically water-scarce country, necessitating increased investment and strengthened management to meet water demand. Over half of freshwater abstractions in Nigeria come from groundwater, while approximately 41% come from surface water. In 2021, the annual water demand was approximately 5.93 billion m^3, which is projected to rise to 16.58 billion m^3/year by 2030 (FMEnv, 2021a). Although 70% of Nigerians reportedly have access to basic water services, over half of these sources are contaminated, and access is uneven across urban households and in rural areas. In 2021, up to 179 million Nigerians lacked access to safely managed drinking water services, contributing

to high morbidity and mortality rates among children under five (Federal Ministry of Water Resources Nigeria et al., 2022; Blanchard, 2023). Among the challenges are ageing infrastructure, insufficient human resources for maintenance, increased demand from water-intensive industries, and a rapidly growing population (MBNP, 2021b).

Economic development: Development indices

Nigeria has made some socio-economic progress in recent years; however, it is ranked 161st among 189 countries on the Human Development Index (HDI), which reflects life expectancy, level of education, and per capita income (UNDP, 2024). The country has a high poverty rate, with over 80 million Nigerians living below the national poverty line, especially in rural regions (WB n.d.). For this reason, the Government of Nigeria, in its current National Development Plan (2021 – 2025), targets an average annual economic growth of 4.6% to lift 35 million people out of poverty and create 21 million full-time jobs (MBNP, 2025). Welfare distribution in Nigeria is also relatively unequal, as reflected by its Gini coefficient (Eurostat, 2021), which stood at around 0.35 in 2022. Nigeria also faces significant gender disparities (Gender Inequality Index, GII rank 165 in 2022), which limit women's participation in political and economic spheres (UNDP, 2024). Despite these efforts, Nigeria continues to face massive developmental challenges, including the need to diversify the economy, address inadequate infrastructure, and build strong and effective institutions. Other challenges include poor governance and inefficient public financial management systems, as well as inequality in income and opportunities and growing poverty levels.

1.3 Current relevant policies

Nigeria has both national and sectoral policies, whose goals and strategies are reinforced and complemented by the NAP process. The National Adaptation Strategy & Plan of Action on Climate Change (NASPA-CCN) of 2011 outlines adaptation measures across major sectors and serves as a baseline for sectoral adaptation priorities, upon which subsequent policies draw and build. The National Adaptation Plan (NAP) Framework of 2020 (2020-2030) is a central adaptation framework that provides direction for coordinating and mainstreaming the NAP process effectively into development planning. NAPF outlines the institutional roles, implementation approach, monitoring & evaluation plan, and funding strategy, all of which are gender-responsive. Other related national policy documents include the National Climate Change Policy and Response Strategy 2021-2030, the Adaptation Communication (ADCOM) of 2021, and the Gender and Climate Change Action Plan. In 2021, Nigeria developed a Climate Change Act, which establishes the National Council on Climate Change (NCCC) and sets the legal basis for climate policies and climate finance mechanisms. The Conflict Sensitivity and Peace-building



A view of stranded people due to floods following several days of downpours in Kogi State, Nigeria, Oct. 6, 2023. (AP Photo/Fatai Campbell, File)

framework was mainstreamed into the NAP process in 2025 to ensure that adaptation planning in Nigeria considers conflict and peacebuilding dimensions, as several climate risks overlap with conflict risks in Nigeria. The NAP process is consistent with the Economic Recovery & Growth Plan (ERGP) and Nigeria's National Development Plan (NDP) 2021-2050. The NDP has the core goal of driving sustainable and inclusive socioeconomic growth, building resilience, improving quality of life, strengthening institutions, reducing poverty and dependency, and diversifying the economy.

1.4 Mandates guiding climate change adaptation planning

As a signatory to the UNFCCC CAF agreement, Nigeria is required to develop a NAP process as contained in Decision 1/CP.16 under the Cancun Adaptation Framework (CAF, 2010), which addresses, among other issues, enhanced adaptation actions in Nigeria. Nigeria's NAP derives its mandate from this supporting agreement as a response to climate change. Five recently developed national policies, namely: Nigeria's National Adaptation Plan Framework of 2020, the Climate Change Act of 2021, Nigeria's Adaptation Communication (ADCOM) of 2021, the National Climate Change Policy (2021), and the Updated NDC of 2025, create the impetus for the elaboration of Nigeria's NAP to actualise its adaptation objectives and measure the effectiveness of planned strategies. The NAP process provides a suitable platform for a broad-based, coordinated, and coherent national adaptation response. Nigeria's NDC 3.0 contains an adaptation component premised on the need to avert potential risks, including significant losses that could arise from sectors such as

energy and agriculture in the absence of planned adaptation (DFID, 2009; UNFCCC, 2015). Similarly, the Adaptation Communication (2021) highlighted the necessity to develop a NAP that is coherent, effective, and offers a sustainable adaptation approach to the consequences of climate change. Nigeria's NAP process aligns with the adaptation components of the recently updated NDC 3.0. The Climate Change Act (2021) prescribes a national assessment of risks, vulnerabilities, and the extremes of climate change impacts on vulnerable communities and populations and ecosystems; setting out actions for mainstreaming climate change responses into sector functions; and identifying actions for adaptation and mitigation against climate change. The prescriptions of the Act are part of the mandate of the NAP Process.

1.5 Policies related to gender and vulnerable groups

Nigeria has made significant advances in addressing gender issues and promoting inclusivity to build climate resilience. Among the key policies developed to support this are the National Gender Policy (2006, revised in 2022); NASPA-CCN (2011), which explicitly requires adaptation measures to respond to gender-differentiated needs; the National Gender Policy on Agriculture (2019), which highlights that women form a larger proportion of the agricultural workforce and requires equitable access to agricultural inputs and services; and the National Adaptation Plan (NAP) Framework (2020), which identifies gender responsiveness as a requirement. More specifically, Nigeria developed the National Action Plan on Gender and Climate Change (NAPGCC) 2020-2025 to ensure that Nigeria's climate change responses mainstream gender considerations, so that women, youth, persons with disabilities, and other vulnerable groups are included in planning, implementation, and policy processes. In 2024, the Ministry of Women Affairs committed to implementing the Women in Green Economy project.

The recently approved NDC 3.0 (2025) recognises the provisions of NAPGCC and the UNFCCC Gender Action Plan, and commits to ensuring the integration of gender-responsive measures across all sectors for mitigation and adaptation, to deliver equitable climate action benefits, including the use of SMART indicators to track performance and gender-responsive budgeting, among others. These policies and programmes share targets to mainstream gender in climate policy design and implementation; ensure the participation of vulnerable groups; collect sex-disaggregated data on climate impacts, monitoring and evaluation; build the capacity of vulnerable groups; and mobilise for gender-sensitive initiatives. Despite these efforts, gaps remain due to under-funding, disconnects at subnational levels, poor coordination, cultural biases and limited data.

1.6 Policies for relevant systems in the NAP

Nigeria already has sectoral policy commitments and statements in agriculture and food security, energy, water resources, forestry, and health that are directly or indirectly connected to the goals of Nigeria's NAP. The Agriculture Promotion Policy (APP) (2014) promotes activities typical of the NAP, such as scaling up climate-smart practices and investing in irrigation, small-scale water harvesting, and storage. These are congruent with strategic actions in the NAP process.

The energy sector has the National Clean Cooking Policy, which explicitly links clean cooking to health, reduces wood fuel demand, and lowers emissions, delivering both mitigation and adaptation co-benefits. The Renewable Energy and Energy Efficiency Policy also supports the deployment of mini-grids as part of adaptation and access objectives. These policies are consistent with the NAP process, which promotes distributed renewable energy to reduce the vulnerability of rural communities and encourages reducing pressure on forests and household vulnerability through clean cooking.

Nigeria's water sector policy identifies watershed management, climate-resilient water supply, integrated water planning, flood control, integrated catchment management, and irrigation as priority measures. These are also emphasised in the NAP Process.

The Ministry of Health recently launched the Health National Adaptation Plans (HNAP) 2025, which highlight vector-borne diseases, heat stress, nutrition security and health system resilience as core adaptation concerns. These are part of the NAP process focus. Similarly, the forestry and coastal systems promote reforestation, agroforestry and sustainable land management to reduce desertification and protect livelihoods. The National Gender and Climate Change Policy advocates gender responsiveness as an important requirement for NAP actions.

Chapter Two

Nigerian NAP Process

2.1 Vision and Goals

Nigeria's NAP is intended to strengthen the effectiveness of adaptation governance and coordination; enhance the capacity to undertake NAP planning in Nigeria; and analyse climate information and prioritise adaptation options. It also aims to strengthen the capacity to mainstream adaptation measures into national and sectoral policies and plans, provide a financing strategy for NAP implementation, and strengthen national capacity for monitoring, reviewing and reporting on the NAP process.

2.2 Goal and Objectives

The elaboration of the NAP goal is to guide Nigeria through adaptation planning in the medium and long term, to integrate adaptation needs into multi-sectoral policies, strategies, and legislative frameworks that reflect the country-specific context, and to strengthen the coordination process.

Objectives

The elaboration of the NAP Process involved vulnerability assessments, considering various ecological contexts to support the improvement of the adaptive capacity of communities, institutions and sectors in Nigeria. The process also enables objective evaluation of climate change risks and creates an avenue for the coherent integration of adaptation planning into national and sub-national policy frameworks over the medium and long term, aligned with sustainable development targets. Nigeria signed the Paris Agreement on September 22, 2016, and ratified it on May 16, 2017.

Article 7(1) of the UNFCCC sets a clear global goal for adaptation, namely to:

- enhance adaptive capacity and resilience;
- reduce vulnerabilities, with a view to contributing to sustainable development; and
- ensure an adequate adaptation response in the context of holding average global warming well below 2 ˚C and pursuing efforts to hold it below 1.5 ˚C



The Federal Government of Nigeria has demonstrated commitment to reducing vulnerability to the impacts of climate change by developing the NASP-CCN, Nigeria's National Adaptation Plan Framework, in 2020 and by incorporating adaptation objectives into its Nationally Determined Contributions. The country has also established institutional and legal frameworks to meet its international obligations on climate action, including the recently promulgated Climate Change Act 2021.

- The specific objectives of the NAP Process are to:
- reduce vulnerability and impacts of climate change, by building adaptive capacity and resilience;
- facilitate the integration of climate change adaptation, in a coherent manner, into relevant new and existing policies, programmes and activities, in particular development planning processes (eg Agenda 2050) and strategies, within all relevant sectors and at different levels, as appropriate;
- clarify Nigeria's approach to its NAP process which includes articulating the country's vision of climate change adaptation, its adaptation objectives, the principles that will guide adaptation actions, roles and responsibilities among relevant stakeholders.

NAP is a reference framework that brings together adaptation planning efforts across sectors and decision-making levels (i.e., national, state, and local

governments) and aligns the NAP process with existing policies (e.g., Agenda 2050, Economic Recovery & Growth Plan [ERGP], NASPA-CCN, National Climate Change Policy Response and Strategy [NCCPRS]), the Climate Change Act, NDC 3.0 strategies, and adaptation research.

2.3 NAP Scope

The scope of Nigeria's NAP comprises activities that reduce vulnerability, build capacity, and promote climate resilience across communities and economic sectors over the medium and long term, and that align with the existing national development agenda and international commitments developed with the participation of all stakeholders. The economic sectors include: agriculture, water resources, energy, forestry, transport, health and sanitation, commerce and industry, and livelihoods.

2.4 Guiding Principles of the NAP Process

The elaboration of Nigeria's NAP adopted the guidelines of the UNFCCC (2012) and those enunciated in the NAPF (2020). They include:

i. Participation and Ownership: This is to ensure that adaptation strategies are not just theoretical, but practical, locally relevant, and sustained over the long term, transforming NAP from a top-down, bureaucratic document into "whole-of-society" approaches that build resilient communities

ii. Cross-cutting approach: The NAP document shall reflect the cross-cutting nature of climate change as it relates to other development issues such as poverty reduction, job creation and sustainable livelihoods;

iii. Gender –Responsiveness: This principle will ensure that marginalised groups such as women, youth and the physically challenged who are more impacted by climate change are properly carried along in planned adaptation measures to reduce their vulnerability.;

iv. Coherence: This promotes coherence and a concerted approach to adaptation and development planning in national response to climate change

v. Iterative and Flexibility: This is critical to ensure that the NAP is not a static, one-time document, but rather dynamic, evolving, and context-specific strategies.

vi. Conflict sensitivity and peace-building inclusion: This will ensure that the country's NAP process integrates conflict sensitivity and peacebuilding from design through implementation and monitoring and evaluation, effectively building climate resilience and anticipating, preventing, and addressing climate-related conflicts

vii. Empowerment: NAP is designed to continuously identify and address gaps in capacity to achieve adaptation results;

viii. Integrated and Comprehensive Approach: This will ensure that climate change adaptation is not treated as a standalone project but is embedded across all levels of development planning and across multiple sectors. This approach is crucial for building national resilience, optimising resources, and reducing vulnerability to climate impacts in a coherent, sustainable manner.

ix. National Agenda Alignment: It ensures alignment with Nigeria's existing agenda regarding sustainable development and promotes multiple co-benefits to support several key environmental and developmental goals and obligations;

2.5 Regulatory frameworks and institutional arrangements for the NAP formulation

2.5.1 Governance structures/Institutions

The Nigeria Adaptation Communication of 2021 outlines the governance structure for Nigeria's NAP Process. It comprises a first-layer Steering Committee that provides leadership and guidance to the NAP Process, ensuring that its implementation remains aligned with its vision and strategic direction. The Steering Committee includes members of the Inter-ministerial Committee on Climate Change (ICCC) and is chaired by the Director of the DCC, representing the Honourable Minister of Environment. The ICCC is primarily made up of representatives from the various MDAs at the national level. Some ICCC members come from academia, the private sector, and the media. Next is the Technical Working Group, comprising experts working on adaptation issues and experienced researchers from universities and research institutes, to support the Steering Committee. The Technical Working Group will work directly with the Steering Committee (Figure 2.1). The third layer in the organogram will consist of three consultative groups comprising the private sector, CSOs, and international technical and financial partners (donors).

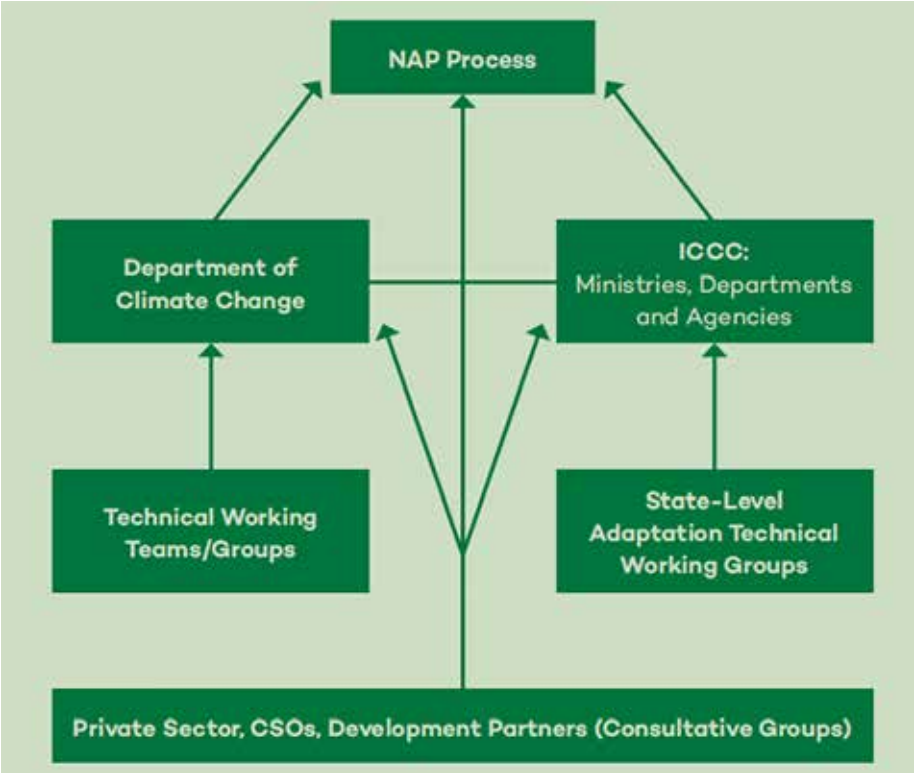


Figure 2.1: Institutional arrangement for Nigeria's NAP process (ADCOM, 2021)

2.5.2 Plans for integrating adaptation and NAPs in development planning and plans

Nigeria's NAP process recognises the existence of the National Development Plan 2050, the Medium-Term Expenditure Frameworks (MTEF), as well as the sectoral policies and strategies. The goal of the NAP aligns with the vision of NDP 2050, which aims to transform Nigeria into an upper-middle-income country, with an average real GDP growth rate of 7 per cent and a nominal GDP of US$11.7 trillion by 2050, by fully engaging all resources to achieve inclusive growth, reduce poverty, achieve social and economic stability, and create a sustainable environment consistent with global concerns about climate change. The NAP is also consistent with the strategies of the MTEF.

The adaptation priorities of the NAP align with the recently submitted NDC 3.0 (2025) recommendations and financing strategy, and therefore their implementation will advance NDC adaptation targets. Among the NAP strategies are supporting the future update of the national development plan to clearly reinforce the NAP within its strategic plan and sectoral targets. The Inter-ministerial Committee on climate change (ICCC), as part of the coordination mechanism for Nigeria's NAP, will, in collaboration with the National Council on Climate Change, facilitate the sectoral integration process. The Federal Ministry of Environment, which coordinates the implementation of the NAP, will also liaise with the National Planning Commission

to effectively integrate and ensure budgetary provision for NAP priorities in the relevant sectors. Already, several MDAs have begun implementing climate-responsive and gender-sensitive adaptation programmes that will facilitate NAP integration. The agriculture sector is introducing drought- and disease-resistant, high-yielding seeds, irrigation resilience, and early warning systems. The health sector has developed the Health-NAP with strategies in disease surveillance, heat, and vector control.

NAP provides for capacity building to support the integration of adaptation priorities into sectoral planning in a cost-effective manner and to demonstrate responsiveness to gender issues, conflict sensitivity, and peace-building. Mobilisation of funds for the implementation of NAP remains a challenge, but the NAP process provides for a public-private adaptation arrangement to complement other finance sources needed for adaptation investments in productive sectors. The NAP process highlights the imperative for gender considerations, conflict-sensitivity, and peace-building in the implementation and monitoring process.

2.6 Processes supporting the development of the National Adaptation Plan

2.6.1 Prioritisation of decision-making processes and adaptation options.

The development of the NAP process in Nigeria was decided through a multi-stakeholder approach across different strata, using science-based data. The Department of Climate Change in the Federal Ministry of Environment was the lead coordinating institution. The exercise involved Ministries, Departments and Agencies (MDAs) at the federal and subnational levels, civil society organisations, academia, private sector actors, and vulnerable community groups—including women, youth and people living with disabilities (PWD)—who were consulted through workshops, interviews and focus group discussions. Decision-making at each stage was guided by the Climate Risk Assessments (Annex 1), the Economic Appraisal (Annex 2), Gender Analysis (Annex 5), the Finance Strategy (Annex 3), Monitoring and Evaluation frameworks (Annex 4), and the Stakeholders' Engagement Strategy and the Conflict Sensitivity and Peace Building Analysis (Annex 8), to ensure that adaptation responses are contextually appropriate. The process took cognisance of Nigeria's broader development priorities as contained in the NDP 2050, the Climate Change Act, and international commitments such as the Paris Agreement, the SDGs and the Sendai Framework for Disaster Risk Reduction. Adaptation options for Nigeria were prioritised using a Multi-Criteria Analysis (MCA), considering the severity of climate risks, economic viability, contribution to poverty reduction, gender equity, alignment with national policies, and potential for environmental sustainability and social co-benefits. Adaptation measures that strengthen climate-resilient agriculture, efficient water resource management, and climate-proofing of the energy sector were given priority. Options

that enhance the resilience of vulnerable populations were also emphasised.

2.6.2 Multi-stakeholder engagement process

Throughout the cycle of elaborating the NAP process, a Stakeholders' Engagement Framework (Annex 7), developed for this purpose, was adopted to ensure that all needs and priorities of the relevant stakeholders were integrated into the NAP. Multi-Stakeholder Focus Group discussions were organised to capture the socio-environmental characteristics of the State/LGA and to ensure the widest possible coverage of issues, as people were able to express themselves freely. Adopting the stakeholders' engagement strategy within the NAP approach enabled the establishment of a wide consultative platform to build buy-in and to garner local and technical knowledge. It also facilitated access to a wide range of personnel with skills, knowledge and experience to enhance climate actions. Among the elements of the stakeholders' engagement framework were setting the engagement objectives; identification and categorisation of stakeholders; stakeholder analysis and mapping; grievance redress mechanism; and monitoring and reporting.

2.6.3 National roadmap and framework

Nigeria's NAP process was developed to capture the national context through a climate vulnerability assessment conducted across all ecological regions, using a gender-responsive approach. This ensures the integration of adaptation planning across national, state and local strata in line with global standards. The guiding principles, which underpin the NAP process, are already outlined in section 2.5. The institutional arrangement describes policies and institutions relevant to the NAP process, including the NASPA-CCN (2011), NAPF (2020), ADCOM 2022, and NDC 3.0 (2025), as well as the coordination structures shown in Figure 2.1. The roadmap of Nigeria's NAP integrates all the Readiness and Support initiatives already described in 2.7.1. Adaptation priorities reflect the strategic and weighted sectoral adaptation plans and measures identified based on relevant data, economic and financial considerations, and the opportunities they offer. The implementation strategy outlines the plan for implementing the adaptation measures to reduce vulnerability across the economic sectors. It identifies the capacity development requirements and the stakeholders' engagement strategy to achieve the adaptation goals. The monitoring and evaluation framework provides tools with gender-responsive indicators for tracking the performance of the adaptation strategies across the sectors.

2.6.4 Identification/stocktaking of desirable and available information

Nigeria's NAP stocktaking provided available information to support evidence-based decision-making. Relevant data were identified and harnessed across the

priority sectors, including local knowledge that reflects Nigeria's diverse ecological zones and socioeconomic peculiarities. Among the documents reviewed were the NDP 2050, NDC 3.0, Climate Change Act, the National Communications, and reports from the National Bureau of Statistics, the Central Bank of Nigeria, NiMET, NIHSA, the World Bank Climate Change Portal, and IPCC, as well as research publications from academia, development partners, and civil society organisations, from which datasets were generated to support climate projections, assess vulnerability conditions, and identify gaps in knowledge. These reviews and engagements generated fact-based reports on CVA, CBA, stakeholder engagement, and gender analysis, providing both climate and socioeconomic data for the NAP process. The stocktaking exercise highlighted the need to build databases, improve coordination among institutions and other stakeholders, and enhance information sharing, guided by the general principle of NAP.

2.6.5 Socio-economic data

Development of NAP is enabled by access to both climate and socio-economic data, which provide insight into future climate risks, the impacts of these risks across communities and sectors, and guidance for identifying development priorities. The climate data accessed included historical climate data, climate projections, hazards and disaster data, and climate-related information on the relevant sectors. Socio-economic data consist of demographic, economic and livelihood, infrastructure and settlements, and governance and institutional data. Subsequently, the climate and socio-economic data were triangulated to produce reliable and logical scenarios; for instance, rainfall projections were linked with agricultural productivity and how this could influence the deployment of climate-smart practices.

2.7 Policies, strategies and plans

In response to the challenge of climate change, Nigeria has in place national and sectoral adaptation-related policies, strategies and plans. This NAP updates the country's medium- and long-term adaptation strategies, drawing on new knowledge to promote technological innovation, align with recent relevant policies' provisions and international commitments, and strengthen coordination and sustainable planning in response to future climate risks and the demand for resilience. Among the key adaptation frameworks are the National Adaptation Strategy and Plan of Action on Climate Change (NASPA-CCN) 2011; Nigeria's National Adaptation Plan Framework 2020; and the National Agricultural Resilience Framework (NARF) 2014.

NASPA-CCN highlighted current and future climate change scenarios in Nigeria and offered sector-specific strategies and actions to address diverse climate change impacts. These sectors include agriculture (crops and livestock), freshwater resources, water resources and fisheries, coastal ecosystems, forests,



Livestock Productivity and Resilience Support Project (Photo Credit: Peoples gazette Farm)

and biodiversity. It provided a platform for integrating adaptation solutions into existing policies and laid the foundation for subsequent adaptation frameworks, such as the NARF NAPF and the National Agricultural Technology and Innovation Policy (NATIP) 2022 - 2027.

NAPF 2020 provided a guide for developing, coordinating and implementing national policies, plans, strategies and legislation to enable Nigeria to address its adaptation needs. The Framework served as a key point of connection for all policies towards climate-resilient development in the country. It was intended to clarify Nigeria's approach to its NAP process; align the NAP process with existing policies; and refocus the NAP process by underscoring the significance of the sectoral approach and the peculiarities of the country's context.

NARF created an avenue for improving resilience in the agricultural sector by mainstreaming adaptation to climate variability and change, using the principles of adaptive management and participatory engagement. It supports measures to reduce deforestation and forest degradation, and to address land degradation and desertification through sustainable land management to increase food security.

NATIP updated the Agriculture Promotion Policy (APP) 2014. The APP was designed to unlock Nigeria's untapped potential by deploying 21st-century knowledge, technology and innovation across the agricultural sector to make Nigeria food and nutrition secure. NATIP is built around 10 intervention pillars and 3 cross-cutting issues, including support for climate-smart agricultural practices using digital agriculture that could increase biodiversity, enrich soils, improve watersheds, promote organic farming and enhance ecosystem support services. This would cumulatively boost productivity and reduce emissions of methane and other GHGs.

The growing losses from extreme weather events and climate hazards, such as floods and drought, are clear indicators that Nigeria has yet to achieve effective

climate resilience across ecological zones and sectors, attributable to weak support for using the NASPA-CCN as an instrument to drive adaptation efforts in the country (NAPF, 2020). In addition, poor coordination and incoherent approaches have been blamed for the slow progress so far and have provided the justification for the approval given to Nigeria by the GCF in 2019 to execute a Readiness and Preparatory initiative for NAP. The development and full implementation of climate adaptation measures are critical for Nigeria to manage the growing challenges of climate change. NAPF elaborated the country's adaptation and mitigation strategies in line with Article 4 of the UNFCCC, towards fostering "low-carbon, high growth economic development path and building a climate-resilient society" (FME, 2012b). It provides a framework for responding to climate change-induced challenges in the country. Other policies, such as the Agricultural Promotion Policy (FMARD, 2016) and the National Gas Policy (Ministry of Petroleum Resources, 2017), do not derive directly from global collaborative responses but reflect best practices from relevant international processes.

The Climate Change Act was enacted to provide the legal framework for achieving low greenhouse gas emissions, including green growth and sustainable economic development in Nigeria. The Act demonstrates the country's commitment to meeting its obligations to combat climate change and provides a comprehensive regulatory framework to achieve its long-term climate goals, encompassing a net-zero target, funding, environmental and economic accountability, and the promotion of climate action. Key highlights of the Act include the establishment of the National Council on Climate Change and the provision for the establishment of a Climate Change Fund; a carbon budget and a national climate change action plan; and the undertaking of vulnerability and risk assessment.

In 2023, Nigeria developed its Long-Term Low Emission Development Strategies (LT-LEDS) – 2060, which supports sustainable social and economic development while reducing greenhouse gas emissions over the medium to long term. Nigeria's LT-LEDS encompasses short-term adaptation measures to address existing climate vulnerabilities and long-term measures to monitor the country's persistent vulnerabilities and address uncertain threats, thereby fostering climate-resilient national development. To achieve climate-resilient development, the LT-LEDS proposes leveraging inter-linkages and jointly undertaking adaptation and resilience efforts alongside mitigation efforts in sectors related to terrestrial and wetland ecosystems, food security and production, and energy.

The number of adaptation policies demonstrates the efforts to mainstream adaptation into national planning in Nigeria. However, the limited advances in the country's coping capacity can be attributed to poor coordination of adaptation actions across Nigerian ministries and agencies at all levels of government. Notably, the quarterly meetings of the Inter-Ministerial Committee on Climate

Change (ICCC), a key coordinating mechanism established to promote engagement on Nigeria's climate response across sectors, have been intermittent and ad hoc.

Nigeria's NDC 3.0 (2025) includes an adaptation component premised on the need to avert potential risks, including significant losses in sectors such as energy and agriculture in the absence of planned adaptation (DFID, 2009; UNFCCC, 2015; NCCC 2025). Nigeria's NAP process aligns with the adaptation components of the NDC 3.0. The range of adaptation options in the NDC is consistent with the NAPF, underscoring the importance of the incremental evolution of Nigeria's adaptation responses.

In the NDP 2050, the environmental and regional development section specifically emphasised the environment and climate change, which is among the seven pillars/ structures of the NDP's blueprint transformational agenda for Nigeria.

2.8 Sectoral Studies

Several reports from preceding empirical exercises in the National Adaptation Plan Readiness Project, such as the Climate Change Modelling and Climate Change Risk Assessment (CRA), Economic Appraisal for the National Adaptation Plan (NAP) for Nigeria; Gender Analysis Report to inform the National Adaptation Plan (NAP) process in Nigeria; and Integrating Conflict Sensitivity and Peace-building Into Nigeria's National Adaptation Plan Process, informed the synthesis of Nigeria's NAP after a critical desk review. The key highlights of these reports are below:

2.8.1 Climate Risk Assessment (CRA).

Information and data from the CRA analysis were extensively integrated into Nigeria's NAP programme. The CRA covered Nigeria's five major agro-ecological zones, namely the Swamp Forest zone, the Tropical Rainforest zone, the Guinea savannah zone, the Sudan savannah zone, and the Sahel zone, as well as urban areas, using nine representative States, on which climate impacts were analysed for certain key sectors across the country (Figure 2.2). The sectors are agriculture (crop production and livestock), forestry, water, health and sanitation, energy and transport. The CRA used impact storylines (ISLs) of previous major climate hazard events in Nigeria to examine the differentiated current and future impacts of the climate hazards (rain-induced flooding, heat waves, drought, and sea-level rise) for each agro-ecological zone and key sectors, with particular focus on future climate projections (Annex 1). The ISLs focused on four past events that took place in different parts of Nigeria, namely: the 2022 flood period in the Central-South region, the 2024 extreme urban heat in Lagos, the drought events in the Sahel region over the last century (particularly since the 1980s), and sea-level rise in coastal areas.

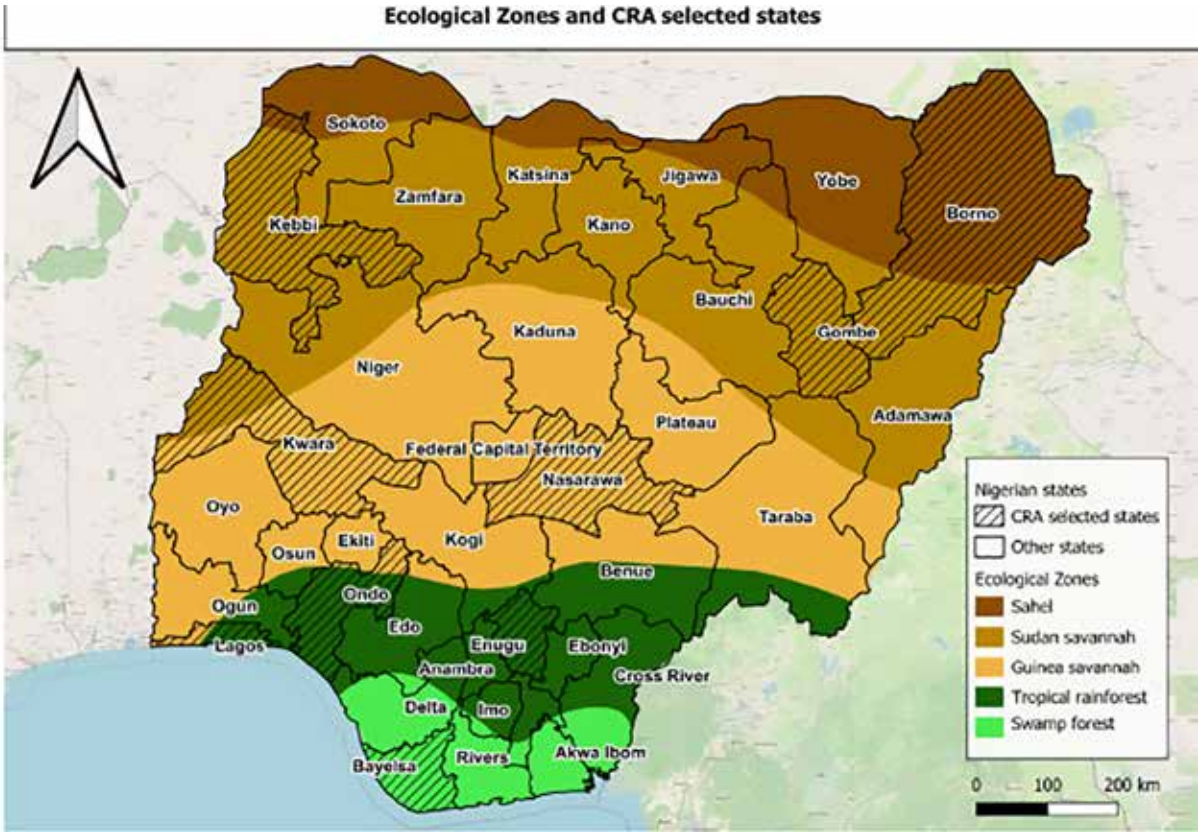


Figure 2.2. Distribution of states studied and ecological zones across Nigeria (FME-CRA, 2025)

The CRA used downscaled or area-based climate change projections to explore a range of potential future conditions under different trajectories of greenhouse gas (GHG) emissions and socio-economic developments. This created scenarios that are part of the Shared Socioeconomic Pathways (SSPs), which integrate climate projections with societal trends (IPCC Sixth Assessment Report on climate change in 2021). Two scenarios within the Shared Socioeconomic Pathways (SSPs) framework were examined to provide a comprehensive understanding of the potential impacts of climate change under different levels of emissions and policy responses, helping to inform strategies for mitigation and adaptation. They are:

- SSP2-4.5: This scenario represents an intermediate pathway for GHG emissions. It assumes the world continues along historical development patterns, with moderate efforts to mitigate climate change. In this scenario, global CO_2 emissions stabilise around mid-century but do not decline significantly, leading to a continued rise in atmospheric GHG concentrations. The "4.5" refers to the radiative forcing level of 4.5 watts per square metre (W/m^2) by the year 2100, which corresponds to a global temperature increase of approximately 2.5°C to 3°C above pre-industrial levels.

- SSP5-8.5: This scenario represents a high-end pathway characterised by very high GHG emissions, driven by fossil-fuel-intensive development and

a lack of effective climate policies. It assumes rapid economic growth, high energy demand, and reliance on carbon-intensive technologies, leading to a significant increase in atmospheric GHG concentrations. The "8.5" refers to the radiative forcing level of 8.5 W/m^2 by 2100, which could result in a global temperature rise of 4°C or more above pre-industrial levels.

By examining these two scenarios, the analysis provided a comprehensive understanding of the potential impacts of climate change across different levels of emissions and policy responses, informing strategies for mitigation and adaptation.

However, some of the sources used in this framework referred to Representative Concentration Pathways (RCP) scenarios (earlier scenarios developed by the IPCC). They represent different levels of greenhouse gas (GHG) concentrations and the resulting radiative forcing (energy imbalance) in the atmosphere by 2100. Each RCP scenario is named after the expected radiative forcing level in watts per square metre by 2100. They are:

- RCP2.6: Very low emissions, strong mitigation efforts.
- RCP4.5: Intermediate emissions and moderate mitigation (aligned with SSP2-4.5), under which global warming stabilises mid-century. For Nigeria, this implies a temperature rise of about +2.0 to +2.5 °C by 2050 relative to pre-industrial levels, with moderate increases in wet-season rainfall (+5%) and more frequent heavy rainfall days.
- RCP6.0: Higher emissions, limited mitigation.
- RCP8.5: Very high emissions pathway, the "business-as-usual" scenario (aligned with SSP5-8.5), representing limited global mitigation. For Nigeria, this scenario suggests +2.5 to +3.2 °C by 2050, and up to +4.5 °C by 2080, with wetter conditions in the south (+10% rainfall), more intense droughts in the north, and sea level rise in the range of 31-39 cm by 2080.

The 30-year period 1981-2010 served as the reference period. For future scenarios, the following periods were used: near-term (2011-2040), mid-term (2041-2070), and long-term (2071-2100). To refine large-scale climate projections, statistical downscaling was applied to the Coupled Model Intercomparison Project Phase 6 (CMIP6) climate scenarios, aligning them with high-resolution reference data (0.05° for precipitation and 1° for temperature). A Cumulative Distribution Function Transform (CDFT) method 2 was used to adjust daily model data and correct the seasonal cycle at the monthly scale. A multi-model mean was adopted to minimise biases inherent in individual climate models, which arise from differing physical representations of the climate system. The accuracy of these corrections

was verified by evaluating annual averages and seasonal cycles of key climate variables. Additionally, a climate screening was conducted in collaboration with NiMet (Nigerian Meteorological Agency) to determine the most relevant climate factors for impact assessment. Seven climate indicators, as defined by the ETCCDI (Expert Team on Climate Change Detection and Indices) (Karl et al., 1999), were selected for further analysis (see Table 2.1).

Precipitation			
R10mm	Number of Heavy rainy days	No. of days for which PR > 10mm/day	Days
R20mm	Number of Very Heavy rainy days	No. of days for which PR > 20mm/day	Days
R50mm	Number of Extremely Heavy rainy days	No. of days for which PR > 50mm/day	Days
Drought			
CDD	Consecutive dry days	Max. annual number of consecutive dry days (PR < 1mm)	Days
SPI	Standardized Precipitation index	Measure of 'drought' on the scale of 3 and 12 months	Days
Heatwave			
WSDI	Warm Spell Duration Index	Annual no. days contributing to events in which 6 or more consecutive days experience Tmax > 90 th percentile	Days
HWD	Heat wave duration	The length of the longest duration of the heat wave	Days

Table 2.1: List of hazards and indicators of interest for the Nigeria's Climate modeling and Climate Risk Assessment (Source: FME-CRA 2025)

2.8.2 Economic Appraisal for the NAP Process

Nigeria's NAP integrated the findings of a rigorous cost-benefit analysis (CBA) of Nigeria's priority adaptation strategies, conducted within the framework of its National Adaptation Plan (NAP) Process. The CBA aims to guide strategic decision-making by policymakers, development partners, and stakeholders involved in climate adaptation planning, identifying cost-effective, high-impact strategies that can enhance community resilience, stimulate economic growth, and promote sustainable development in Nigeria.

The CBA analysis focused on the agriculture, water resources, and energy sectors, identified as highly vulnerable to the impacts of climate change (Annex 2). The analysis adopted the same two Representative Concentration Pathways, RCP 4.5 (aligned with SSP2-4.5) and RCP 8.5 (aligned with SSP5-8.5), used in the Climate Risk Assessment (FMEnv-EA, 2025); used 2024 as the reference year for valuation;

set the exchange rate at ₦1,472.1475/US$ (the average exchange rate of 2024); and applied a 10% social discount rate with an 8% sensitivity. It also used 3-4% per annum GDP growth; a 12% long-term average inflation rate; and population projections of 223 million in 2024, projected to reach 263 million by 2035 and over 300 million by 2050 as baseline economic parameters.

The CBA considered 5 adaptation strategies across the agriculture, water resources, and energy sectors, identified through the CRA impact-chain process, sectoral policy reviews, and stakeholder consultations, as shown in Table 2.2. Following identification of adaptation options, a structured cost estimation process was undertaken, using upfront costs (CAPEX), operating costs (O&M), and expected economic, social, and environmental benefits over a 35-year horizon. The cost-benefit analysis (CBA) applied two core profitability indicators - the Net Present Value (NPV) and the Benefit-Cost Ratio (BCR) - to assess the economic viability of the prioritised climate adaptation strategies in Nigeria (CBA, 2025). The relative performance and attractiveness of the strategies were also ranked separately using Benefit-Cost Ratios (BCRs) and Net Present Values (NPVs) as yardsticks under both the 2050 and 2080 horizons.

Nr.	Adaptation Strategy	Adaptation Options	Type of Cost	Type of Benefit
1	Floods management Agricultural Adaptation Strategy	Flood management and control systems (e.g. flood-resistant ranching, community radios) Post-flood livestock relocation and recovery	Investment (CAPEX) O&M (annual management/maintenance)	Avoided damage to crops, livestock, and rural infrastructure
2	Heat impacts management Agricultural Adaptation Strategy	Livestock cooling & climate management Post-harvest cooling and storage systems Irrigation expansion Climate-resilient crops (including R&D) Comprehensive fish and aquaculture adaptation	Investment (CAPEX) O&M (annual) R&D (variety development, breeding, extension)	Productivity gains Avoided post-harvest losses Additional crop output Avoided drought-related losses Increased aquaculture revenues
3	Cross-cutting Agriculture Adaptation Strategy	Smart agriculture production management Agricultural insurance	Investment (CAPEX) O&M (annual)	Ecosystem services (soil fertility, water regulation) Social co-benefits (migration reduction, empowerment) Risk transfer and indemnities for farmers
4	Water Resource Management Adaptation Strategy	Water supply augmentation (rainwater harvesting, groundwater, treatment, recycling) Flood protection & demand management (infrastructure, early warning, emergency response)	Investment (CAPEX) O&M (annual maintenance)	Additional population with safe water access Improved public health and hygiene Avoided infrastructure damages Improved water quality Reduced erosion and land losses
5	Energy Sector Adaptation Strategy	Renewable energy expansion (solar, wind) Grid hardening and modernization Energy storage & backup systems	Investment (CAPEX) O&M (annual maintenance) Grid modernization costs	Fuel cost savings; Increased energy revenues Avoided outage losses; Direct and indirect job creation; Avoided GHG emissions Grid stability and efficiency gains

Table 2.2: Types of cost and benefits analysed for each adaptation strategy (Source: FMEnv EA, 2025)

2.8.3 Climate Mobility Analysis for the NAP

Climate mobility refers to the movement of people motivated by the adverse effects of sudden- or slow-onset climate impacts. It occurs both within and across national borders and involves varying levels of constraints, agency, and vulnerability, encompassing various forms of movement, including forced displacement, migration and planned relocation, as well as voluntary and involuntary immobility. Climate mobility occurs over different distances and can be temporary, recurrent,

or permanent.

The inclusion of climate mobility in the NAP is based on an analysis of modelling projections from the Africa Climate Mobility Initiative (ACMI), a partnership between the African Union Commission, the World Bank, and the United Nations, supported by the Global Centre for Climate Mobility (GCCM). Launched in 2021, the ACMI aims to create a common evidence base and policy agenda to address climate-forced migration and displacement in Africa while harnessing climate mobility to strengthen the continent's collective resilience and development (Amakrane et al., 2023, p. 16). The ACMI used a gravity model, drawing on ISIMIP climate-impact data, to project future shifts in population distribution under combined development (SSP1, SSP3) and climate (RCP2.6, RCP6.0) scenarios, accounting for factors such as flooding, crop and rangeland productivity, water security, sea-level rise, and conflict.

The analysis of climate mobility for this NAP is further based on primary data collected in the Ajegunle suburb of Lagos, a waterside informal settlement of around 550,000 residents, mostly internal migrants, where research was conducted in 2021, including 110 survey interviews and three focus group discussions (Mixed Migration Centre, 2023). It also draws on a literature review of the existing academic and grey literature on climate risks as a driver of human mobility in Nigeria, the impacts of climate mobility on those who move and the communities involved, and the consideration of the issue in existing policy frameworks (GCCM, 2026).

Reference to climate mobility in the NAP complements Nigeria's inclusion of climate mobility dynamics in other key climate strategies, such as the NDC 3.0, the Just Transition Plan, and subnational plans such as Lagos' Resilience strategy. It further aligns Nigeria's climate policy framework with the country's obligations under the Kampala Declaration on Internal Displacement and with commitments made in the Kampala Ministerial Declaration on Migration, Environment and Climate Change and the Global Principles for Addressing Climate Mobility. By including climate mobility in the NAP, Nigeria can strengthen policy integration across sectors to support positive adaptation journeys for some of its most vulnerable communities. While most Nigerians are likely to have a strong attachment to their land and homes, acknowledging and proactively supporting communities in exploring climate mobility as an adaptation strategy will be critical for social cohesion and positive community transition in the face of adverse climate impacts.

2.8.4 Gender Analysis for NAP

Vulnerability to climate change at different spatial levels is exacerbated by inequity and marginalisation linked to gender, ethnicity, income, or combinations thereof (high confidence), especially for many Indigenous Peoples and local communities

(high confidence) (IPCC, 2023). Gender issues play a crucial role in shaping the effectiveness and success of climate change adaptation strategies because women and men have distinct roles and responsibilities in Nigeria; have limited access to productive resources, including land, credit, and technology; and have limited access to education and other economic benefits – which exacerbate pre-existing inequalities and vulnerabilities, hampering women's ability to adapt effectively to climate change impacts (Akinbami et al., 2019; Chingarande et al., 2020). To this end, Nigeria incorporated gender dynamics into the National Adaptation Plan process to foster inclusive climate resilience, promote social justice, and ensure sustainable development (Jangbrand, 2022; Dazé, 2020; Sriram, 2018; Nelson, 2015). The gender analysis adopted a desk review of relevant literature, national policies, laws, strategies, and plans, and consultations with key stakeholders, comprising state and non-state actors.

The gender analysis drew on the National Action Plan on Gender and Climate Change (NAPGCC, FMEnv, 2020b) and on experiences recorded during the implementation of gender-responsive climate actions in Nigeria and across the globe to identify weaknesses and integrate country-specific recommendations. Nigeria still has a great deal of work to do to achieve gender equality, given the strong influence of cultural, religious, and traditional beliefs on gender norms and roles in the country. It also took into account relevant existing policies in Nigeria, including: National Climate Change Policy for Nigeria (2021-2030); Climate Change Act 2021; Nigeria's Adaptation Communication to the UFCCC (2021); NAPF (2020); LTV-2050; LT-LEDS (2021); and NASPA-CCN (2011). Other pivotal frameworks that guide gender decisions in Nigeria and were acknowledged in the NAP Process are the National Gender Policy (2021-2026) and NAPGCC (2020). Key gender issues identified in NAPGCC for consideration in the elaboration of the NAP Process include:

i. Recognition of gender differences in adaptation needs, opportunities and capacities

ii. Equitable participation and influence by women and men in adaptation decision-making processes;

iii. Equitable access to financial resources and other benefits resulting from investments in adaptation between women and men (FMEnv-GE, 2025).

Some highlighted recommendations from the Gender Analysis which were adopted in the NAP Process to address gender considerations, include:

i. Applying gender considerations in selecting climate experts and researchers for conducting climate scenario modelling;

ii. Determining exposure, sensitivity, and adaptive capacity of men and women and other vulnerable groups during vulnerability assessment;



View of Idanre Town from Idanre Hill (https://en.wikipedia.org/wiki/Idanre)

iii. Conducting a comprehensive gender review of existing policies, plans, and budgets to identify gaps and opportunities for integrating gender-responsive adaptation into various sectors;

iv. Commissioning gender dimension studies to assess the economic costs and benefits of different adaptation options and measure and utilise them to inform decision-making processes;

v. Conducting a comprehensive stakeholder mapping exercise to identify and continue to involve key stakeholders;

vi. Ensuring participation of gender actors, women-led focus groups/rights organisations, youths, and PWDs in the validation of NAP;

vii. Adopting Gender-responsive approaches in developing tracking and monitoring tools for public expenditure for change adaptation in Nigeria, including national budgets and all funding sources (Annex 5).

2.8.5 Conflict Sensitivity and Peace-Building in NAP

Climate change could lead to or exacerbate conflict through various cascading impacts. Therefore, Nigeria developed a framework for integrating conflict sensitivity and peacebuilding into Nigeria's National Adaptation Plan Process, with support from the NAP Global Network (Annex 8). Understanding these cascading impacts is crucial for developing effective adaptation strategies, reducing the risk of conflict, and promoting peace and stability. The initiative began by integrating conflict analysis tools and methods into the CRA approach, including participatory risk assessments and the organisation of CRA fieldwork and conflict analysis workshops in some States. Consultations and counselling were organised for the

NAP Team and Nigeria's NAP Secretariat on ways to integrate conflict sensitivity and conflict resolution into the process, and on entry points for mainstreaming peacebuilding and conflict-sensitivity issues in the overall NAP process. According to the Framework, climate impacts such as droughts, desertification, and changing weather patterns may lead to water and food scarcity, creating tensions and conflicts over access to these resources. Five logical steps recommended in the Integrating Conflict Sensitivity and Peace-building document and mainstreamed into Nigeria's NAP process are to:

i. strengthen existing and develop new institutional and policy frameworks for conflict-sensitive adaptation;

ii. integrate conflict sensitivity into CRA and NAP formulation processes;

iii. ensure NAP implementation is conflict sensitive by providing climate resilient infrastructure and building social cohesion;

iv. include peacebuilding provisions as part of NAP implementation by prioritising adaptation measures that reduce conflicts and support peacebuilding; and

v. include peacebuilding markers into the monitoring and evaluation (M&E) framework (FMEev-ICPB, 2025)



People walk through floodwaters after heavy rainfall in Hadeja, Nigeria, Sept. 19, 2022 (AP Photo, File)

Chapter Three

Climate Risks and Vulnerabilities

3.1 Conceptual Framework

Nigeria is highly vulnerable to the impacts of climate change. Consequently, climate-related extreme weather events are severely affecting Nigeria's economy. The aim of the NAP process is to identify effective measures that will reduce the country's vulnerability to climate change and enhance its adaptive capacity across sectors and communities, thereby protecting lives and livelihoods and sustaining development gains. Given resource constraints, the NAP process is streamlined to focus on priority sectors, including highly vulnerable sectors and cross-cutting sectors such as agriculture, energy, health, water resources, ecosystems, and biodiversity, as well as conflict-sensitive adaptation. Achieving the NAP goal depends heavily on effective finance mobilisation, which could come from the public sector, the private sector, international finance, or a blended finance mechanism.

Phasing the implementation of the NAP into medium- and long-term strategies supports financial mobilisation and strengthens national capacity to integrate flexibility into the process. Cross-cutting themes such as gender, youth, indigenous knowledge, disaster risk reduction, and governance were integrated to achieve wider participation and ownership. Defining the institutional arrangements and coordination framework is crucial for successful implementation and for achieving the adaptation goals. Monitoring, Reporting and Verification (MRV), with appropriate indicators, is built into the NAP process, allowing tracking of progress and financial flows. An effective monitoring and evaluation plan, with tools such as results-based frameworks, is useful for periodic reviews, learning loops, and updating the NAP. Reporting patterns, communication, and stakeholder engagement strategies are also useful for successful planning and implementation of the NAP.

3.2 Boundary Conditions for the Assessment

Nigeria's NAP process focuses on agriculture, water resources, and energy as the priority sectors for adaptation actions. However, the process does not neglect the high vulnerability and adaptation needs of other key economic sectors, such as

health, transport, commerce and industry, and education. The NAP also highlights the relevance of other critical cross-cutting issues, such as women and gender equality, youth, persons with disabilities, and private sector engagement. Relevant strategies were identified on the basis of IPCC SSP2-4.5 and SSP5-8.5 and in alignment with national aspirations. The national context was first examined in relation to the physical implementation of the adaptation strategies, strengthening institutional capacities, the need to mobilise resources, social inclusion and equity, attracting transformational investments, and creating the enabling environment to support the implementation of the NAP. Women, youth, older people, persons with disabilities, ethnic communities, socially disadvantaged groups, and the private sector are integral considerations across all sectors to ensure participatory NAP implementation.

3.3 Synergy with SDGs, Sendai Framework for DRR, and other relevant regional and national frameworks

Nigeria's NAP aligns with and strongly complements the Sendai Framework 2015-2030, whose goal is to "prevent new and reduce existing disaster risks through the implementation of integrated and inclusive economic, structural, legal, social, health, cultural, educational, environmental, technological, political and institutional measures that prevent and reduce hazard exposure and vulnerability to disaster, increase preparedness for response and recovery, and thus strengthen resilience". Nigeria further demonstrated its commitment to the Sendai Framework by developing a National Disaster Risk Management Policy (NDRMP) in 2018 to strengthen its capacity to reduce vulnerability, prevent the creation of new disaster risks, and promote the accountability needed at all levels. The NAP approach is consistent with objective 3 of the NDRMP, which aims to reduce the vulnerability of the poor by building capacity and resilience to disaster risk.

Nigeria is committed to the UN Sustainable Development Goals (SDGs) for 2015-2030, which envision a future that is economically sustainable, socially inclusive and environmentally resilient. Nigeria's SDG framework was established through the Nigeria Governor's Forum. The NAP aligns with the 17 SDGs, which represent a universal call to action to end poverty in all its forms, safeguard the planet and ensure that all peoples enjoy peace and prosperity by 2030. It particularly aligns with SDGs 1 (No Poverty), 2 (Zero Hunger), 5 (Gender Equality), 6 (Clean Water and Sanitation), 7 (Affordable and Clean Energy), and 13 (Climate Action). Nigeria's NAP is a national instrument for achieving these goals through adaptive pathways to reach SDG targets by 2030. All the Adaptation Priorities emphasised in Nigeria's NAP directly or indirectly align with the SDGs' strategic interventions.

3.4 Baseline Climate Information (1981-2010 reference period)

3.4.1 Greenhouse gas (GHG) concentration and emission

Nigeria has submitted three (3) National Communications (NCs) to the UNFCCC, all containing Greenhouse Gas (GHG) inventories. These include the Initial National Communication (2003), Second National Communication (2014), and Third National Communication (2020). In addition to NCs, Nigeria has submitted at least two Biennial Update Reports (BUR1 in 2018 and BUR2) to provide regular, updated GHG inventories. A detailed National Inventory Report (NIR) for 2000-2017 was submitted in 2021. Nigeria has also submitted three Nationally Determined Contributions (NDCs) with updated GHG data. The latest NDC.3 was submitted in 2025. These submissions indicate a steady rise in the concentration of GHGs in Nigeria's atmosphere. Total net national emissions (Table ES 1) for the three direct GHGs, including removals, amounted to 609,783.8 Gg CO2-eq across the four IPCC sectors. AFOLU led the sectors with 366,733.9 Gg CO2-eq (60.1%) of total aggregated emissions, followed by Energy with 206,452 Gg CO2-eq (33.9%), Waste with 23,330.3 Gg CO2-eq (3.8%), and the remaining 13,267.1 Gg CO2-eq (2.2%) from IPPU. CO2 was responsible for 72.9% of the emissions, CH4 for 21.2%, and N2O for 5.8%. Specifically, the total emissions from the AFOLU sector were 371,022 Gg CO2-eq in 2016, with a removal of 4,288 Gg CO2-eq under HWP, giving net emissions of 366,734 Gg CO2-eq, which is the highest for the entire period under consideration. The net emissions increased by about 25.6% in 2016 compared with the year 2000 emissions. In the energy sector, the total aggregated emissions increased from 116,057.44 Gg CO2-eq in the year 2000 to 206,452.45 Gg CO2-eq in 2016. Of the two main sources, fuel combustion activities recorded an increase of 206% in average carbon dioxide (CO2).

3.4.2 Temperature

Temperature, rainfall and sea-level rise are important drivers of climate change. In Nigeria, there's significant evidence that these climate elements have continued to change over time. Observed temperature trends suggest that mean annual temperature has increased over time across the country's coastal, rainforest, Sudan, Guinea savanna and Sahel areas (Onyeneke et al., 2020; Ayanlade et al., 2018). The observed pattern of Nigeria's average temperature also indicates an increasing trend over a period of 122 years (1901 – 2022), using data from the World Bank Climate Change Knowledge Portal (see Figure 3.1).

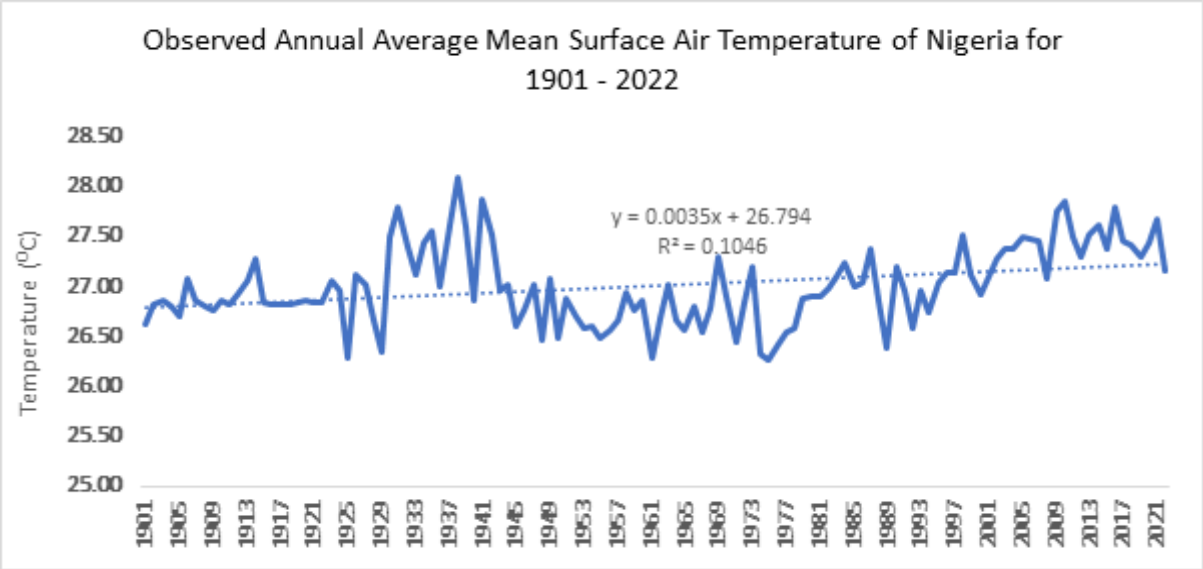


Figure 3.1: Observed Annual Average Mean Surface Air Temperature of Nigeria for 1901 – 2022
Source: World Bank Group (2024a). World Bank Climate Change Knowledge Portal https://climateknowledgeportal.worldbank.org/country/nigeria/climate-data-historical

As shown in the CRA report, Nigeria experienced an average of about 16 heatwave days per year during the reference period (1981-2010). Furthermore, the number of heatwave days is highest in some ecological zones, such as the Sahel, and lowest in the south, such as the mangrove/swamp forest zone. During the reference period, the longest heatwave of the year lasted an average of 25 days. However, in contrast to the number of heatwave days, the heatwave duration was longer in the southern ecological zones, such as the Guinean savannah, tropical rainforest and mangrove swamp forest.

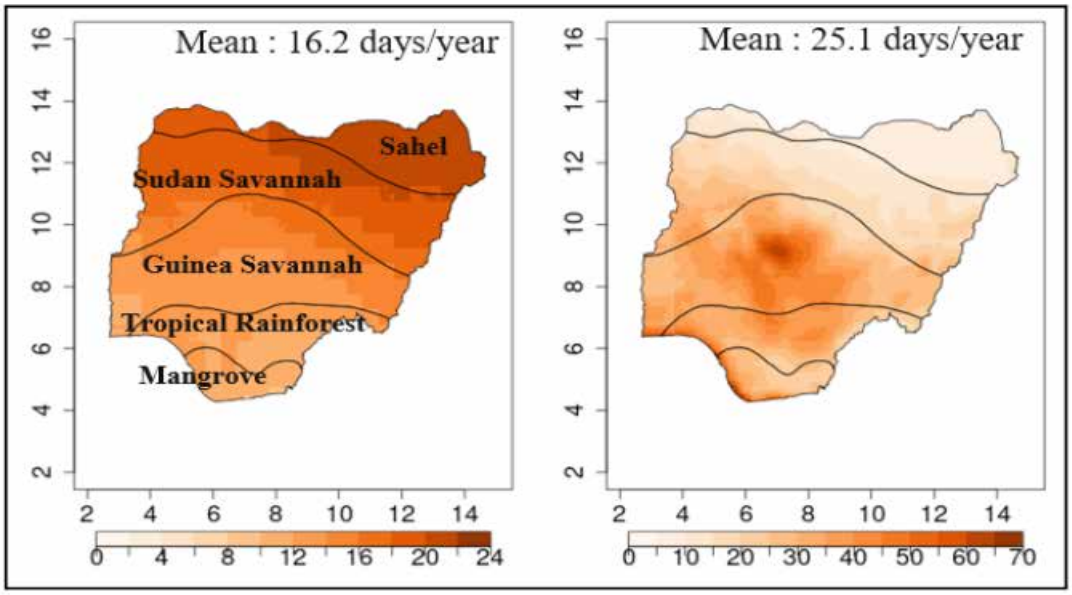


Figure 3.2 Average and highest heatwave days per year during the reference period (1981-2010) Source: CRA Report 2025



The Nigerian government is assessing the vulnerability of sectors, regions and communities to identify the extent to which they are affected by the effects of climate change. (Nov. 2019, NAP Global Network)

3.4.3 Rainfall

Rainfall has shown significant variability in amount and pattern over time, including its distribution across the different agroecological zones. Ayanlade et al. (2018) observed that the rainforest agro-climatic zone has higher rainfall (both in volume and number of rainy days) than the savanna regions. All the major agroecological zones in Nigeria show interannual variability in rainfall amount (World Bank 2024a; Onyeneke et al., 2020) (Figure 3.2).

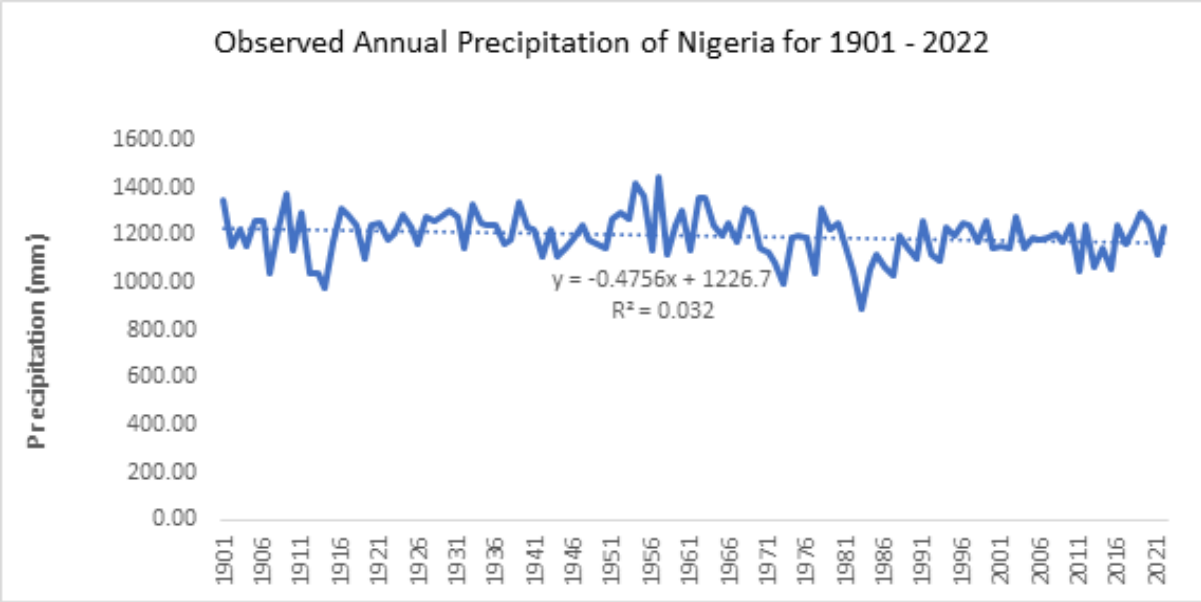


Figure 3.3: Observed Annual Precipitation of Nigeria for 1901 - 2022
Source: World Bank Group (2024a). World Bank Climate Change Knowledge Portal https://climateknowledgeportal.worldbank.org/country/nigeria/climate-data-historical

Nigeria records an average of around 46 heavy rainy days (R10 mm) per year, mostly in the tropical rainforest and mangrove/swamp forest (southernmost ecological zones), within the reference period, according to the FMEnv-CRA (2025). The same applies to heavy rainy days (R20 mm), with an average of around 14 days per year. Extremely heavy rainy days (R50 mm) are very low in this country. They were mainly recorded in the wettest ecological zone, notably the mangrove/swamp forest, during the reference period. According to FMEnv EA (2025), rainfall-induced floods destroyed ₦700 billion worth of agricultural investments in 2022 and affected 9 million people across 29 states in 2024, many of them farming households.

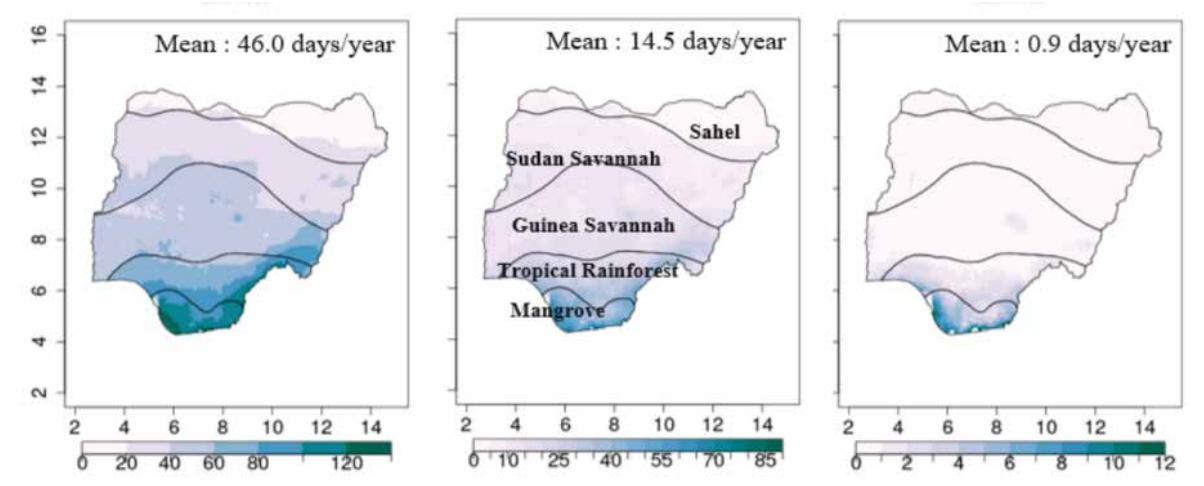


Figure 3.4: Annual Number of heavy rainy days -R10mm (left), very heavy rainy days - R20mm (middle) and extremely heavy rainy days - R50mm (right) during 1981-2010 reference period. Source: FMEnv-CRA 2025

The drought hazards were analysed in the CRA study at different scales. The number of consecutive dry days (CDD) in the country during the 1981-2010 reference period averaged 107 days per year. This period is also marked by drought duration at the 3-month scale (SPI-3), corresponding to the short accumulation period, and at the 12-month scale (SPI-12), corresponding to the medium accumulation period, with around 3 months per year at both scales. Sudan Savannah and Sahel recorded the longest dry days (CDD) and the most months for SPI-3 in the country during the reference period. At the 12-month scale (SPI-12), they also recorded one of the widest dry months (Figure 3.5). This latter is more pronounced in the central part of the Guinea Savannah, where the topography is more pronounced.

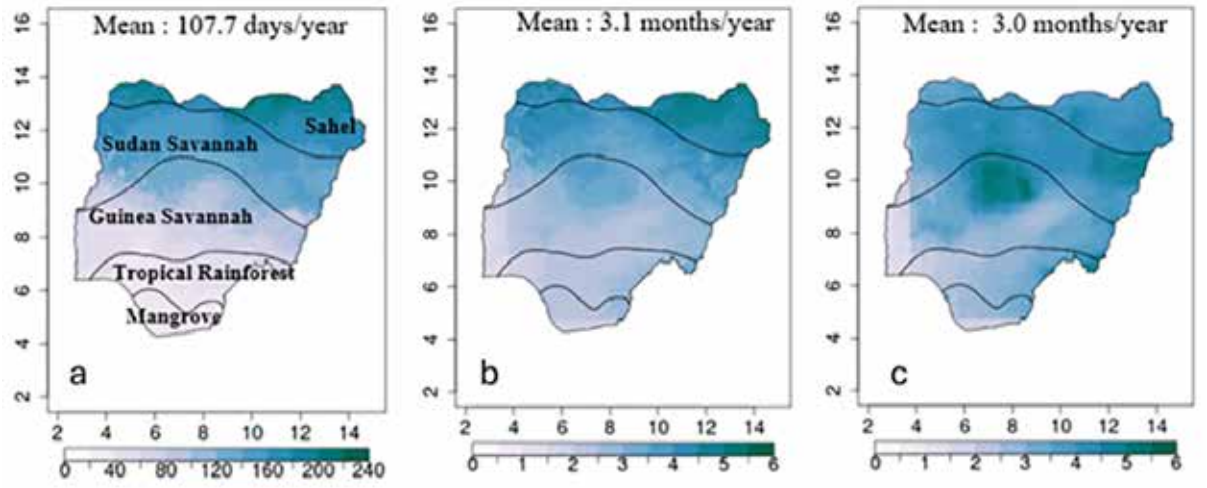


Figure 3.5 SEQ Figure 1 Arabic 14 Annual number of consecutive dry days -CDO (days), number of months in moderated, severe and extreme drought (months) based on (b) SPi-3 and © SPi-12 for the 1981-2010 reference period (Source FMEnv-CRA 2025)*

3.4.4 Sea-level rise

Nigeria's coastline, approximately 853 km long, is low-lying and highly vulnerable to sea-level rise (Danladi et al., 2017; Oloyede et al., 2021). Even small increases in sea level could have major effects and permanently alter coastlines. For coastal ecosystems and communities, sea-level rise poses an additional threat of flooding. This rise mostly stems from oceanic thermal expansion and the melting of Antarctic ice, which could cause salinisation, flooding and erosion (IFO Institute for Economic Research and Climate Analytics, 2021). As a result, low-lying coastal areas are projected to be submerged by 2100 (Merem et al., 2019). Along its vast coastlines, diverse forest ecosystems in the South, including the Niger Delta area with its 1,900 km^2 mangrove swamps and cities within, have experienced climatic hazards ranging from heavy rains to coastal inundation in recent decades (Merem et al., 2019). Between 1960 and 1970, the southern coastline saw a sea-level rise of 0.462 m, which resulted in the flooding of communities, the erosion of low-lying areas, and the loss of mangroves, amid the growth of human settlements and infrastructure (Merem et al., 2019). Baba et al. (2023) observed that the average sea-level rise anomaly in Nigeria between 2015 and 2020 is 15.94 cm. Nigeria's coastline, originally endowed with a fragile mangrove system, has been depleted over time due to human activities such as crude oil exploration, use of wood as fuel, timber, fishing, sand mining, etc (Oloyede et al., 2021).

Period	Heatwave	Rainfall-induced floods	Drought	Sea-level rise
Historical (1981-2010)	The country experienced an average of 16 heatwave days per year, with the Sahel recording the highest number of heatwave days. The longest heatwave lasted 25 days on average, primarily affecting the Guinea Savannah, Tropical Rainforest, and Mangrove zones.	The country recorded an average of 46 heavy rainy days (R10mm) per year, with most occurring in the Tropical Rainforest and Mangrove/ Swamp Forest zones. Heavy rain (R20mm) averaged 14 days/year.	The Sudan Savannah and Sahel recorded the longest consecutive dry days (CDD) and months. The country experienced 107 CDD per year and three months/year of drought (SPI-3 and SPI-12).	Southern (is there any northern coastline? coastline saw a 0.462m sea level rise between 1960-1970, resulting in flooding, erosion, and mangrove loss. The current global estimate of sea level rise is 0.2m.

Table 3.1: Summary of Some Notable Historical Climate Trends

3.5 Climate Risks and Vulnerabilities Projections

3.5.1 Climate scenarios projection

Future climate projections for Nigeria indicate significant warming and shifts in precipitation patterns over the next century.

3.5.1.1 Expected temperature levels in variability (at medium and long term for both national and agroecological zones)

Projections from two Representative Concentration Pathways (RCPs) – the high-emission scenario (RCP 8.5) and the medium-emission scenario (RCP 4.5) – indicate an increase in mean annual temperature across Nigeria (TNC), with increases ranging from 1.48°C to 3.48°C in the near future (Federal Ministry of Environment, 2020). The CRA report, using the SSPs assessment, also shows changes in mean annual temperature across all agroecological regions. The CRA (2025) projects that Nigeria will experience significant warming under both RCP 4.5 and RCP 8.5, with warming of +2.5 to +3.2 °C by 2050.

3.5.1.2 Expected rainfall levels in variability (at medium and long term for both national and agroecological zones)

Nigeria is likely to see changes in the number of heavy rainy days across the near-, mid-, and long-term, with a strong projection of an average increase in the number of heavy rainy days within the reference period under SSP2-4.5 and SSP5-8.5. The CRA (2025) projects that Nigeria will experience significant rainfall variability under both RCP 4.5 and RCP 8.5, with more frequent heavy rainfall events. The number of days with extreme rainfall (≥20 mm) is expected to increase by ~20% by 2080, raising concerns about a higher frequency of floods. Projections of future increases in precipitation are particularly significant for the northern ecological zones, such as the Sahel and Sudan savanna, from the near-term to the long-term. An increase in the number of very heavy rainy days is likely to be recorded in the country, with the lowest increase in the near-term and the highest in the long-term under both SSPs.

Similarly, the Third National Communication (TNC) indicated a likely significant increase in precipitation across the country up to 2070, with annual rainfall projected to rise by 9 to 45.8%. In particular, the southern part of the country is projected to see more areas with high annual rainfall, with seasons of up to 3000mm/year, and rainfall expected to increase by 20 to 30mm per year in susceptible regions by 2050 under RCP 8.5 in Nigeria (Federal Ministry of Environment, 2020). This will be due to the expected warmer atmosphere, which could hold more evaporated water, making rainfall heavier and more intense and causing flash floods. A slight

increase in the number of extremely heavy rainy days from near-term to long-term is expected under both SSPs, except for the long-term under SSP5-8.5, where the increase is very large and spreads across all ecological zones (Figure 3.7). Specifically, the Sahel and Guinea Savannah would likely see the highest increase in the future (FMEnv-CRA 2025). There is also the likelihood of a decrease in the near-term in the mangrove/swamp forest zone, where the highest number of extremely rainy days occurred. One of the direct consequences of the increased occurrence of heavy rain is flooding. The findings of the CRA report appear contrary to TNC predictions under RCP 8.5, which indicate a significant decrease in rainfall (10-20% more dry days) in the northern areas (-20 to -25mm rainfall) by 2050 (Climate Analytics, info Institute and Potsdam Institute for Climate Impact Research, 2021). Nevertheless, these projections would impact different socioeconomic sectors such as agriculture, water resources, health, energy and so on.

Flood risk for the current period indicates that southern Nigeria is the area with the highest flood risk. For the remaining parts of the country, areas with high flood risk are those located near major rivers (Figure 3.6).

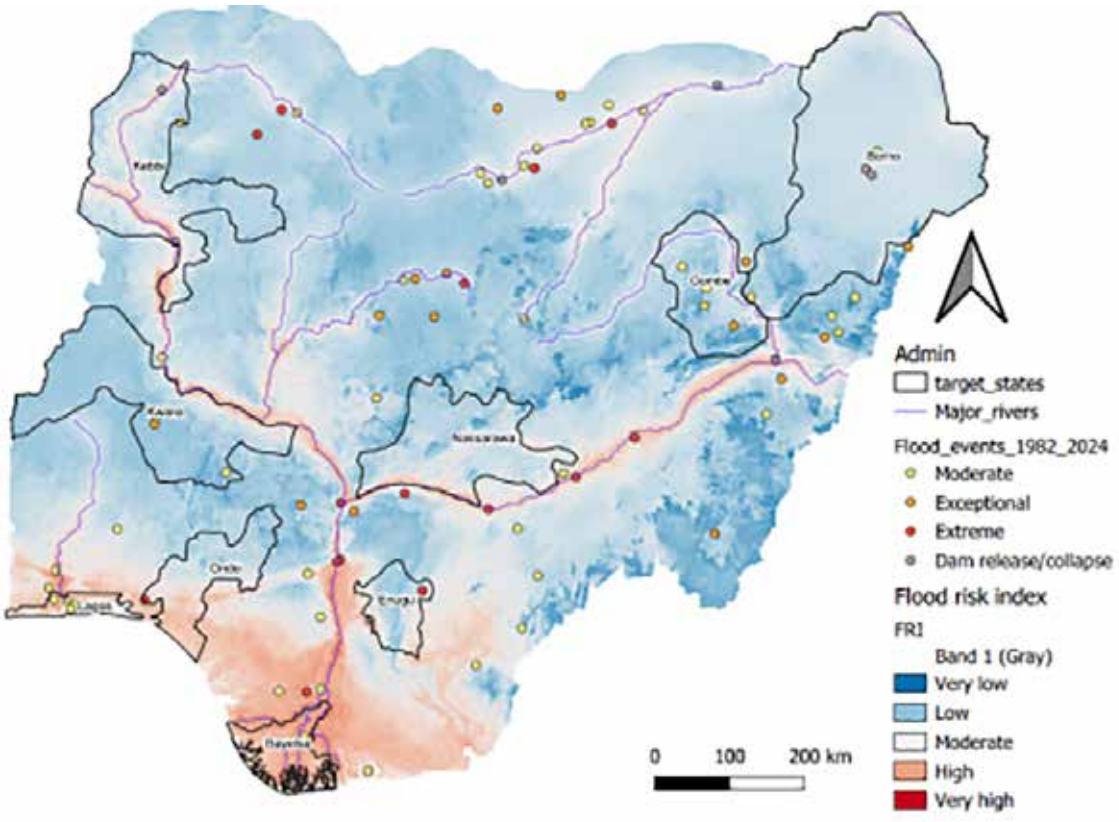


Figure 3.6. Projection of flood risk index (Source FMEnv CRA, 2025)

Increases in heavy rainfall are projected to trigger landslides and mudslides, and consequently lead to the formation of gully erosion in sedimentary terrains, with the gully erosion area in the region increasing to 172.65 km2 by 2030 (Enemuoh et al., 2023).

3.5.2 Increased Extremities of Hazards

The major climate hazards experienced in the country include flooding, drought and aridity, extreme heat waves, landslides and erosion, and sea-level rise. These hazards have varying degrees of impact on the national economy. The occurrence of these hazards is also projected to affect different sectors of the economy, as well as communities and individuals, in the future.

3.5.2.1 Rain-induced flooding

Nigeria has experienced several major flood events in the past, with their frequency and severity increasing over time. Nigeria's coastal states face extensive risks from storm surge along the entire coast, inland flooding and wildfires in the Niger Delta region, and negative rainfall anomalies in the southeast. Notable flooding events include those in 2012, 2015 and 2022. Severe flooding in 2012 affected seven million people and caused economic damage estimated at $500 million, while severe flooding in 2015 affected one million people and resulted in damage of approximately $25 million (GERICS, 2015). The impact assessment of the 2022 flood showed extensive loss of life (with over 600 deaths), damage to livelihoods, and the disruption of essential services and economic activities (National Bureau of Statistics, National Emergency Management Agency and United Nations Development Programme). The flood affected 4.4 million people across Nigeria, including 2.6 million children, and some 2.4 million people were displaced (UNICEF, 2023). The Nigeria Bureau of Statistics reported in 2023 that the devastating impact of the 2022 flood was in the range of US$3.79 billion to $9.12 billion (Oke et al., 2023).

Climate change trends in Nigeria are expected to increase the risk and intensity of flooding through more frequent and intense heavy rainfall events (Figure 3.8). Future projections indicate an average increase in the number of heavy rainy days from near-term to long-term across Nigeria, especially in the northern ecological zone (Sahel and Sudan savannah). An increase in very heavy rainy days could also be recorded in the country, with the lowest increase in the near-term and the highest in the long-term under both SSPs (FNEnv-CRA, 2025). For extremely heavy rainy days, Nigeria is expected to see a slight increase in the number of days from near-term to long-term under both SSPs, except in the long-term under SSP5-8.5, where the increase is very large and spreads across all ecological zones. Increases in the number of extreme rainy days will directly lead to flooding.

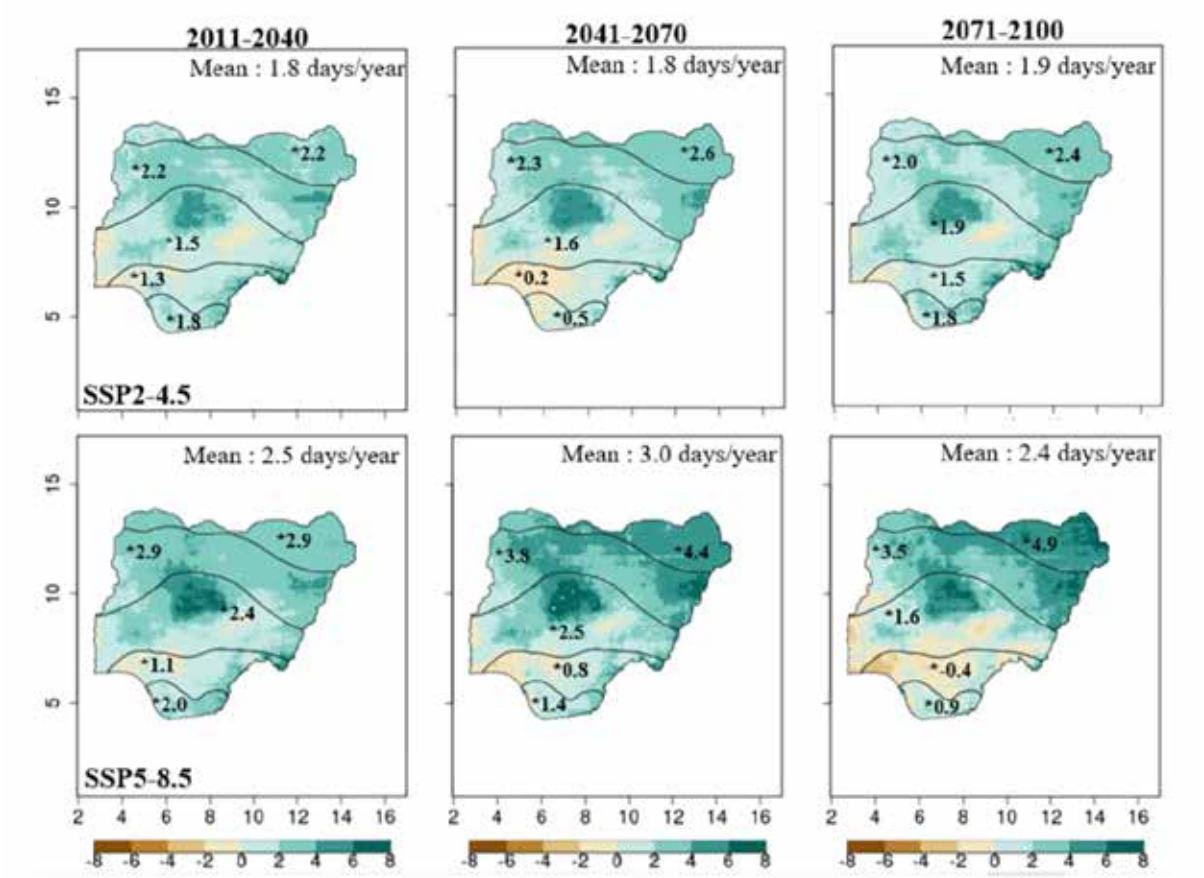


Figure 3.7. Evolution of the annual number of heavy rainy days – R10mm (days) in near-term – 2011-2040 (left), mid-term –2041-2070 (middle) and long-term – 2071-2100 (right) compared to the reference period for SSP2-4.5 (top) and SSP5-8.5 (bottom) (Source FMEnv CRA, 2025)

The TNC noted that the frequency and severity of floods will increase. This could be due to projections of 1 to 1.5 additional days of heavy precipitation in 2080 (Climate Analytics, IFO Institute and Potsdam Institute for Climate Impact Research, 2021). Flood-exposed regions in Nigeria are projected to double in size, or even more, by 2080, and surface water runoff is expected to increase in the north and decrease in the central regions (Climate Analytics, IFO Institute and Potsdam Institute for Climate Impact Research, 2021). An increased incidence of flooding may also result in soil erosion and waterlogging of crops, thus decreasing yields and increasing food insecurity. Given the projected trend, Nigeria may become a hotspot of food insecurity in the future, resulting in significant rain-induced economic losses, damage to arable land and infrastructure, and loss of lives and livelihoods, as heavy flooding destroys or blocks infrastructure, decreases soil quality, and has adverse effects on agricultural production. This will increase economic hardship among a significant population of Nigerians who depend on agriculture as their main means of income and economic livelihood. Moreover, land degradation and soil erosion, exacerbated by recurrent flooding, will adversely impact agricultural production and disproportionately affect the livelihoods of the rural poor.

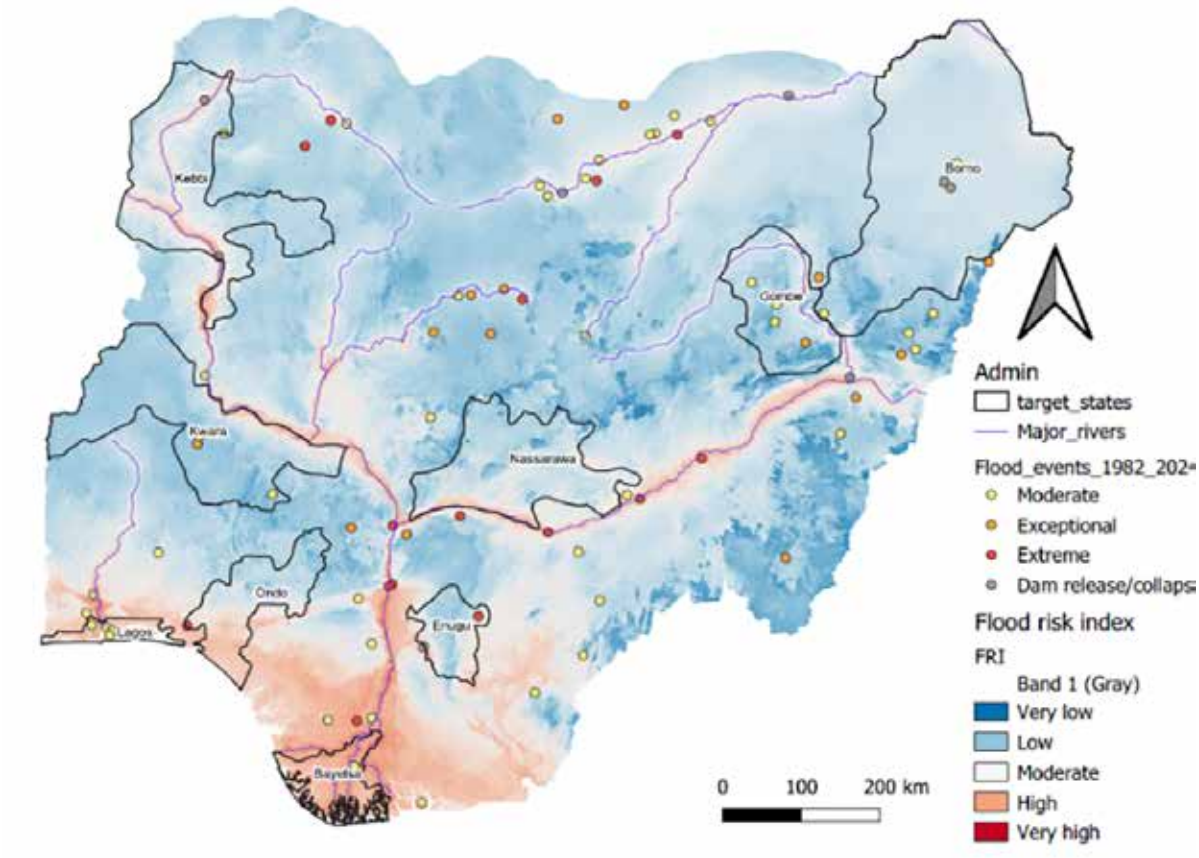


Figure 3.8. Projection of flood risk index

3.5.2.2 Drought and Aridity

The future evolution of consecutive dry days (CDD) in the country is shown in Figure 3.9. On average, the longest period without precipitation will gradually decrease across the country under SSP2-4.5 and SSP5-8.5. This decrease is expected to be most pronounced in the northernmost ecological zones, notably the Sahel and Sudan Savannah. However, an increase in the longest period without precipitation should be recorded in southern zones such as Mangroves and Tropical Rainforest. Moreover, this increase should be progressive in these zones from near-term to long-term under both SSPs.

Compared with the reference period, the dry duration for SPI-3 will also gradually decrease from near- to long-term across the country under both SSPs: -0.6 to -0.8 months/year under SSP2-4.5 and -0.8 to -1.2 months/year under SSP5-8.5. The same pattern holds for SPI-12, but with a greater decrease. This decline for SPI-3 and SPI-12 is expected across all the country's ecological zones, with a particularly sharp drop in the Guinea Savannah, Sudan Savannah and the Sahel.

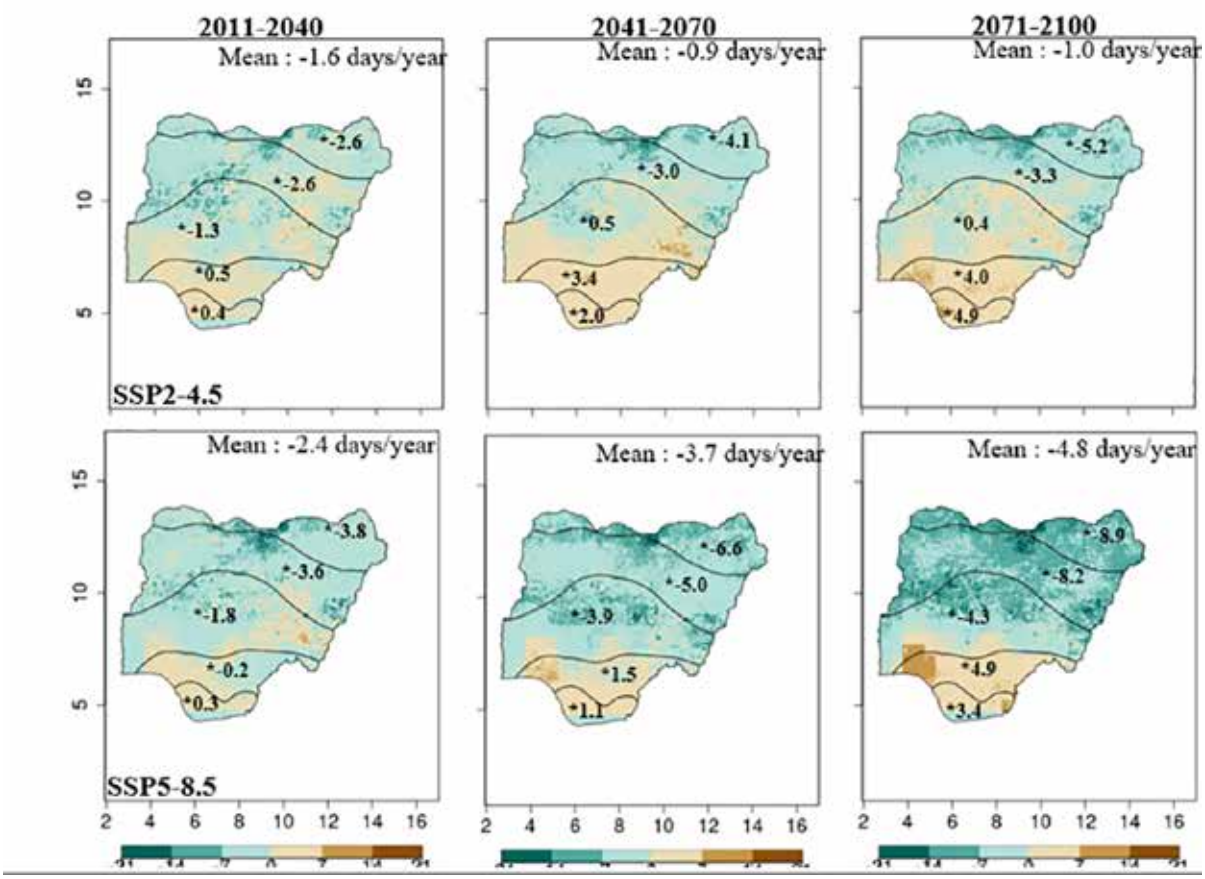


Figure 3.9. Evolution of the annual number of consecutive dry days - CDD (days) in near-term – 2011-2040 (left), mid-term – 2041-2070 (middle) and long-term – 2071-2100 (right) compared to the reference period for SSP2-4.5 (top) and SSP5-8.5 (bottom).(Source FMEnv CRA, 2025)

Drought is a recurring event in Nigeria, particularly in many northern States, where there is an increased number of dry days, reduced annual rainfall volume and rainy days, and a lowering of the groundwater table. Drought has historic significance in Nigeria, especially in the Sahelian States. Insights from the literature suggest an expansion of the Sahara Desert by approximately 3600 square kilometres, including the drying up of Lake Chad due to a southward shift in precipitation (Haider, 2019; Hwang et al., 2013). The trend has been towards a generally drier climate, which is worsening food insecurity in the country (Haider, 2019).

In 1972-1973, a severe drought (largely caused by decreasing rainfall) in the Sudano-Sahelian areas of West Africa led to catastrophic loss of human life and livestock (FMEnv, 2020). Ogunride et al. (2023) predicted an increase in drought frequency and intensity in Nigeria. In their drought analysis under the RCP 8.5 scenario, the frequency of severe/extreme drought would intensify towards the end of the 21st century. Water availability will be affected by possible periods of drought (Ayanlade et al., 2018).

Rainfall has continued to decline since the 1970s to date. This period of drastic

rainfall decline is marked by a sharp increase in temperature, resulting in chronic aridity in the extreme northern part of the country (Echendu, 2020), and the middle areas are at increasing risk of aridification, compounded by farmer-herder conflicts and water access (USAID, 2018). Water availability will be affected by possible periods of drought. Considering population growth for the country in the coming years, water availability is projected to decrease by more than three-quarters – from 3 300 m^3 per capita in 2000 to about 800 m^3 per capita in 2080 – below the UN threshold for water scarcity (Climate Analytics, IFO Institute and Potsdam Institute for Climate Impact Research, 2021) – making it increasingly difficult to provide enough water to the entire population.

At the aggregate country level, soil moisture is projected to decline slightly by about 1.1% as 2080 approaches (Climate Analytics, IFO Institute and Potsdam Institute for Climate Impact Research, 2021). However, regional disparities exist, and soil aridity could increase by 10% in 2080 in some northern regions under the RCP6.0 scenario (Climate Analytics, IFO Institute and Potsdam Institute for Climate Impact Research, 2021).

3.5.2.3 Variations in Sea-level rise

The current global estimate of sea- level rise is 0.2. 1m, with projections to increase to 1. 32 m by the year 2100 and 5. 61 m by 2300 under the Representative Concentration Pathway (RCP) 8. 5 (World Economic Forum, 2024; Horton et al., 2020). This implies that, with the present 0. 2 m sea- level rise, the inundated area of 3,400 km^2 in Nigeria' s coastal regions will increase to 18,400 km^2 under the projected 1 m scenario (Federal Republic of Nigeria, 2023). The projected sea- level rise values based on the RCP 8. 5 scenario for the years 2025, 2050, 2075, and 2100 are 11. 92, 29. 07, 53. 09, and 84. 25 cm, respectively (Oloyede et al., 2021). Empirical evidence suggests that 40. 1% to 58. 9% of the Niger Delta coastline is highly or very highly vulnerable to sea- level rise (Dike et al., 2024). Coastal settlements such as Bonny, Forcados, Lagos, Port Harcourt, Warri, and Calabar, which are less than 10 m above sea level, will be seriously threatened by a 1 m sea- level rise under SSP 5- 8. 5 (Federal Republic of Nigeria, 2023). Additionally, the country's coastal zone and low- lying islands in the Gulf of Guinea are vulnerable to sea-level rise (Figures 3. 10, 3. 11 and 3. 12). An estimated 27 to 53 million people in the country may need to be relocated with a 0. 5 m increase in sea level (USAID, 2018). Nigeria's coastal and marine areas are also home to the country' s economically important petroleum and fisheries industries (Federal Republic of Nigeria, 2023). Some studies suggest that a 1- metre rise in sea level would inundate about three-quarters of the Niger Delta area, which could lead to substantial losses in Nigeria' s GDP and property (Oyegun, 2018; Danladi, 2017). The lives and livelihoods of Nigeria' s coastal communities are vulnerable to sea- level rise. The oil and gas and fisheries industries will be severely affected because these industries are based in

coastal communities. Nigeria' s revenue and foreign exchange earnings would also be threatened, as petroleum is a major foreign exchange earner for Nigeria. These predictions underscore the importance of coastal planning and management to enhance the adaptation of the inhabitants of the Nigerian coastal zone.

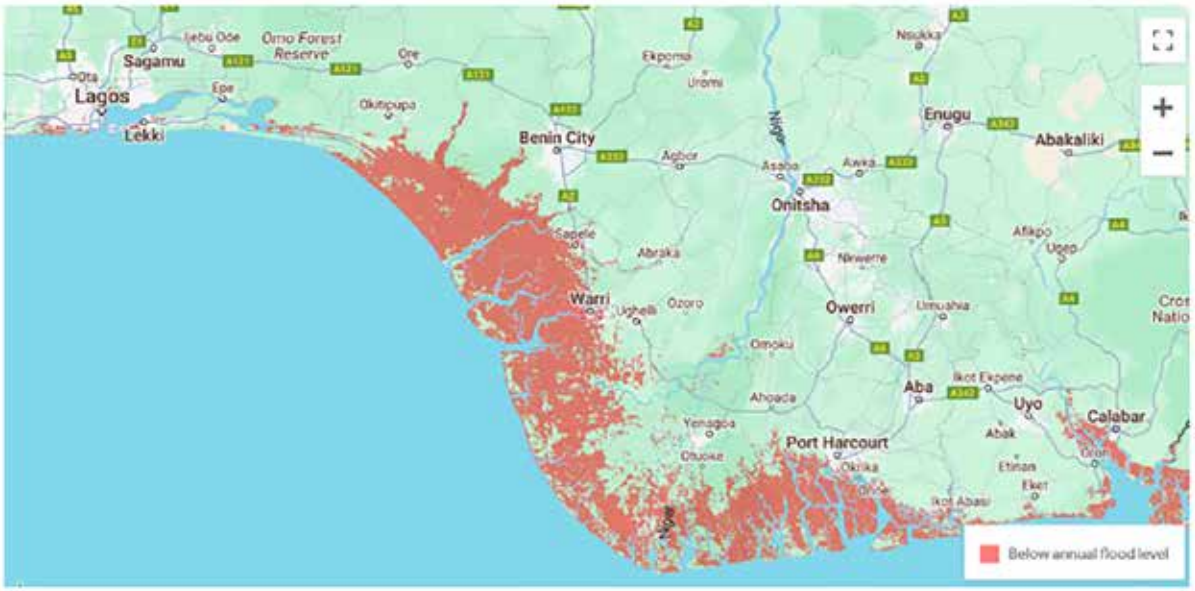


Figure 3.10. Flooded area on the coast of Nigeria by 2050 according to the SSP2-4.5 scenarios. Source: coastal.climatecentral.org

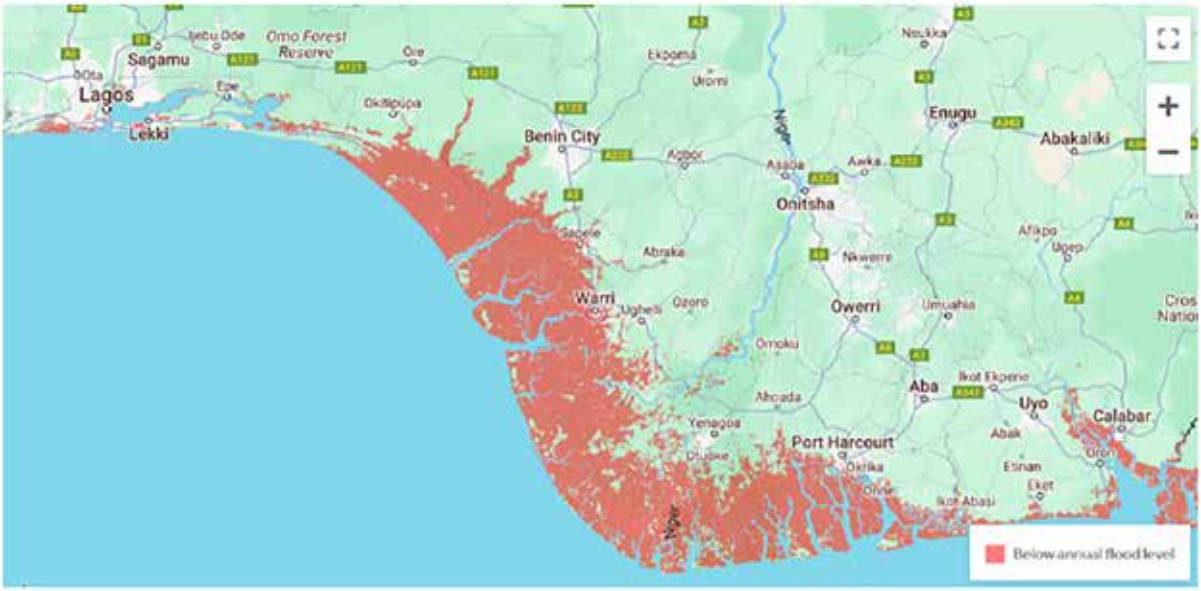


Figure 3.11: Flooded area on the coast of Nigeria by 2100 according to the SSP2-4.5 scenarios. Source: coastal.climatecentral.org

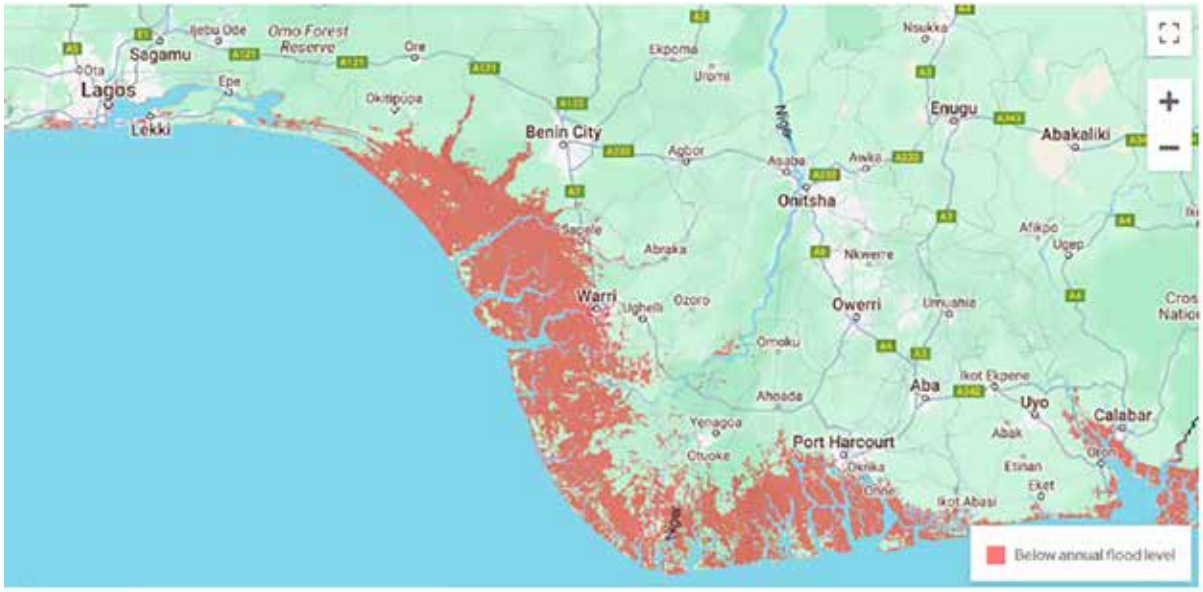


Figure 3.12: Flooded area on the coast of Nigeria by 2050 according to the SSP5-8.5 scenarios
Source: coastal.climatecentral.org

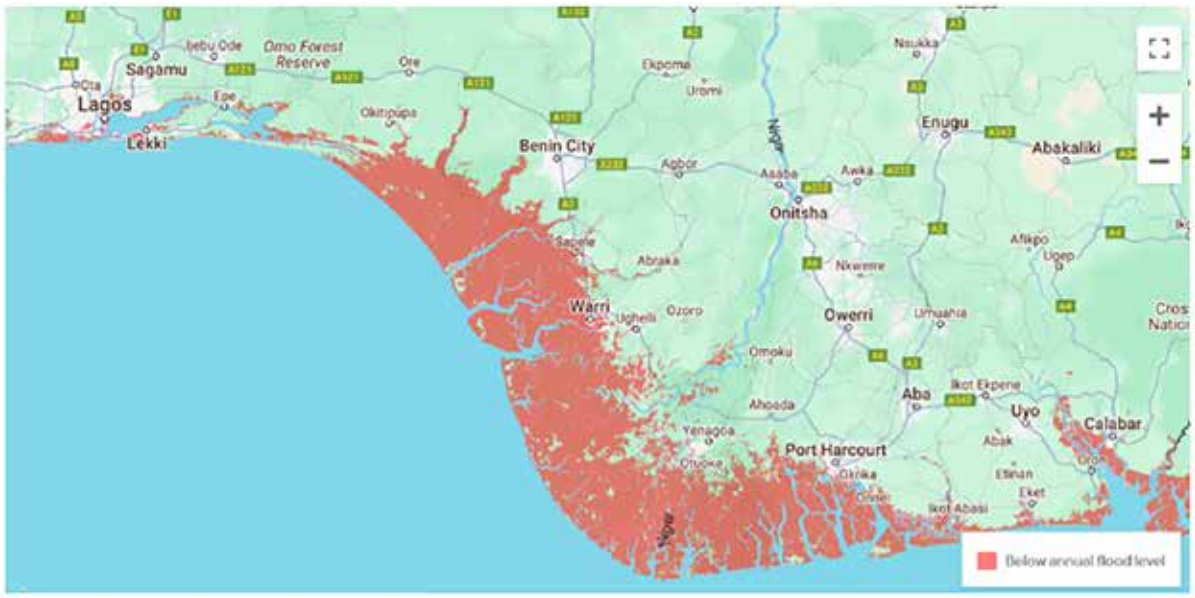


Figure 3.13: Flooded area on the coast of Nigeria by 2100 according to the SSP5-8.5 scenarios.
Source: coastal.climatecentral.org (3.5. Summary of climate trends in Nigeria)

3.5.2.4 Extreme heat waves

As shown in Figure 3.14, heatwaves in Nigeria are most frequent between March and May, preceding the rainy season (World Bank 2024b). Projections of changes in the number of days and duration of heatwaves indicate a gradual increase in the number of heatwave days across the country from the short term to the long term (Figure 3.15). This increase is expected to be greatest in the southern ecological zones, such as tropical rainforest and mangrove (FMEnv-CRA 2025). The states in the southern ecological zones are likely to experience the greatest increase in the

duration of the longest heatwave from the short term to the long term under both SSPs; this duration will increase every year across the country, rising from 11 days in the short term to 39 days in the long term under the SSP2-4.5 scenario, and from 10 to 75 days under the SSP2-4.5 scenario (Figure 3.16).

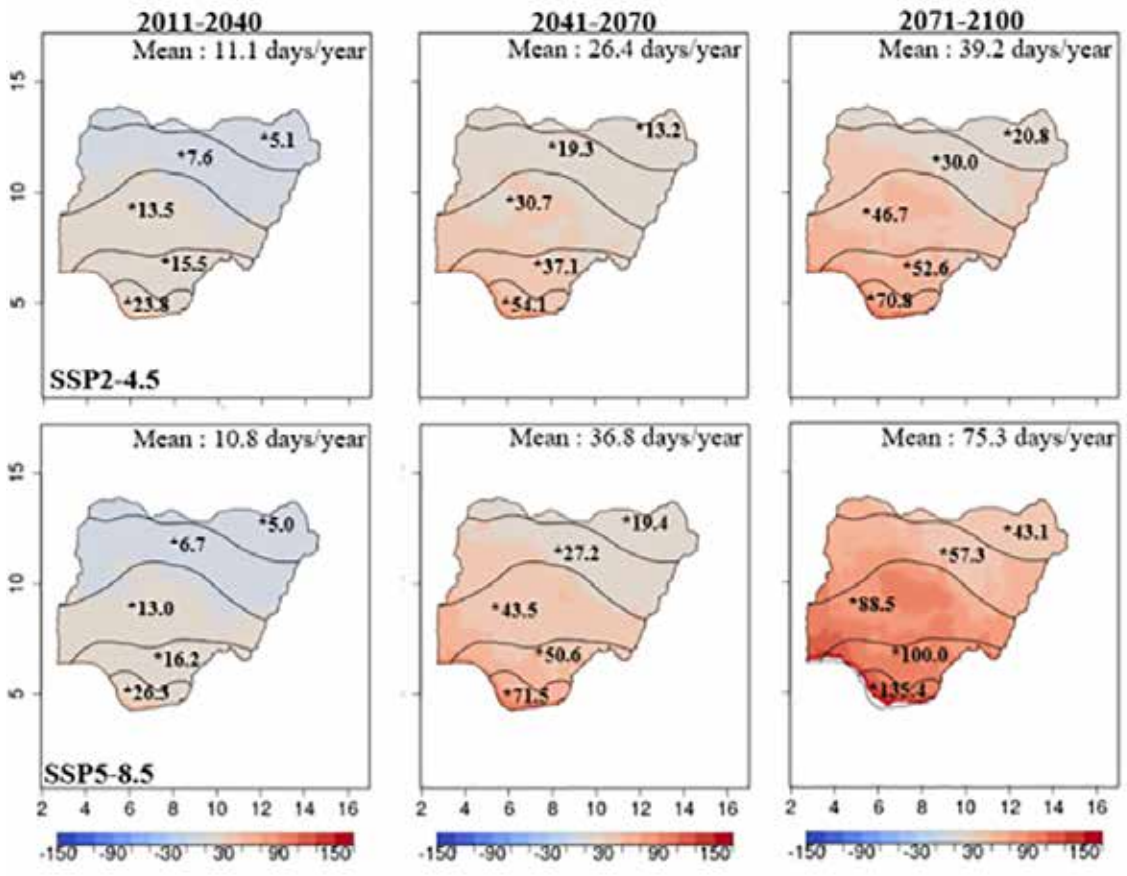


*Figure 3.14 SED Figure*Arabic 13 Projection of changes in the duration of the longest heatwave - HWD (days) each year across Nigeria, near term - 2011-2040 (left), mid-term - 2041-2040 (middle) and long-term - 2071-2100 (right), compared with the reference period for ssp3-4.5 (top) and ssp5-8.5 (bottom) scenarios. Source FMEnv-CRA, 2025.*

Nigeria's temperature is projected to change in the future, and the results may negatively impact several economic sectors, such as agriculture, health, and natural resources. For instance, projections based on both emission scenarios (RCP4.5 and RCP8.5) indicate that crop or agricultural yield or productivity will decrease by about 7 per cent in the short run and up to 25 per cent in the long run by 2070 (TNC - Federal Ministry of Environment, 2020).

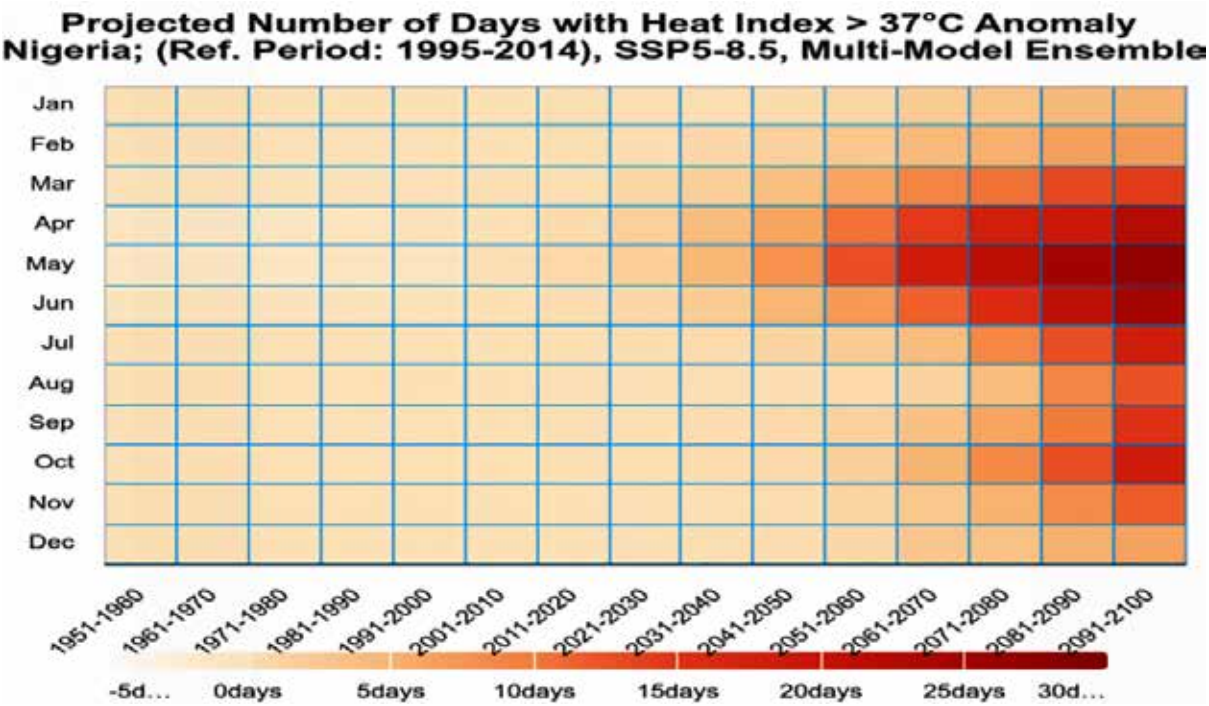


Figure 3.15: Projected number of days with heat index > 370C anomaly
Source: World Bank Group (2024b)

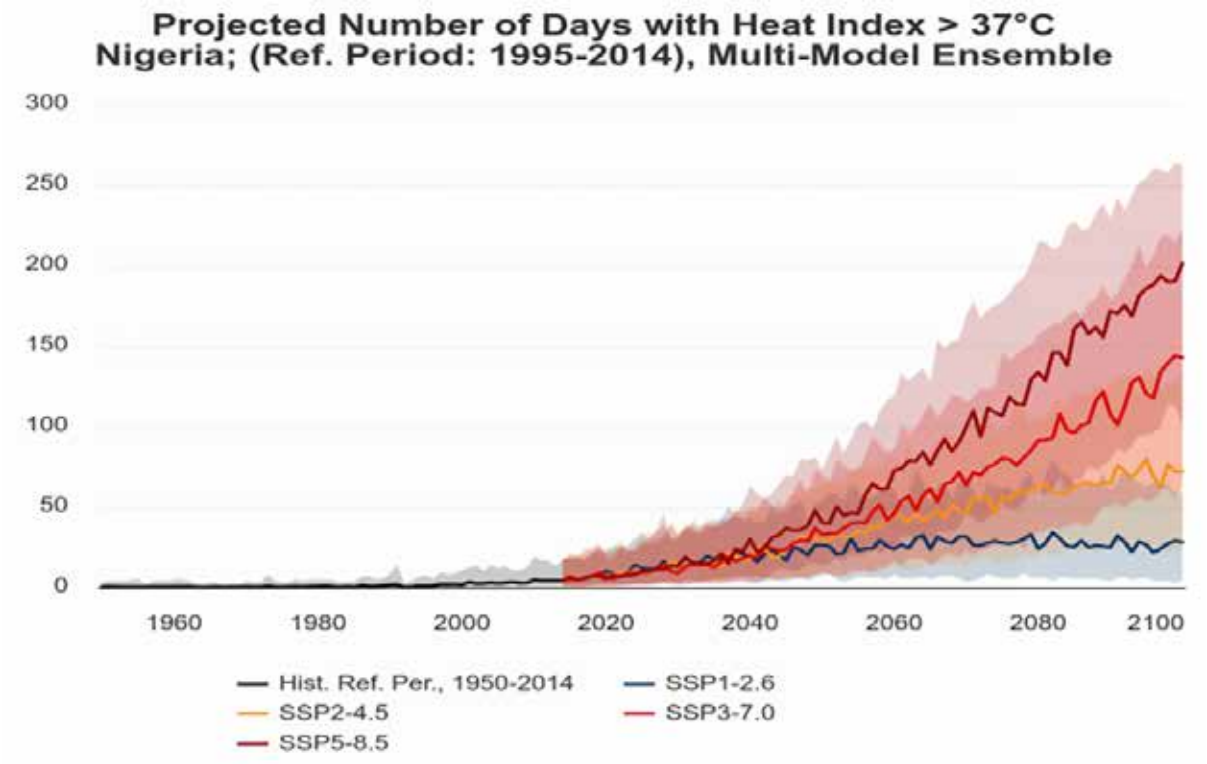


Figure 3.16: Projected number of days with heat index > 370C
Source: World Bank Group (2024b)

3.5.2.5 Landslides and erosion

Heavy rainfall could trigger landslides, mudslides, and consequently gully erosion in sedimentary terrains. An increased incidence of extreme rainfall may also result in soil erosion and waterlogging of crops, thereby decreasing yields and increasing food insecurity. This may lead to significant economic losses, damage to agricultural land and infrastructure, and human casualties. Land degradation and soil erosion, exacerbated by recurrent flooding, adversely affect agricultural production, disproportionately affecting the livelihoods of the rural poor. More extreme weather events, such as intense rainfall after prolonged dry spells, can lead to erosion and flash flooding, damaging roads and infrastructure, wiping out crops, and putting additional lives at risk (USAID, 2018).

In 2015 and 2020, there were about 99.21 km2 and 148.77 km2 of gully erosion sites, respectively, in southeast Nigeria – Abia, Anambra, Ebonyi, Enugu, and Imo States (Enemuoh et al., 2023). This represents a significant increase in the area affected by gully erosion in the region. Enemuoh et al. (2023) projected that the area affected by gully erosion in the region would increase to 172.65 km2 by 2030. This prediction highlights the urgency of proactive measures to counteract the adverse effects of gully erosion on the environment, infrastructure, and communities. Anticipatory adaptation measures are therefore needed to respond to the threat of gully erosion in Nigeria. Planning, resource allocation, policy formulation, and collaborative efforts are needed to address the escalating threat of gully erosion in South-East Nigeria.

Landslides are more pronounced in southern Nigeria than in the northern part of the country. The loose nature of sedimentary rocks and gully erosion triggered by heavy rainfall and subsequent flooding are the primary causes of most landslides in southern Nigeria (Bamisaiye, 2019; Nnanwuba et al., 2022). Furthermore, Nnanwuba et al. (2022) listed other factors and their levels of contribution to landslides in southeastern Nigeria. These factors include rainfall (0.860), slope (0.566), aspect (0.553), distance to fault (0.512), elevation (0.348), distance to drainage (0.333), Normalised Difference Vegetation Index (NDVI) (0.250), general curvature (0.117), land use (0.074), and geology (0.038) (see Figure 7). Heavy rainfall is the main factor contributing to landslides in the region. Heavy rainfall is also implicated in gully erosion (Figure 3.17).

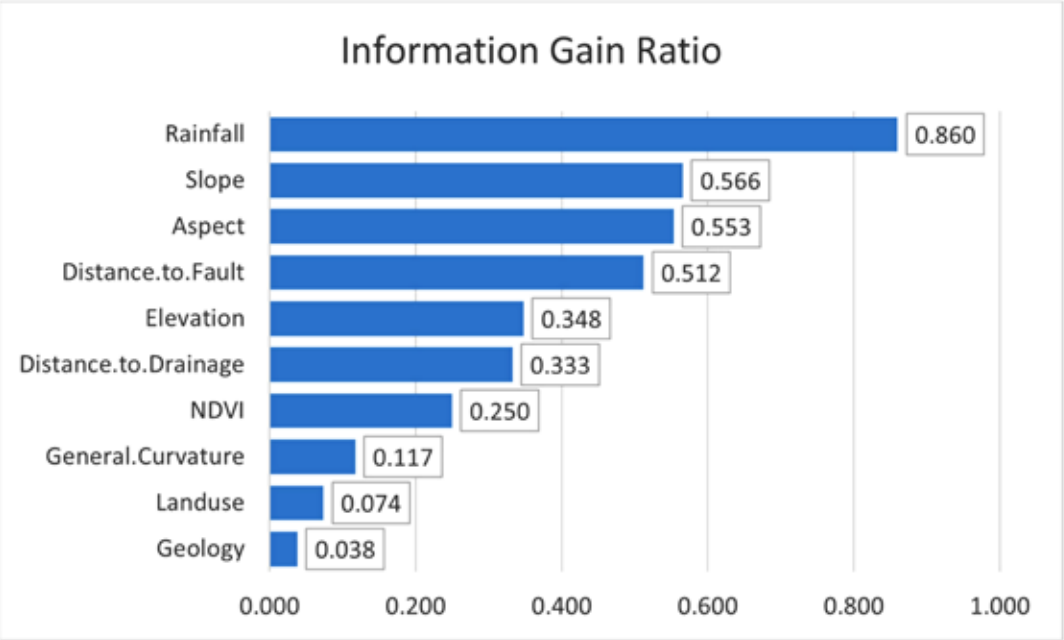


Figure 3.17. Information gain ratio of factors contributing to landslide in southeast Nigeria
Source: Nnanwuba et al. (2022)

In southeastern Nigeria, landslides are also caused by gully erosion. Widespread occurrences of large, deep gullies in the southeastern part of Nigeria, together with the annual impact of rainfall-induced floods, have been reported to weaken the ground around gully sites and slopes, resulting in landslides in many locations (Egboka et al., 2019; Nwankwo & Nwankwoala, 2018). Landslides negatively impact the topography/morphology, settlement areas, agricultural lands, and vegetation/land cover of parts of southeast Nigeria (Igwe & Una, 2019).

The urgency of erosion and landslide crises necessitates adaptation measures to manage the risks. Anticipating future trends in gully and landslide development is crucial for effective planning, resource allocation, and the formulation of adaptation and mitigation strategies.

Period	Heatwave	Rainfall-induced flood	Drought	Sea-level rise
Near-term (2011-2040)	Heatwaves begin to increase, particularly in southern zones (Tropical Rainforest and Mangrove), with a gradual rise in the number and duration of heatwave days.	Increase in heavy rainy days, especially in northern ecological zones (Sahel and Sudan Savannah). The Mangrove zone sees a slight decrease in extreme rainy days. Flood risks expand, particularly in coastal and riverine areas.	No major changes in drought patterns, but early indications of decreasing dry months in the north and increasing dry months in the south.	The coastal inundated area of 3,400 km^2 is expected to grow, threatening Lagos, Port Harcourt, Warri, Calabar, and Bonny.

Mid-term (2041 - 2070)	Heatwaves increase significantly. Under SSP2-4.5, heatwave days rise by 11-39 days/ year, and under SSP5-8.5, by 10-75 days/year. The south continues to experience the most prolonged heatwaves.	Increase in heavy rainfall intensity (+1 to +2 days), especially in the Sahel and Sudan Savannah. Flood events become more frequent, particularly in July and August. The rainy season extends in some areas.	Drought conditions start decreasing, particularly in the sahel, Sudan Savannah, and Guinea Savannah. Overall, dry periods reduce by 0.6 to 0.8 months/ year (SSP2-4.5) and 0.8 to 1.2 months/year (SSP5-8.5).	Sea levels expected to rise to 0.5m, potentially displacing 27 to 53 million people
Long-term (2071-2100)	Heatwaves become a dominant climate stressor. The country could experience +150 to +250 additional heatwave days per year under SSP5-8.5. The longest heatwaves could exceed 75 days in the south.	Heavy rainfall events increase further (+3 days/ year in the west). Flood-prone areas expand significantly, affecting southern states and major riverine regions. The most extreme rainfall increases under SSP5-8.5.	Drought conditions reduce significantly across the country. SPI-3 decreases by 1.2 months (SSP5-8.5), with the greatest reduction in Sahel, Sudan Savannah, and Guinea Savannah.	Sea levels projected to rise by 1m, submerging 18,400 km^2 of coastal land. Low-lying areas on the coast projected to be permanently underwater.

Table 3.2: Summary of Climate Hazards on Near-term, Mid-term, and Long term

3.5.3 Climate stress areas

All sectors of the Nigerian economy, including various development initiatives, are vulnerable to the impacts of climate change. Natural, human, and physical assets are vulnerable to climate change. Climate change affects the ability of Nigeria's natural capital to deliver its wide range of goods and services. Extreme weather events negatively impact Nigerian cities, roads, drainage systems, power plants, ports, and other types of infrastructure. Nigeria's human capital is also threatened by climate change, as it affects the ability to build and improve the country's health and education. For example, floods and heat waves affect children's access to school and hinder conducive learning.

Nigeria's diverse ecological zones have created opportunities for multiple livelihoods, agricultural practices, and commodities, all of which are affected by climate change and shocks (Table 3.2). Vulnerability, which varies from region to region, reflects existing variation in the climates and socioeconomic features of Nigeria's ecological regions. Rising sea levels threaten coastal communities, with flooding, saltwater intrusion, and waterborne disease becoming more frequent. Drought and aridity, combined with rising surface air temperatures, limit the country's hydropower systems in the north and frustrate agricultural production and

fishing, endangering food security and negatively impacting health and nutrition (USAID, 2023). In the central and southern regions, high-intensity rainfall events cause floods and erosion, resulting in significant loss and damage to infrastructure. Floods and other climate-change-related hazards in Nigeria have also increased the incidence of diseases, particularly malaria, which was responsible for about 200,000 deaths in 2021 in the country (WHO, 2022). A combination of frequent climate-change-related hazards, a large population, poor infrastructure, poverty, limited access to technologies, and low adaptive capacity heightens Nigeria's vulnerability.

3.5.3.1 Risks and vulnerabilities of different sectors

3.5.3.1.1 Agriculture

Agriculture is a significant contributor to Nigeria's economy and food security. The sector contributed 21.07% to the gross domestic product in the first quarter of 2024 and employed over 70% of Nigerians (FAO, 2024; NBS, 2024). Over 80% of Nigerian farmers are smallholders, accounting for 90% of the country's agricultural production, yet they have limited access to improved production technologies and mechanisation (Oyaniran, 2020). Nigeria's agricultural land area is about 686,440 km2, representing 75.4% of the country's total land area (World Bank Group, 2024c; World Bank Group, 2024d). The country's agriculture can be divided into four sectors: crop production (accounting for around 87.6% of the sector's total output) – with cash and food (or subsistence) crops – livestock (8.1%), fishing (3.2%) and forestry (1.1%) (Oyaniran, 2020). Gender roles in Nigeria's agricultural sector are often differentiated, with men mainly involved in ploughing, planting, and the more lucrative cash crops, while women are more often involved in growing food crops, processing agricultural products, and marketing surpluses. Although women have limited access to resources such as land and credit, they still make a significant contribution to Nigeria's food security by growing subsistence crops and participating in post-harvest activities such as storing and processing agricultural products.

3.5.3.1.1.1 Crop production

Nigeria's agricultural production covers almost 70 million hectares, with diverse food crops cultivated by farmers, including rice, maize, yam, cocoyam, beans, sorghum, millet, melon, and soya beans, which are mainly grown by small-scale farmers (FAO, 2024a). The most widespread food crops are maize and cassava, grown by almost 50% of households, while cocoyam and sesame beni-seeds are grown by 7.7% and 3.6% of Nigerian households, respectively (Figure 3.18). Important cash crops grown in the country include cocoa, palm oil, rubber, groundnut, and cotton.

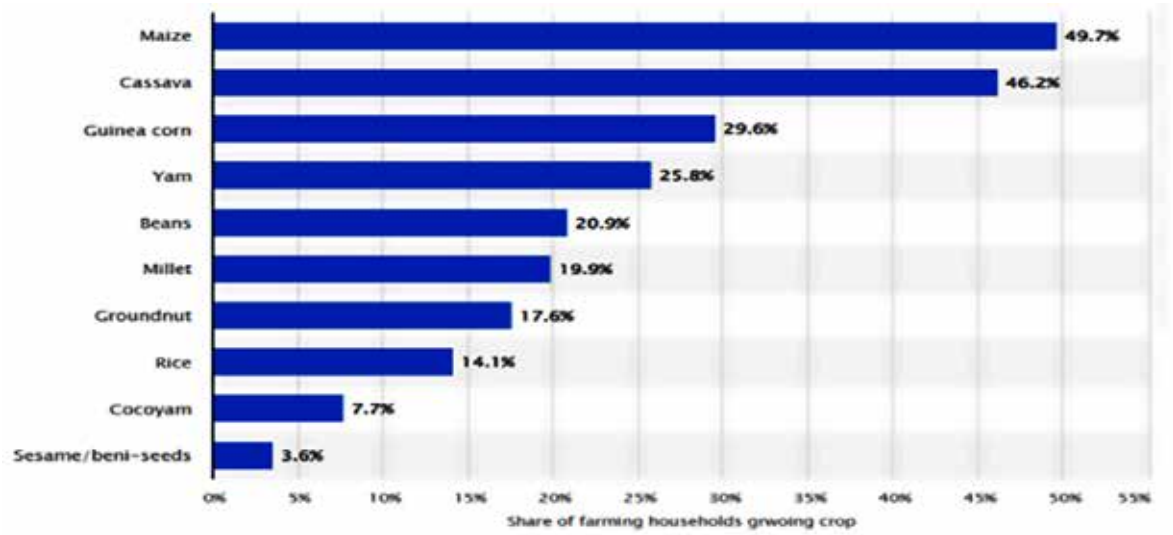


Figure 3.18. Ten major crops among households in Nigeria as of 2019 (Source: Statista, 2023

Nigeria's agricultural sector faces a myriad of problems, including poor soil fertility, land fragmentation, an inefficient tenure system, a weak agricultural extension system, low adoption of improved technologies, substantial post-harvest losses, and limited access to inputs and finance, all of which contribute to low productivity in the sector. Climate change exacerbates these challenges. Agricultural yields in Nigeria are increasingly at risk from climate change, as shifting weather patterns, rising temperatures, and more frequent extreme events such as droughts and floods disrupt crop production cycles. Climate change affects agricultural production in several ways. Currently, erratic rainfall and prolonged dry seasons limit water availability for crops, while rising temperatures stress plant growth, reduce soil moisture, and increase evapotranspiration, leaving crops and livestock vulnerable. Unpredictable rainfall complicates farmers' planning and shortens the cropping season and growing days. Numerous crops are susceptible to even minor variations in rainfall and temperature.

The cascading impacts of these changes affect the population's livelihoods and threaten food security, as lower agricultural yields directly affect Nigeria's food supply, increase prices, and reduce household incomes, especially among smallholder farmers who rely heavily on seasonal harvests (Figure 3.19). Nigerian agriculture is mainly rain-fed (less than 1% is irrigated) and is therefore intrinsically sensitive to climatic hazards. Farmers are also particularly vulnerable due to limited access to irrigation, climate-resilient seeds, and crop insurance, all of which could buffer against climate impacts.

Increased heat and drought in the Northeast have been linked to crop failures and declines in agricultural output. Along the Southern coast, rising sea levels are flooding cropland, making soils unsuitable for planting, and in the Southeast, erosion is further reducing yields. This vulnerability is compounded by insufficient

infrastructure, such as storage facilities to protect crops during extreme weather, and by inadequate access to financial resources and technical assistance that could help farmers adapt. As climate change progresses, Nigeria's agricultural sector may face heightened economic losses, increased poverty, and food insecurity, all of which threaten broader socio-economic stability in the country.

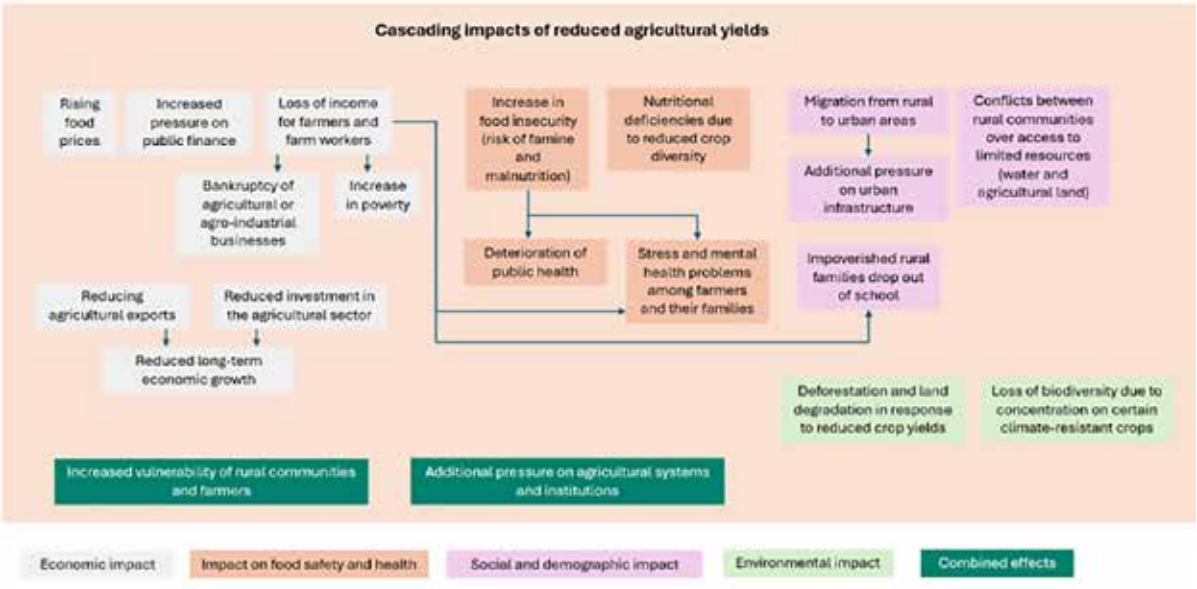


Figure 3.19: Cascading Impacts of climate hazards on agricultural crop yields

3.5.3.1.1.2 Livestock

Livestock production in Nigeria is a crucial component of the agricultural sector and plays a significant role in the country's economy and food security. The livestock sector contributes about 17% to Nigeria's agricultural GDP and 5% to the national GDP (ILRI, 2023). Nigeria's livestock sub-sector includes the production of poultry (180 million), goats (76 million), sheep (43.4 million), cattle (18.4 million) and pigs (7.5 million) (FAO, 2021). The vulnerability factors and primary impacts are shown in Figure 3.20.

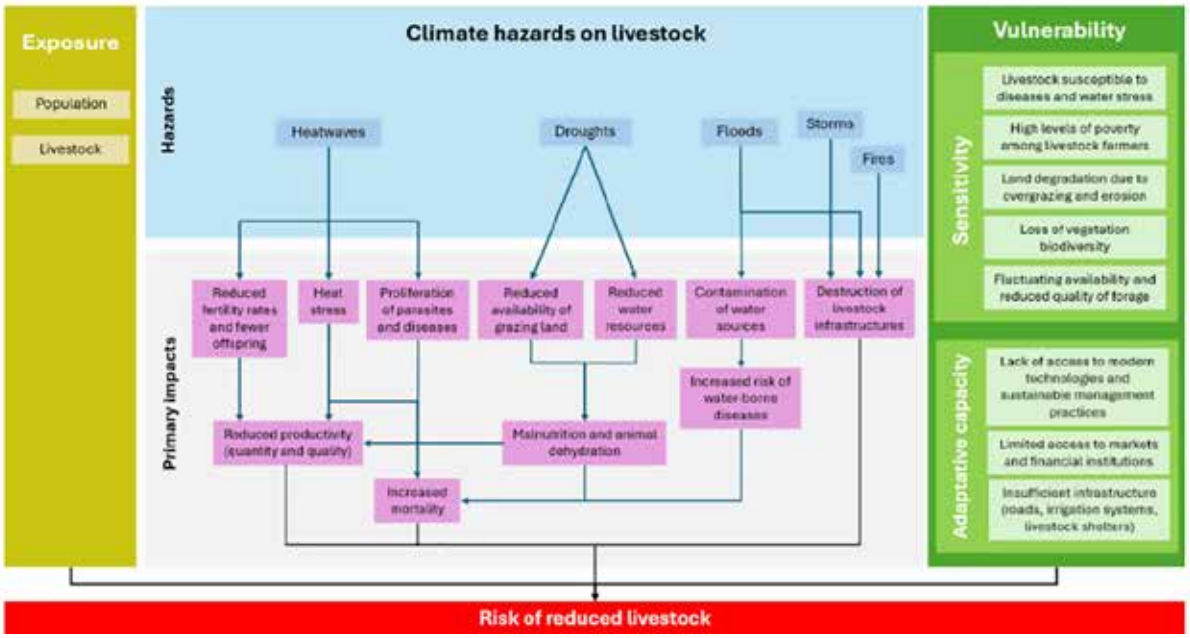


Figure 3.20: Cascading Impacts of climate hazards on livestock production

Nigeria's livestock sector faces significant and growing risks from climate change, primarily from rising temperatures, unpredictable rainfall patterns, prolonged droughts, and extreme weather events such as floods. Higher temperatures and increased heat stress reduce livestock productivity by affecting health, fertility, and milk production, while prolonged droughts deplete water sources and degrade grazing land, leading to poor-quality fodder. Droughts also limit access to critical feed resources, weakening livestock health and increasing mortality rates. As droughts become more frequent, they reduce pasture availability and force herders to travel greater distances in search of water, increasing the likelihood of conflict over resources, particularly between pastoralists and farmers in northern Nigeria. Weight loss in animals can reduce meat and dairy production. Additionally, floods damage infrastructure, contaminate water sources, and spread diseases such as foot-and-mouth disease, further stressing livestock populations. Key cascading impacts on the national economy include increased prices of animal products, job losses across the livestock value chain, reduced income, increased rural-to-urban migration, and shortages of livestock products in the market.

Figure 3.20 shows the key vulnerability factors, including reliance on rain-fed grazing lands and traditional herding practices; lack of adaptive infrastructure, such as drought-resilient pasture management and efficient water storage systems; limited access to veterinary services and to climate-adapted breeds that are more resistant to heat stress and disease; low financial resources; limited access to technology; and inadequate support for livestock farmers.

3.5.3.1.2 Fishery

Nigeria is the second-largest producer of farmed fish on the continent and the fifth-largest consumer. Fishing enterprises operate across a range of habitats nationwide, including marine, brackish and freshwater, with most activity concentrated along Nigeria's 960-km coastline. About 90% of fish produced in Nigeria is sold in local markets, partly constituting 40% of the country's dietary protein consumption for the growing population (Tokpo, 2024, CRA, 2025). With an estimated annual fish demand of 2.0 million metric tons, only about 42.5% is met by domestic production; the rest is supplemented by fish imports (Idowu et al., 2011). Nigeria's domestic fisheries production comprises four main segments: artisanal (coastal) fishing, which contributes 80%; industrial coastal (trawl) fishing, 10%; artisanal inland fishing, 6%; and aquaculture, 4% (Idowu et al., 2011). In 2022, 613,800 people were employed in inland fishing, of which 27% were women; 120,000 in sea fishing, of which 20% were women; and 13,600 in aquaculture, of which 2% were women (FAO, 2024b). Aquaculture has received a boost from private-sector investment in recent years, making Nigeria a major producer in Sub-Saharan Africa. However, this sub-sector is increasingly affected by climate change and further hampered by inherent vulnerabilities.

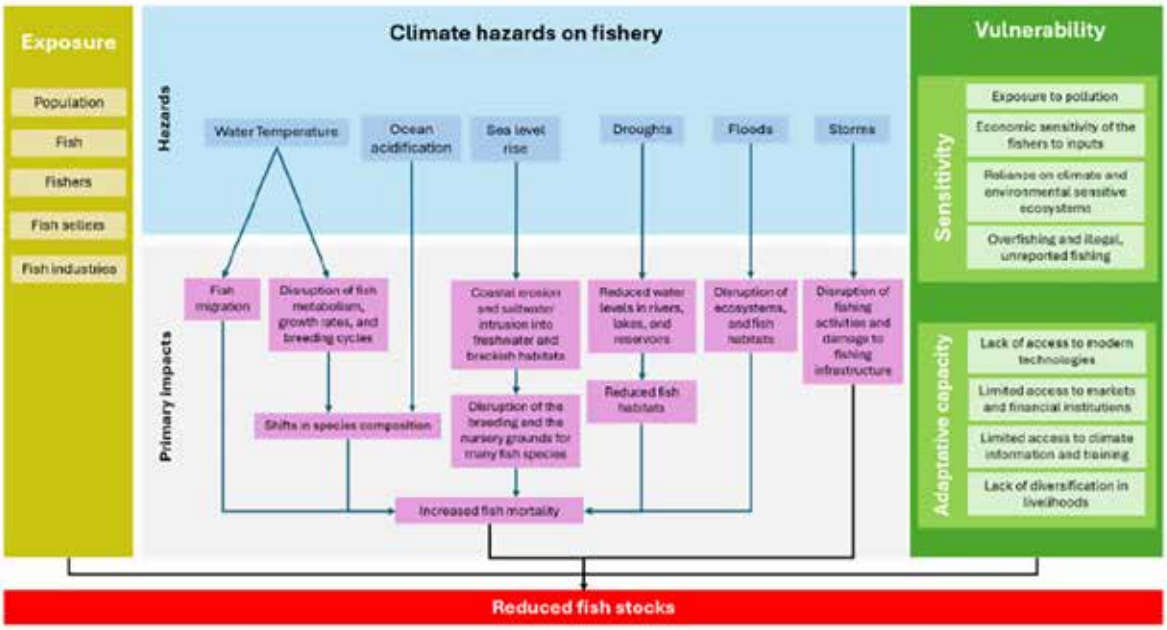


Figure 3.21 Cascading Impacts of climate hazards on fishery sector

The country's fisheries sector faces immediate and future climate risks from rising water temperatures, altered rainfall patterns, ocean acidification, sea-level rise, and more frequent extreme weather events. Figure 3.21 indicates that rising water temperatures, which affect both inland and marine fisheries, will disrupt the natural habitats and migratory patterns of fish species, alter reproductive cycles, and push certain species towards extinction or force them to migrate to cooler

waters. Unpredictable rainfall patterns and prolonged dry seasons are altering the flow of rivers and lakes, disrupting breeding grounds, particularly for freshwater fish, and resulting in seasonal water scarcity. Ocean acidification, caused by increased carbon dioxide absorption, is reducing the availability of calcium carbonate, essential for shellfish and coral growth. This degrades coral reefs and other critical habitats that serve as breeding and feeding grounds for various fish species, thereby affecting fish populations. Rising sea levels exacerbate these challenges by causing saltwater intrusion in coastal and freshwater areas, making certain habitats unsuitable for freshwater species and harming the fishing communities that depend on these ecosystems. Extreme weather events, such as storms and floods, damage fishing infrastructure, from boats to ports, and increase sedimentation in water bodies, reducing water quality and causing fish kills due to oxygen depletion.

Other key vulnerabilities of the fishery sector include (i) limited access to modern equipment, advanced fish-finding technologies, and sustainable fishing methods among the traditional and artisanal fishing population, who constitute the majority in the sector; (ii) overfishing driven by high demand and weak regulatory enforcement, which weakens fish populations and reduces their resilience to climate stressors; and (iii) limited access to climate data and early warning systems, which would otherwise help inform decisions about safe fishing times and locations (Figure 3.21). Overall, the impacts on Nigeria's economy include decreased fish stocks, higher prices for fish products, loss of livelihoods for many fishing-dependent communities, and greater food insecurity across Nigeria, posing a substantial threat to both local economies and national food supply chains.

3.5.3.1.3 Forestry

Nigeria has a valuable forest stock that covered 7.7% of the country's land area in 2015. The forest sector contributes about 2.5% to the Gross Domestic Product and employs around 5 million people through the supply of timber and non-timber forest products (NTFP) (UNEP, n.d.). These forests provide important ecosystem goods and services for the population, including forest-based farming, household equipment, furniture and building materials, timber logging, fuel wood for cooking or charcoal production, fodder (grasses and leaves for livestock), materials for craft, medicinal plants, NTFP, including leaves, honey and fruit, as well as hunting of animals (Fadairo et al., 2017). According to Nigeria's Fifth National Biodiversity Report, 70% of Nigerian households and 90% of the country's rural population use firewood for their energy needs (UNEP, n.d.). Nigeria's forest resources are divided into five distinct ecological zones, namely: the freshwater and mangrove ecological zones, the lowland rainforest, the derived savannah, the Guinean savannah and the Sudan and Sahel ecological zone. There are also a few strips of mountain forest on the plateaux and highlands (Federal Ministry of Environment, Nigeria, 2020). The

Niger Delta in southern Nigeria has the largest mangrove system in Africa, with an estimated 1 million hectares (UNEP, n.d.).

Of an estimated 10 million hectares (about 10% of Nigeria's surface area) originally set aside as forest reserves and conservation areas, only 6% of the total area remains, partly due to deforestation, forest land degradation, climate change and infrastructural development (Federal Ministry of Environment, Nigeria, 2020). Protected areas in Nigeria include 994 forest reserves, 7 national parks, 32 game reserves and sanctuaries, 11 Ramsar sites, 27 Important Bird Areas, 2 World Heritage sites and 5 biosphere reserves.

Climate change poses a significant current and future risk to the forest sector in Nigeria, exacerbating existing challenges such as deforestation, illegal logging, and land conversion for agriculture. As shown in the figure below, increased temperatures, changing rainfall patterns, and more frequent extreme weather events have complex, cascading impacts on forest ecosystems, biodiversity, and human communities, both in the immediate term and over the long term.

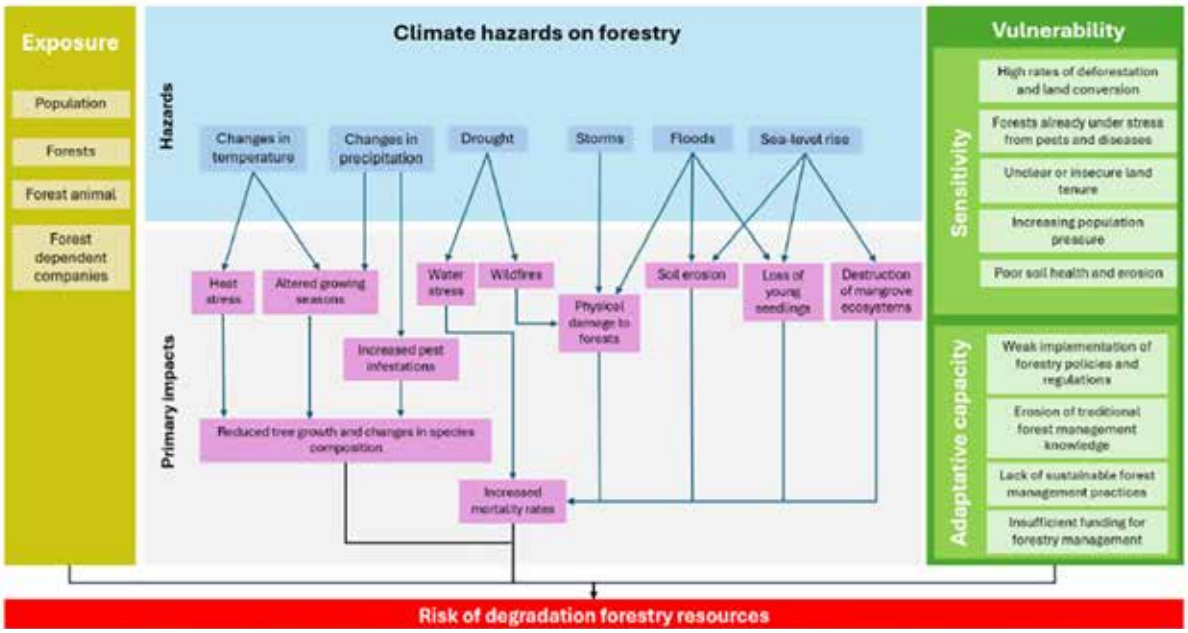


Figure 3.22 Cascading Impacts of climate hazards on forestry resources

Rising temperatures and changing rainfall patterns are stressing some tree species, disrupting their growth cycles, reducing their resilience, and causing shifts in forest ecosystems. Prolonged droughts are worsening these conditions, drying out soils and increasing tree mortality, particularly among species that depend on consistent rainfall. This loss of vegetation cover also exposes soils to erosion, reduces soil fertility, and diminishes forests' ability to regenerate, which can lead to gradual desertification, particularly in the northern regions of Nigeria. Increased wildfires, fuelled by drier conditions and accumulated dead vegetation, are causing widespread forest destruction, releasing vast amounts of

carbon dioxide and accelerating biodiversity loss by damaging habitats critical to various plant and animal species. Sea-level rise is another factor affecting coastal mangrove forests, as saltwater intrusion damages the roots and structure of these ecosystems, weakening their role as protective barriers against coastal erosion and as nurseries for marine life.

These cascading impacts severely reduce the productivity and resilience of the forest sector, threatening the ecosystem services forests provide, such as water regulation, carbon sequestration and soil protection, all of which are vital to Nigeria's environmental and economic stability (Figure 3.22). These climate risks are compounded by the sector's inherent vulnerabilities, including high rates of deforestation driven by unsustainable logging, agricultural expansion and firewood collection. Unsustainable practices not only reduce forest cover but also weaken the adaptive capacity of forest ecosystems by disrupting biodiversity and ecosystem resilience, making forests more vulnerable to the impacts of climate change. In addition, the sector's adaptive capacity is limited by weak enforcement and a lack of community involvement in forest management, leading to unchecked illegal logging and inadequate reforestation efforts. Many rural communities rely heavily on forests for their livelihoods, from timber and fuelwood to non-timber forest products, creating a high level of dependency that is difficult to shift without viable economic alternatives. Limited financial resources and access to climate-resilient forest management practices also hamper efforts to develop adaptation strategies, such as reforestation with drought-resistant species or the establishment of community-managed protected areas. As climate change accelerates, Nigeria's forest sector is at risk of significant degradation, which would undermine local economies, reduce biodiversity, increase carbon emissions and reduce natural defences against climate impacts such as erosion and flooding. This degradation not only threatens the forest sector but also has wider socio-economic and environmental impacts, as forest loss will reduce the resilience of surrounding communities and ecosystems, leading to increased vulnerability across the country.

3.5.3.1.4 Water Resources

Nigeria is endowed with a diversity of water systems, with significant regional variations due to the country's climatic zones and geography, ranging from the arid Sahel in the north to the humid tropical rainforests in the south. These include substantial annual rainfall, large surface water bodies, rivers, streams and lakes, as well as plentiful reserves of groundwater. The Niger and Lake Chad Basins are the two most prominent water systems in the country, though several smaller coastal river basins exist. In addition to these two major basins, Nigeria also includes the South-western and South-eastern littoral basins, which play a significant role in the country's hydrology.

Surface Water basin	Area (Km²)*	% of Total Country Area	Main rivers
Niger basin	584,194	63%	Niger and its tributaries (Benue, Sokoto and Kaduna)
Lake Chad basin	179,282	20%	Komadougou Yobe and its tributaries (Hadejia, Jama'are and Komadougou Gena
South-Western Littoral basin	101,802	11	Rivers from the hilly area south and west of the Niger River
South-eastern littoral basin	54,493	6	Cross , Imo

Table 3.3: Nigeria's main surface water basins (data Aquastat – FAO)

**The total area is 919,770km2 slightly different from the official total area of the country of 923,768 km²*

Despite these resources, Nigeria remains vulnerable because the water supply is unable to meet domestic, industrial and agricultural demand in recent times. A very large proportion of the population is at risk of water stress, as less than 40% have direct access to drinking water. Climate change is leading to increased variability in precipitation, causing flooding in certain wetlands in the south of the country and reduced rainfall in the savannahs of the north, which may lead to droughts and a reduction in surface water resources in the north.

As the country's population rises to 258 million by 2030, with a commensurate increase in demand for food, demand for agricultural water is expected to rise. National water consumption for irrigation in Nigeria stood at 1926 MCM/day (million cubic metres per year) in 2010 and will rise to 6245 MCM/day (million cubic metres per year) in 2030. About 88% of rural Nigerian households use surface water, and 83% of these households are among the poorest in the country (Abubakar, 2019). The demand for municipal water in Nigeria will also rise to 8,852 MCM/year (million cubic metres per year), which represents 53% of Nigeria's total water demand by 2030, or 16,585 MCM/year (Ojo et al., 2020). This situation is already putting considerable pressure on Nigeria's water resources, and the effects of climate change will only exacerbate the problem further (Ojo et al., 2020).

Water resources in Nigeria face serious and growing risks from climate change, driven by rising temperatures, shifting rainfall patterns, more frequent droughts and floods, and other climate-induced phenomena.

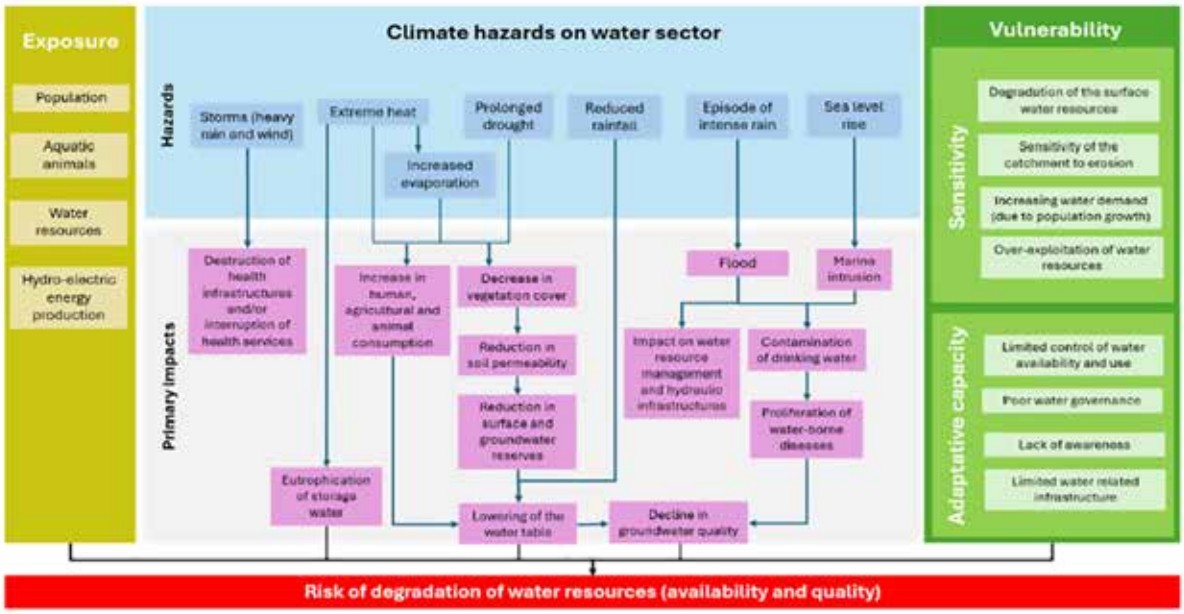


Figure 3.23: Cascading Impacts of climate hazards on water resources

Changing rainfall patterns are causing water scarcity in many regions, particularly in the northern parts of the country, where dry seasons are becoming longer and more intense, reducing river flows and groundwater recharge. Higher temperatures increase evaporation from surface water sources such as rivers, lakes and reservoirs, further reducing available water supplies while increasing demand from agriculture, industry and households.

Droughts, exacerbated by climate change, reduce access to water for drinking, irrigation and livestock, affecting food security and rural livelihoods, particularly in the northern part of the country. Meanwhile, in southern and coastal areas, rising sea levels are causing saltwater intrusion into freshwater aquifers, contaminating vital water sources for people and agriculture. Intense rainfall and flooding events, worsened by climate change, are overwhelming drainage systems and damaging water infrastructure, contaminating drinking water sources with pollutants and increasing the spread of waterborne diseases across all ecological zones of Nigeria. The health and sanitation sector faces significant challenges because of water shortages, with some communities still practising open defecation.

The cascading impacts not only limit access to safe and clean water but also increase the burden on public health systems, reduce agricultural productivity, and drive up costs for water treatment and infrastructure repairs (Figure 3.23). These climate-related water challenges are exacerbated by Nigeria's inherent vulnerabilities, such as (i) indiscriminate waste disposal, (ii) poor management of water resources, (iii) dependence on rain-fed agriculture and over-reliance on surface water resources, and (iv) limited access to infrastructure for efficient water storage and distribution, particularly in rural and underserved urban areas. In addition, rapid population growth

and urbanisation are increasing demand for water, putting additional pressure on already strained water systems and accelerating the depletion of both surface and groundwater resources. The adaptive capacity of Nigeria's water sector is also limited by inadequate community awareness, lack of access to climate-resilient technologies, inadequate investment in resilient infrastructure such as reservoirs, pipelines and treatment plants, and the lack of comprehensive water management to address issues such as equitable water allocation and groundwater regulation. As climate change progresses, these vulnerabilities will worsen, leading to higher incidences of water scarcity, contamination and infrastructure failure, which in turn will exacerbate social inequalities, increase poverty and threaten food security.

3.5.3.1.5 Health

Nigeria's health system and infrastructure are highly vulnerable to climate hazards. In Nigeria, climate change is not only exacerbating existing health problems but also compounding them through deficiencies in water and sanitation, energy availability, waste management and food systems (FMH&SW, 2024). Climate change also threatens health services, water and food security, infrastructure and social protection frameworks.

According to Nigeria's Climate Change and Health Vulnerability and Adaptation (V&A) Assessment Report, published in October 2024, the severity and scale of climate change-related health impacts are expected to worsen across all geographical areas of the country. The report shows that, under all scenarios, the risks associated with climate change in the country will increase, straining healthcare capacity, increasing the vulnerability of communities, and creating conditions conducive to the spread of disease.

Nigeria will experience a 21% increase in disease burden due to climate change by 2030, with significant increases in water-borne diseases, non-communicable diseases and mental health disorders (Nigeria Climate Change and Health, 2024). The level of risk and impact varies by population and geographical zone. The assessment also identifies children, the elderly and pregnant women as being at increased risk from climate impacts, while the north-east and north-west of Nigeria are likely to be more affected. Nigeria still has a high prevalence of communicable diseases and a growing number of non-communicable diseases. Communicable diseases (malaria, acute respiratory infections (ARI), measles, diarrhoea, tuberculosis, HIV/AIDS and neglected tropical diseases) account for 66% of the total burden of disease in the country. An estimated 97% of the country's population of 160 million is at risk of malaria, with children and pregnant women most vulnerable to malaria-related illness and death in Nigeria. Similarly, the burden of non-communicable diseases (e.g. cardiovascular disorders, diabetes, cancers and chronic obstructive pulmonary disease) remains a major challenge in

Nigeria. Malnutrition and nutrition-related diseases continue to be a major public health problem in Nigeria and remain the underlying cause of 53% of under-five mortality in the country.

Climate change exacerbates these health challenges and will continue to do so as climate hazards intensify.

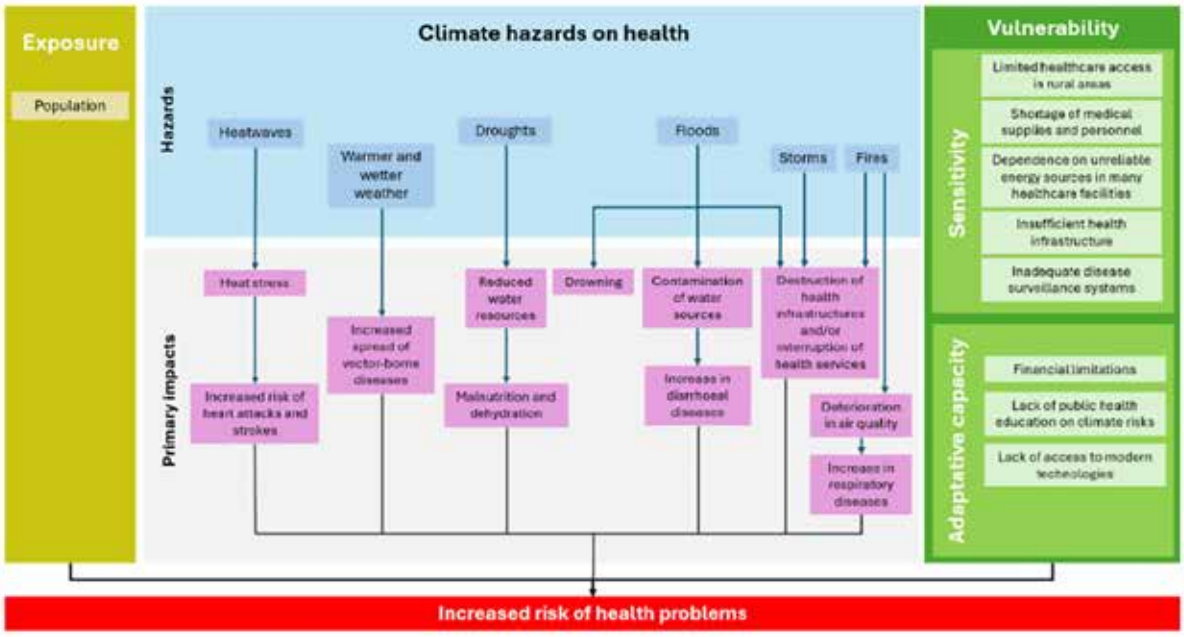


Figure 3.24: Cascading Impacts of climate hazards on health sector

As climate change intensifies, the cascading impacts are becoming more severe and frequent. Rising temperatures and shifting rainfall patterns foster the spread of vector-borne diseases such as malaria and dengue fever, while extreme weather events, such as floods, increase the prevalence of waterborne diseases, including cholera. Rising sea levels, increasing temperatures, and unpredictable rainfall have led to an increase in cholera cases, particularly in the coastal cities of Nigeria.

Rising temperatures, erratic rainfall patterns, and an increase in extreme weather events are driving a surge in heat-related illnesses, respiratory conditions, waterborne infections, and vector-borne diseases. Higher temperatures contribute to heat stress and worsen cardiovascular and respiratory diseases, particularly in densely populated urban areas where access to cooling infrastructure is limited. Changes in rainfall patterns and warmer climates also create favourable conditions for the spread of vector-borne diseases such as malaria and dengue fever, as mosquitoes and other disease-carrying vectors thrive and expand their habitats into regions previously less affected. More frequent and intense flooding, coupled with inadequate sanitation systems, heightens the risk of waterborne diseases such as cholera, typhoid, and dysentery, especially in vulnerable communities with limited access to clean water and proper sanitation. Prolonged droughts further

reduce water availability and quality, exacerbating malnutrition and dehydration, which weaken immune systems and increase susceptibility to illness.

Key vulnerabilities in Nigeria's health sector include (i) limited access to healthcare in rural areas, (ii) shortages of medical supplies and personnel, and reliance on unreliable energy sources that undermine the effectiveness of cooling systems, water and sanitation, and disease prevention efforts; and (iii) limited investment in climate-resilient infrastructure, such as flood-proof health facilities, robust emergency response systems, and early warning mechanisms for disease outbreaks (Figure 3.24). A lack of public health education on climate risks also leaves communities ill-prepared to address climate-related health challenges. As climate change intensifies, these vulnerabilities are expected to worsen, leading to longer and more severe health crises that strain the sector's capacity to deliver timely and effective care.

3.5.3.1.6 Energy

Nigeria's main sources of energy include oil and gas, and biomass. Nigeria is Africa's largest producer of crude oil, the world's 14th-largest oil producer, and the 6th-largest exporter of liquefied natural gas (U.S. EIA, 2020). Despite being one of the world's largest oil producers, access to reliable electricity remains limited, with only 55% of the population having access in 2020 (Environmental and Economic Resource Centre, 2024). In particular, most rural communities remain off-grid and are trapped in an energy-poverty loop. Most of the energy produced and exported comes from oil and gas, whose coastal production is vulnerable to sea-level rise, storm surges and coastal flooding.

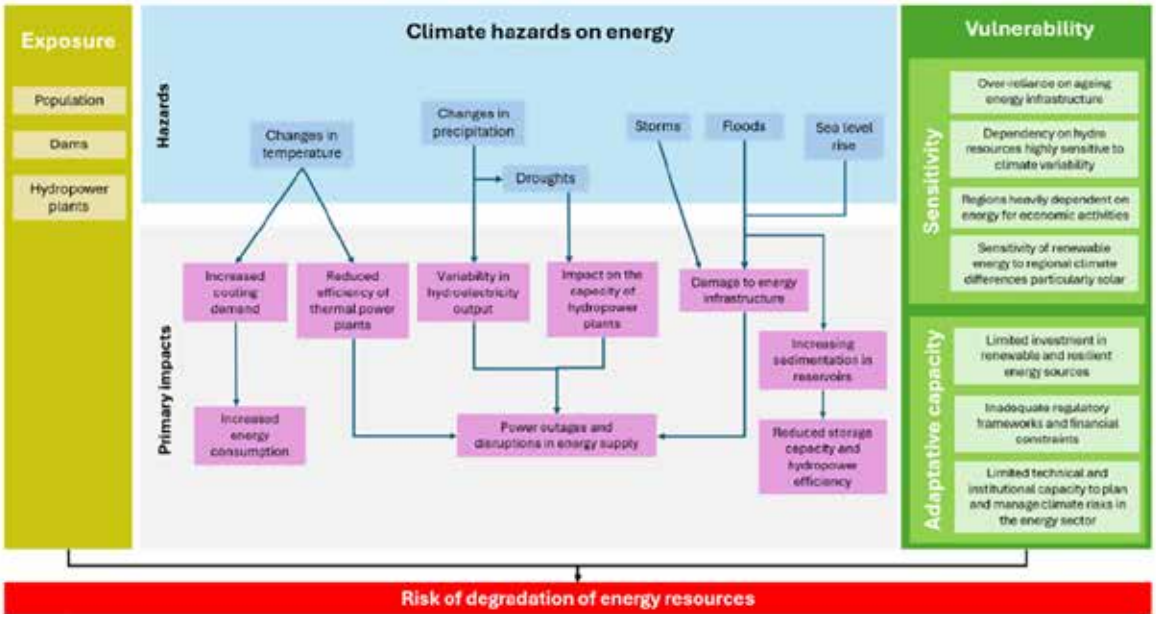


Figure 3.25: Cascading Impacts of climate hazards on energy resources

Nigeria's energy sector is increasingly vulnerable to climate change and faces immediate and long-term climate risks from rising temperatures, unpredictable rainfall patterns and extreme weather events that affect energy production, distribution and demand. Hydropower, which provides about 22% of Nigeria's electricity, is particularly sensitive to changes in rainfall and river flows. Prolonged dry spells reduce water levels in rivers and reservoirs, thereby lowering hydropower generation capacity. Rising temperatures increase electricity demand, particularly for cooling in urban areas, putting additional strain on the grid and forcing an increase in energy imports or reliance on inefficient back-up generators that are costly and often run on fossil fuels. Higher temperatures also reduce the efficiency of thermal power plants, which are another key component of Nigeria's energy mix. Cooling systems in these plants become less effective at higher temperatures, reducing output and increasing operating costs. Extreme weather events, such as storms and floods, exacerbate these risks by damaging energy infrastructure, including transmission lines, substations and pipelines, resulting in widespread service disruptions and costly repairs. Flooding, particularly in coastal and riverine areas, can lead to erosion and the collapse of infrastructure, while increasing sedimentation in reservoirs, reducing storage capacity and hydropower efficiency over time.

The cascading impacts of climate hazards on energy resources are enormous. Nigeria's energy sector is increasingly vulnerable to climate change and faces immediate and long-term risks from rising temperatures, unpredictable rainfall patterns and extreme weather events that affect energy production, distribution and demand (Figure 3.25). Hydropower, which provides a significant proportion of Nigeria's electricity, is particularly sensitive to changes in rainfall and river flows. Prolonged dry spells reduce water levels in rivers and reservoirs, thereby lowering hydropower generation capacity. This reduction in power output strains an already under-resourced and overburdened energy system, leading to more frequent power outages and reduced reliability. In addition, rising temperatures increase electricity demand, particularly for cooling in urban areas, putting further strain on the grid and forcing an increase in energy imports or reliance on inefficient back-up generators that are costly and often run on fossil fuels.

Higher temperatures will also reduce the efficiency of thermal power plants, another key component of Nigeria's energy mix, as cooling systems become less effective, lowering output and raising operating costs. Extreme weather events, such as storms and floods, exacerbate these risks by damaging energy infrastructure, including transmission lines, substations and pipelines, leading to widespread service disruptions and costly repairs. Flooding, particularly in coastal and riverine areas, can cause erosion and the collapse of infrastructure, while increasing sedimentation in reservoirs, reducing storage capacity and hydropower efficiency over time.

Nigeria's energy sector is vulnerable, compounded by inherent sensitivities, including an over-reliance on ageing infrastructure that lacks climate resilience, dependence on fossil fuels and hydropower, both of which are highly sensitive to climate variability, and the fact that regions are heavily dependent on energy for economic activity. The sector's adaptive capacity is constrained by limited investment in renewable and resilient energy sources, such as solar and wind, which could diversify energy production and reduce dependence on water-intensive sources such as hydropower. In addition, inadequate regulatory frameworks and financial constraints limit the sector's ability to implement large-scale climate adaptation measures, such as upgrading infrastructure to withstand extreme weather events or developing early warning systems for at-risk facilities. Rural areas and low-income urban neighbourhoods are particularly affected by these vulnerabilities, as they are often the last to receive electricity during shortages and have limited access to alternative energy sources, thereby exacerbating social inequalities. As climate change intensifies, these challenges are expected to worsen, leading to reduced energy security, higher costs for consumers, and an increased burden on industries that rely on a consistent supply of electricity. The cumulative effects of these cascading impacts could slow economic growth, reduce industrial productivity, and exacerbate poverty and social inequality.

3.5.3.1.7 Transport

Though Nigeria has significantly developed road, air, rail and water transport, the road transport system is dominant for moving goods and services across the country and accounting for 86.4% of the national GDP in 2016. With an estimated road network of 195,000 km, the country's road network has been ranked the largest in West Africa and the second largest south of the Sahara (Federal Republic of Nigeria, 2020). The road infrastructure is poorly maintained particularly in the rural communities, making them susceptible to flood events while many of the roads witness traffic congestion due to inadequacy of the infrastructure especially in the urban areas.

Nigeria's rail system is relatively underdeveloped compared to other countries; however, the ongoing modernization efforts to improve this sub-sector is expected to ease congestion on the roads and provide a more sustainable mode of transportation. The rail networks remain limited to certain regions, and flooding is a challenge to rail tracks and disrupt Services.

The country's waterways, which span about 8,600 km, are another essential transport mode. Rivers like the Niger and Benue are key for moving bulk cargoes, particularly to inland areas. Yet, flooding can lead to the disruption of inland navigation routes, making goods transport vulnerable, especially during the peak of rainy seasons.

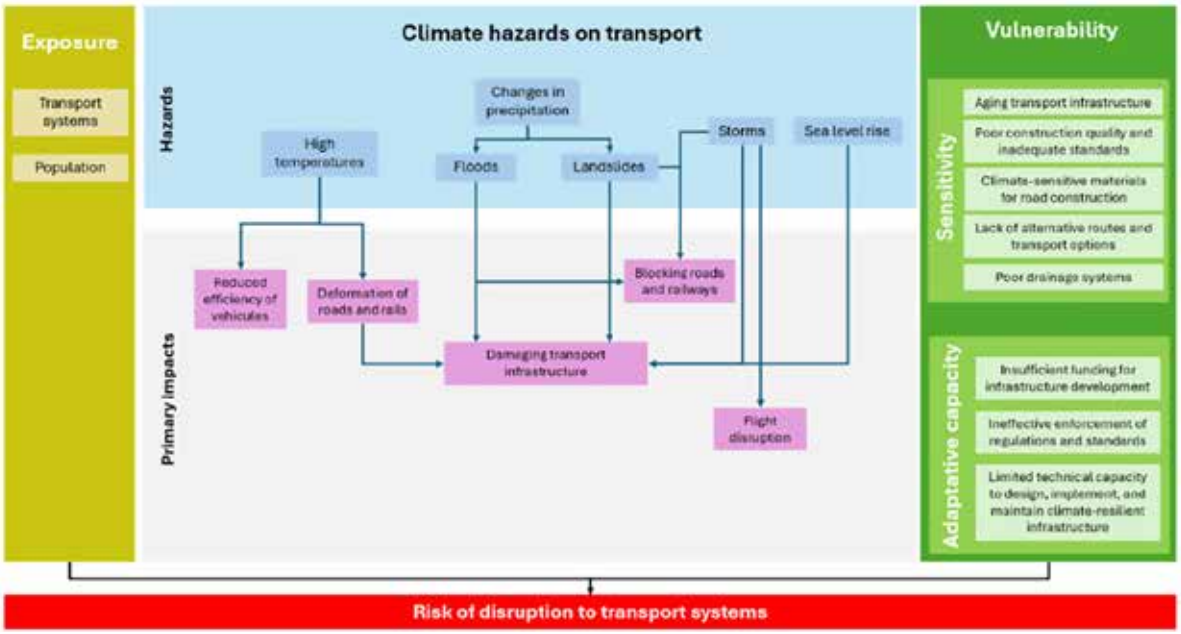


Figure 3.26: Cascading Impacts of climate hazards on transport systems

The transport sector in Nigeria is increasingly vulnerable to the impacts of climate change, with significant risks posed by rising temperatures, erratic rainfall patterns, sea-level rise, and more frequent extreme weather events, all of which disrupt transport infrastructure, services, and mobility. Rising temperatures are accelerating the degradation of road surfaces, causing asphalt to soften and crack, particularly in areas with outdated or poorly maintained infrastructure. This degradation raises maintenance costs, shortens the lifespan of roads, bridges, and railways, and complicates transport logistics. Intense and unpredictable rainfall frequently leads to flooding, particularly in low-lying urban and coastal regions such as Lagos, where drainage systems are often insufficient (Figure 3.26). Flooding erodes soil, undermining road foundations and increasing the risk of landslides, which leads to extended road closures and traffic disruptions. In addition, rising sea levels and coastal erosion threaten Nigeria's port infrastructure, with saltwater intrusion corroding metal structures and compromising the stability of docks and jetties, thereby affecting the flow of goods.

Extreme weather events, such as storms and heatwaves, disrupt air travel by delaying flights and airport operations, increasing operating costs and affecting air safety. Vulnerabilities within Nigeria's transport sector only amplify these risks. Much of the infrastructure is ageing, lacks resilience to climate change, and is concentrated in areas highly exposed to climate hazards. The transport sector's heavy reliance on road transport, which handles over 80% of goods and passengers, means that road disruptions significantly affect national mobility and economic stability.

Limited financial resources further constrain the sector's ability to invest in resilient infrastructure or to improve drainage systems to mitigate flood damage.

There is also a lack of comprehensive climate adaptation planning and effective enforcement, which hinders proactive measures such as strengthening coastal defences around ports or elevating roadbeds in flood-prone areas. Rural areas, with their limited transport infrastructure, are particularly vulnerable, as disruptions often leave these regions isolated and hinder access to essential services. As climate change intensifies, these vulnerabilities will worsen, leading to greater infrastructure damage, higher maintenance costs, and longer service disruptions. These effects will ripple through the economy, limit social mobility, and reduce resilience to future climate impacts. These compounded challenges highlight the urgent need for climate-adaptive transport policies and resilient infrastructure investments to safeguard Nigeria's transport sector in the face of a changing climate.

3.5.3.1.8 Human Mobility

Climate change is reshaping patterns of human mobility and population distribution across Nigeria, generating a complex spectrum of outcomes that range from forced displacement to involuntary immobility. As the NAP's analysis of climate impacts across Nigeria's ecological zones reveals, displacement risk related to climate stressors is not uniformly distributed. It is concentrated in frontline communities exposed to climate impacts and among populations whose livelihoods depend on climate-sensitive resources and who face compounding socio-economic and governance vulnerabilities.

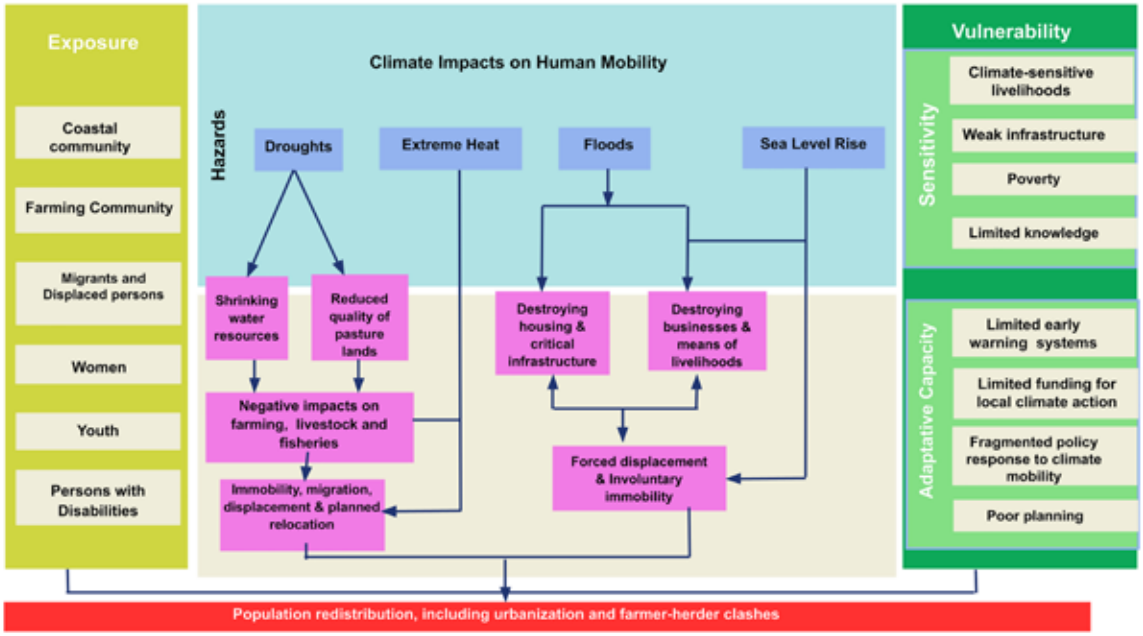


Figure 3.27: Cascading Impacts of Climate on Human Mobility

3.5.3.1.8.1 Regional Climate Stressors and Mobility Pressures

Nigeria's diverse geography exposes different regions to distinct but interconnected climate hazards. In the north, advancing desertification, prolonged droughts, extreme heat, and land degradation (Agu et al., 2020; Vande, 2025) are progressively reducing the availability of grazing land and freshwater resources, undermining agropastoral livelihoods that support millions of people. In southern and coastal areas, sea-level rise, intensifying rainfall variability, coastal erosion, and recurrent flooding threaten settlements, infrastructure, and coastal economies (Ndimele et al., 2024; Nkwunonwo et al., 2020; Yusuf, 2019; Obiefuna, 2021; Amakrane et al., 2023). Across the country, more frequent and severe extreme weather events are disrupting the agricultural calendar and eroding productive assets, pushing climate mobility from a historically adaptive strategy toward an increasingly reactive and distressed response.

3.5.3.1.8.2 Populations at Heightened Risk

If vulnerability to climate risks is unevenly distributed and shaped by livelihood dependence, socio-economic status, past and ongoing displacement experiences, gender, age, and geographic exposure, the following groups face particularly acute risks:

- Pastoralists and farming communities: Fulani herders and other pastoralists face livelihood collapse as water and grazing resources contract, driving southward mobility into the Middle Belt and Benue River Basin and heightening conflict risk with farming communities (Udoh et al., 2025; Agu et al., 2020; Akinyemi & Olaniyan, 2017).
- Coastal communities: Densely populated coastal settlements, particularly informal communities in the south, face escalating flood risk, erosion, and sea level rise. Lagos could see sea levels rise up to three meters by 2050, displacing up to 1.4 million people (Ministry of Environment and Water Resources, 2024); over a meter of sea level rise could displace over 80% of the Niger Delta without adequate adaptation, including planned relocation.
- Women: Structural inequalities — limited land rights, unequal access to finance and training, and gendered household decision-making — reduce women's adaptive capacity and ability to pursue migration as a planned strategy (Nnadi et al., 2023; Yusuf, 2019; Amakrane et al., 2023)
- Persons with disabilities: Socio-economic marginalisation, exclusion from formal adaptation planning, and infrastructural gaps constrain the adaptive capacity of persons with disabilities (Abikoye & Abikoye, 2025).
- Youth: Young people show high mobility aspirations but often lack the skills and capital to secure stable livelihoods at destination, leaving them vulnerable to precarious employment and marginalisation (Amakrane et al., 2023; Agu et al., 2020).

- Migrants and displaced persons: Those who previously moved often encounter new vulnerabilities — inadequate housing, limited services, exclusion from labour markets — in receiving communities. Without planning and support, mobility can reproduce vulnerability rather than resolve it (Owusu et al, 2025; Agu et al., 2020).

3.5.3.1.8.3 Trapped Populations and Involuntary Immobility

A critical but often overlooked dimension of climate risk is involuntary immobility: the situation of households that wish to move but lack the financial, social, or institutional resources to do so. Evidence from Ajegunle, Lagos, indicates that nearly half of the surveyed residents expressed a desire to relocate but were unable to finance the move (Mixed Migration Centre, 2023). In conflict-affected areas of the Middle Belt, immobility may also reflect deep attachments to ancestral land and collective identity, meaning that some populations remain in high-risk environments not solely due to material constraints but also due to culturally grounded decisions (Olumba, 2024; 2025). In both cases, immobile populations remain exposed to escalating climate hazards without viable adaptation pathways.

3.5.3.1.8.4 The Climate–Conflict–Mobility Nexus

In northern and central Nigeria, climate mobility interacts with conflict dynamics in ways that amplify overall vulnerability. Environmental degradation has been documented to intensify competition over land, water, and pasture, fracturing traditional conflict-resolution mechanisms, and, in some contexts, contributing to recruitment into armed groups (Albert, 2025; Vande, 2025; Udoh et al., 2025; Agu et al., 2020; Obioha, 2008). Farmer–herder conflicts in the Middle Belt represent a particularly significant hotspot, where ecological pressures, displacement, contested land tenure and identity-based tensions converge.

Poorly designed adaptation interventions—including large-scale infrastructure projects and coastal protection schemes—may also generate displacement (Rice et al., 2021; Ajibade, 2017) if communities are relocated without adequate consultation, compensation, or livelihood support, further creating maladaptive outcomes that deepen existing inequalities.

3.5.3.2 Future Climate Mobility Projections

By 2050, modelling projections by the Africa Climate Mobility Initiative (ACMI) (Amakrane et al. 2023) suggest that up to 1.5% of Nigeria's total population — or 6.6 million people — could be forced to move internally due to climate impacts under a high-emissions, unequal development scenario (the "Rocky Road" RCP 6.0/SSP3). Under a high-emissions, inclusive development scenario (the "High Road" RCP 6.0/

SSP1), this drops to a high of 4.1 million people moving internally by 2050.

These projections indicate that when development is unequal (SSP3)—including higher population growth and lower economic growth, urbanisation rates and educational outcomes—more Nigerians could be compelled to move to escape climate impacts and seek better living conditions. Climate mobility is lower under a more inclusive development scenario (SSP1)—even when maintaining a high emissions trajectory (RCP6.0)—suggesting that lower population growth and improved economic conditions reduce people's exposure to climate risk and increase their capacity to absorb shocks and adopt climate-adaptive solutions.

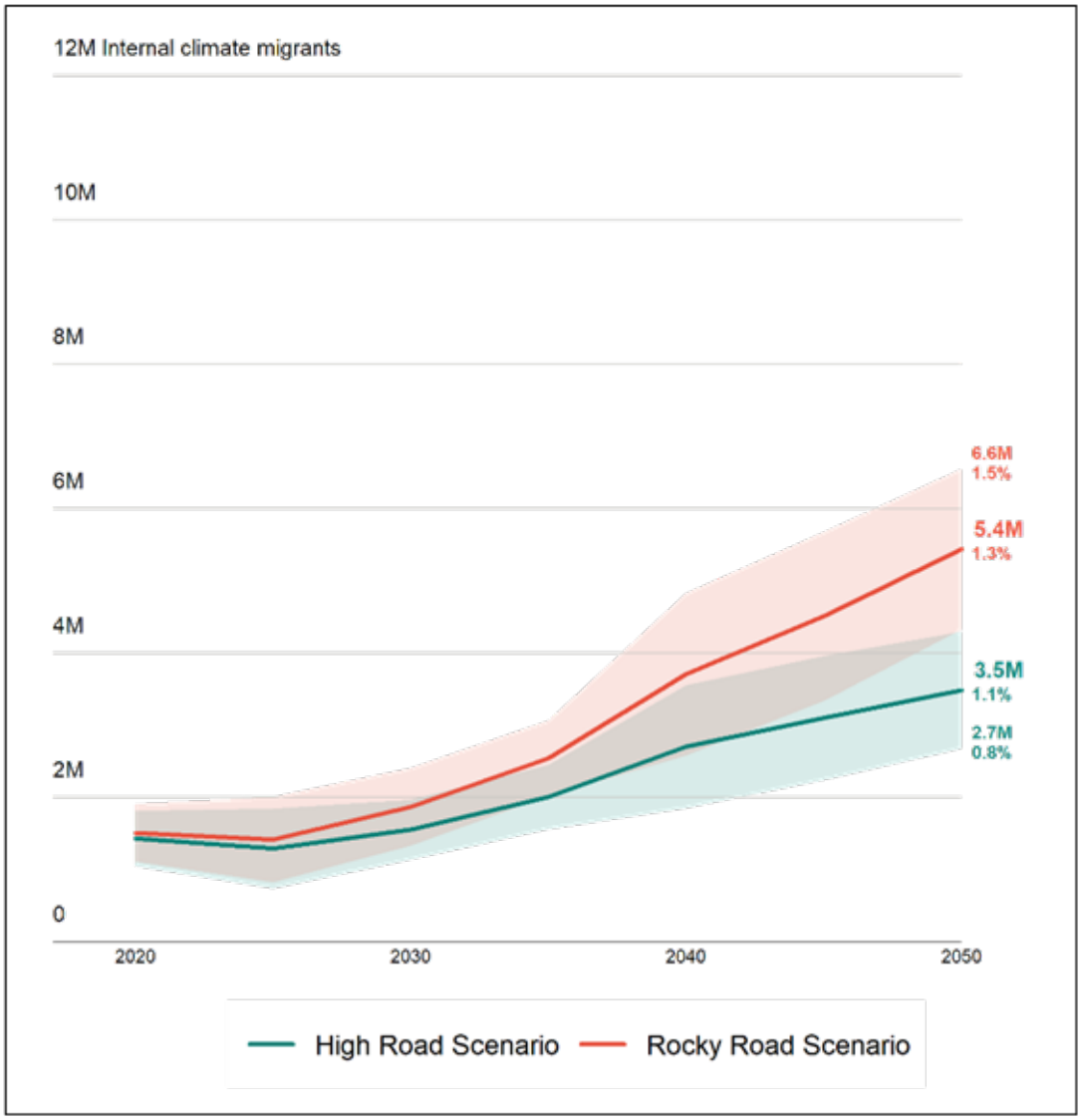


Figure 3.28: Projected internal climate mobility within Nigeria from 2020 to 2050 (shown as the total number of internal climate migrants and as a percentage of the population)
Source: Center For International Earth Science Information Network-CIESIN-Columbia University, CUNY Institute For Demographic Research-CIDR, C. U. O. N. Y., & Global Center Of Climate Migration-GCCM. (2025). African Climate Mobility Initiative (ACMI): Internal Migration Projections [Dataset]. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC). https://doi.org/10.7927/5TV3-FF20

The modelling projections also indicate where hotspot areas of in- and out-mobility are likely to emerge. They signal that people are likely to move out of the coastal zone into the hinterland in the southern region by 2050, especially in Lagos, where sea-level rise and changing rainfall patterns are already causing coastal erosion and severe flooding. They also suggest the likelihood of out-mobility across the Middle Belt region, where flood risk is high, while signalling the possibility of climate in-mobility into the northern parts of the country, where water resources are projected to decline.

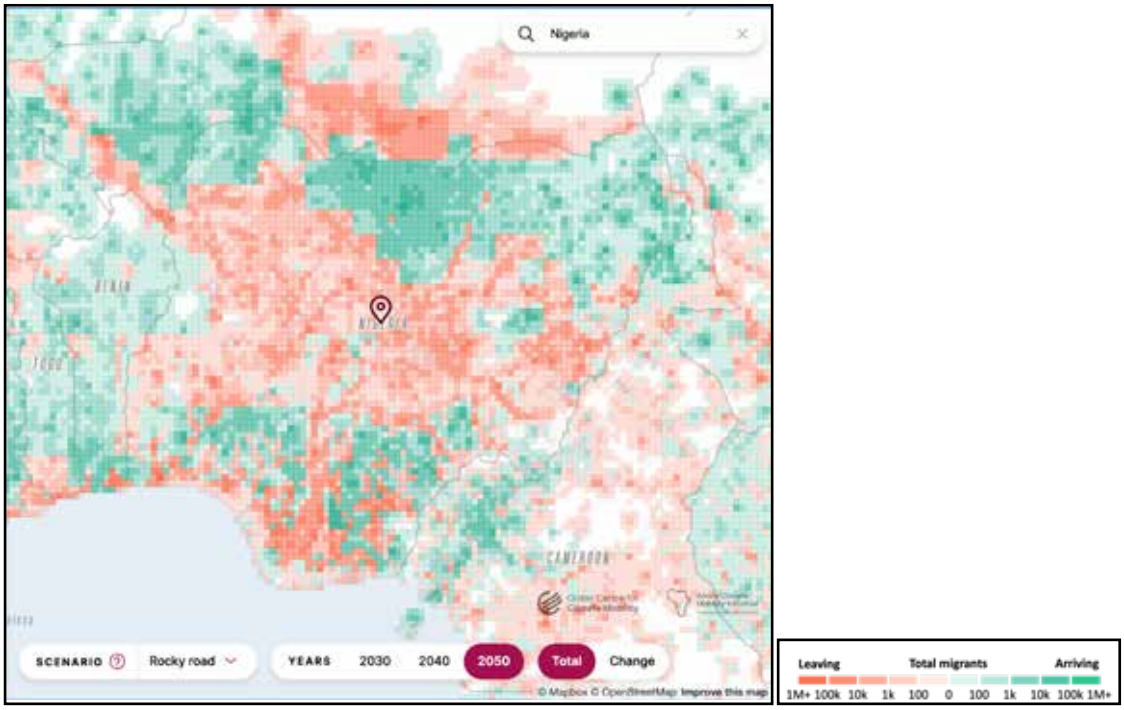


Figure 3.29: Total projected internal mobility by 2050 (RCP6.0/SSP3)

3.5.3.1.8.6 Stock-Taking: Policies, Practices, and Governance

Several existing policy frameworks acknowledge the growing impact of climate change on human mobility in Nigeria. The National Climate Change Policy (2021–2030) highlights increasing climate-driven mobility, particularly through a security lens. The third Nationally Determined Contribution (NDC 3.0) goes further, recognising emerging climate mobility patterns and calling for a standardised definition of climate migration while explicitly embracing mobility as an adaptation strategy. At the subnational level, Lagos State has identified planned relocation as a component of climate resilience in its Climate Adaptation and Resilience Plan (LCARP).

Despite these foundations, Nigeria's approach to climate mobility remains fragmented. No overarching policy framework currently harmonises priorities across sectors, institutions, or levels of government, or positions mobility systematically as part of the national climate resilience effort. This NAP addresses

that gap. Building on existing commitments and grounded in climate projections, it establishes clear policy objectives that identify Nigeria's climate mobility challenges, including the often-overlooked problem of involuntary immobility, and recognises mobility itself as a legitimate and underutilised adaptation strategy for frontline communities.

3.5.4 Risks and vulnerabilities of different Ecological Zones in Nigeria

3.5.4.1 Swamp Forest Zone

The swamp forest region of Nigeria is a unique and ecologically rich zone characterised by dense mangroves, freshwater swamps, and flood-prone areas along the lower Niger Delta and the coast. It spans 5 States (Akwa Ibom, Bayelsa, Cross River, Delta and Rivers States). The vegetation of the swamp forest region consists of four ecological zones: coastal forests, mangrove forests, freshwater swamp forests and lowland rainforests. This ecological zone is characterised by high annual rainfall, ranging from 2,000 mm to over 3,000 mm. The climate is consistently warm, with average temperatures between 26-28°C, and is marked by mild winds. The geography and economy of the region are deeply intertwined with the swamp forest ecosystem. Communities depend heavily on fishing, small-scale agriculture, and forest resources, all of which are influenced by the region's unique hydrological and ecological characteristics (Tables 3 and .4). Agricultural practices in the zone are heavily influenced by abundant water resources, with a focus on fishing and aquaculture.

Climate Hazards	Key Vulnerabilities	Key Impact
Rain induced Flood, High temperature, Sea-level Rise	Environmental Factors: Riverine and estuarine nature; zone partly situated under sea level; clay-rich soil with slow drainage; Pollution from oil exploration and industries; Rain-fed agriculture; Fragile mangroves and freshwater swamp; Deforestation from illegal logging, agricultural expansion, competing land use and infrastructure development Socio-economic factors: Heavy dependence on natural resources for their livelihoods; High poverty; Poor infrastructure poor water and sanitation facilities; Insecurity and Conflicts in the region; poor policy implementation	Rain-induced flooding cause Loss of biodiversity, threatens food security; sea level rise lead to loss of economic activities livelihoods for fishermen, aquaculturist and farmers; destruction of breeding grounds for marine species with loss of biodiversity; displacement of indigenous people; coastal erosion
Key Adaptation Options for Agriculture	Promote Soil Conservation Techniques; Introduce flood and salt tolerant varieties;Improve Flood Control Infrastructure; Enhance Coastal Protection Measures e,g. mangrove buffers; Improved storage facilities, Transportation and Market Access; Weather Forecasting and Early Warning Systems	

Key Adaptation Options for Water Resources	strengthen Water Treatment Facilities; Improve Urban Drainage and Waste Management Systems; Improve Watershed Management ; Early Warning and Monitoring Systems; Expand Climate-Resilient Water Infrastructure
Key Adaptation Options for Energy Sector	Relocate or Elevate Energy Facilities; Invest in Renewable Energy; Strengthen Coastal Defences; Promote Energy Efficiency Measures; Strengthen Pipeline Infrastructure; Improve Spill Response Mechanisms
Key Adaptation Options for Health Sector	Improve Wastewater Treatment Infrastructure; Implement Coastal Wetland and mangrove Restoration against saltwater intrusion; Promote Water Purification and Safe Storage; Strengthen Disease Surveillance and Response; increase Access to Preventive Healthcare;improve Early Warning Systems for Disease Outbreaks; Increase Hygiene and WASH Awareness
Key Adaptation Options for Transportation Sector	Develop Flood-Resilient Infrastructure; Improve Drainage Systems; Improve River and Creek Dredging Practices; Improve Public Transport Cooling Efficiency against heatwaves; Expand Investment in Resilient Transport Systems; Restore and Protect Mangrove Ecosystems

Table 3.4: Summary of Climate Risk Assessment of the Swamp Forest Ecological Zones

3.5.4.2 Tropical Rainforest Zone

The tropical rainforest agroecological zone is characterised by a densely packed evergreen forest with tall trees and thick undergrowth. While some rainforest areas remain pristine and undisturbed, with a dense canopy, most have been impacted or degraded due to conversion to farmland, indiscriminate logging, and wood extraction. In some cases, tropical rainforests are now limited to protected areas such as forest reserves, national parks, and game reserves, which still face encroachment. However, human activities have drastically reduced this once densely forested area (Alahira, 2022). The tropical forest zone experiences high rainfall and warm temperatures year-round, creating favourable conditions for the cultivation of crops such as cocoa, oil palm, yams, and cassava (Table 3.5). The states covered by the tropical forest include Abia, Anambra, Enugu, Imo, Benue, Edo, Ebonyi, Ondo, Ogun, and Lagos.

Climate Hazards	Key Vulnerabilities	Key Impact
Rain induced Flood, High temperature, Sea-level Rise	Environmental factors: Dependence on forest resources; Topography of the rainforest zone includes low-lying areas prone to flooding; Soil degradation due to Unsustainable agricultural practices and population pressure; Socio-economic factors: Population Growth and Pressure; Insufficient and Poor Infrastructure in critical sectors; Improper construction and blockage of natural waterway; Dependence on rain-fed agriculture; Poor Waste Management Practices; Weak Governance and Poor enforcement of regulations and laws	Flooding associated with more frequent/intense rainfall and sea level rise cause displacements and disrupt livelihoods; threatens infrastructure; and undermine food security and local economies by damaging crops, livestock, and fish populations and biodiversity populations; heat stress on ecosystems and communities, impacting agricultural yield, cause water scarcity, and impact public health; increased flooding; poor yield of tree and arable crops; high incidence of diseases; pollution of surface water from farmlands
Key Adaptation Options for Agriculture	Promote flood and heat-tolerant crop species; Improve Irrigation Techniques; Promote Climate-Smart Agriculture and Agroforestry Practices; Strengthen Early Warning Systems; Strengthen Flood-Control Infrastructure; Use Integrated Pest Management	
Key Adaptation Options for Water Resources	Strengthen watershed protection through Reforest and restore wetland; Improve water treatment infrastructure: Improve water-use efficiency; Expand rainwater harvesting; Diversify water sources; Implement desalination technologies	
Key Adaptation Options for Energy Sector	Deploy decentralized microgrids with renewable energy; Diversify energy sources; Relocate critical infrastructure with high flood risks; Use flood-resistant pipeline materials; Build power stations and substations on elevated platforms on flood-prone areas; Implement advanced cooling systems,	
Key Adaptation Options for Heaalth Sector	Strengthen disease surveillance eg early-warning systems and response; Strengthen wastewater treatment; improve solid waste management; Increase investment in flood resilient infrastructure; Monitor water quality	
Key Adaptation Options for Transportation Sector	Regular dredging of rivers and channels; Use Climate-Resilient Construction Materials to Build roads and waterways; Improve Drainage Systems; Upgrade resilience of transport Infrastructure (roads, bridges); Enhance Weather Monitoring and Early Warning Systems; Develop Alternative Transport Strategies by Investing in hybrid transport systems	

Table 3.5: Summary of Climate Risk Assessment and Adaptation Options for the Tropical rainforest Zones

3.5.4.3 Guinea Savannah

The Guinea Savannah is Nigeria's most extensive agroecological zone, covering nearly half of the country's land area. It occupies the central region, often referred to as Nigeria's Middle Belt. The Guinea Savannah extends across Kwara, Kogi, Nasarawa, Niger, Plateau, Taraba, Kaduna, Osun, Oyo, Ekiti and Ogun States. The region has a unimodal rainfall pattern, with a wet season lasting 6 to 8 months and a dry season marked by lower humidity and higher temperatures. The Guinea Savannah is characterised by a mix of scattered trees and tall grasses. The soils are generally fertile, supporting diverse agricultural activities (Table 3,8). Thus, the Guinea Savannah is a vital agricultural hub in Nigeria, where crops such as maize, cassava, yam, and rice are cultivated. The region's climate and soil conditions also favour livestock rearing, making it a significant belt for both crop and animal production.

Climate Hazards	Key Vulnerabilities	Key Impact
Rain induced Flood, High temperature, Sea-level Rise	Environmental factors: Deforestation and Land Degradation: Rapid population growth, agricultural expansion, and unsustainable farming practice; Habitat destruction and overexploitation with loss of biodiversity Socio-economic factors: High poverty levels and limited access to resources; Heavy dependence on Agriculture by a large proportion of the population; lack of basic infrastructure such as drainage and robust transport networks and reliable access to energy; Limited Political Representation and weak governance	Erratic rainfall patterns have exacerbated food insecurity; Heat stress impairs the growth of staple crops; decline in livestock productivity with heat stress; devastating income and financial losses; Shift in climate pattern causing desertification; escalate farmers and herder clash; high temperature cause increased prevalence of agricultural pests and diseases; spread of waterborne diseases; increases waterlogging and soil erosion; Loss of biodiversity; water contamination due to overflow of sewage systems, runoff, and industrial pollutants
Key Adaptation Options for Agriculture	Promote drought- and heat-tolerant crop varieties; Improve Water management and Irrigation Systems eg drip irrigation; Adopt Climate-Smart Agriculture: Promote Soil Conservation Techniques against soil erosion; adopt Agroforestry and Sustainable Land Management; Promote Integrated Pest Management: Enhance Early Warning and Monitoring Systems	
Key Adaptation Options for Water Resources	Strengthen Wastewater and Sewage Management against floods; Improve Drainage and Flood Control Measure; enhance Water Quality Monitoring and early Warning Systems; Promote Safe Water Storage and Treatment	

Key Adaptation Options for Energy Sector	Develop Alternative Renewable Energy Source; Upgrade dam storage capacity and develop drought-resilient hydropower; Elevate and Flood-Proof Energy Infrastructure; Diversify Energy Sources for Resilience; Implement flood-resilient Energy Planning;
Key Adaptation Options for Heaalth Sector	Strengthen Healthcare Preparedness for Heat-Related Illnesses; Improve Drainage and Flood Management Systems against flood; enhance Access to Clean Water and Sanitation; Develop Early Warning and Emergency Response Systems; Encourage Nature-Based Solutions; Promote Safe Water Storage and Treatment strengthen Air Quality Monitoring over heat-induced pollution
Key Adaptation Options for Transportation Sector	Strengthening Climate-Resilient Transport Systems eg roads and bridges; Invest in improving drainage systems on roads to prevent floods; Establishment of Early Warning Systems for Transportation Disruptions; Create flexible Transport Networks with alternative routes for emergency

Table 3.6: Summary of Climate Risk Assessment and Adaptation Options for the Guinea Savannah Zones

3.5.4.4 Sudan Savannah

The Sudan Savannah of Nigeria is a flat, high-rainfall plain with lush green vegetation, neither a rainforest nor a semi-desert such as the Sahel Zone. It is a dry Savannah woodland where trees and grasses co-exist, making it well suited to animal and crop production. The climate is marked by a wet season from May to September, with annual rainfall ranging between 500 and 1000 mm. Sudan Savannah covers Adamawa, Yobe, Zamfara, Kebbi, Gombe, Katsina, Kano, Bauchi, and Jigawa States. The dry season extends from October to April, with high temperatures averaging between 25°C and 35°C. The region is mostly inhabited by peasant farmers without formal training in modern farming techniques. These farmers form a significant percentage of the farming population in Nigeria. The Sudan Savannah is the major source of grains (millet, wheat, maize, etc.) and livestock in Nigeria (Ugwu et al., 2023). Its distinct wet and dry seasons make it particularly vulnerable to the adverse effects of climate change, such as desertification, erratic rainfall patterns, and an increased frequency of extreme weather events (Table 3.7).

Climate Hazards	Key Vulnerabilities	Key Impact
Rain induced Flood, High temperature, Sea-level Rise	Environmental factors: acute water shortages due to declining rainfall; Deforestation and Land Degradation from overgrazing, unsustainable farming practices; Loss of Biodiversity due to destruction of natural habitats Socio-economic factors: Poverty and Food Insecurity; Limited Access to Education and Healthcare because of poor infrastructure; Conflict over Land by Farmers and herders due to frequently clashes; lack of irrigation, and dependence on rain-fed farming; Insurgency and Banditry	Frequent crop failures and livestock losses; Shifts in the crops cultivated by farmers; worsening scarcity of water and pastureland; reduced agricultural productivity and heightened food insecurity; more frequent droughts and floods will reduce water availability and quality; outbreaks of waterborne diseases such as cholera due lack of clean water; Increased food prices cause malnutrition; disrupt energy infrastructure due to flooding and heat wave;
Key Adaptation Options for Agriculture	Promote Climate Resilient Crop Varieties; Promote Crop Diversification; Introduced controlled Irrigation like Install drip or wick irrigation systems; Water Harvesting Techniques; Weather Forecasting and Early Warning Systems; Pasture Rehabilitation; Adoption of Improved Land Management Practices: Implement Integrated Pest Management Practices; Promote Agroforestry Systems:	
Key Adaptation Options for Water Resources	Develop Alternative Water Sources eg Invest in boreholes; Improve Water Storage Infrastructure against water scarcity; Upgrade Water Treatment Facilities against flooding; Enhance Integrated Water Resource Management (IWRM) to prevent conflicts; Support Sustainable Grazing and Farming Practices; Strengthen Early Warning and Monitoring Systems	
Key Adaptation Options for Energy Sector	Upgrading and Modernizing the Energy Grid to withstand extreme weather Event; Promote energy-efficient technologies; Expand Renewable Energy Systems; Install community-level energy storage systems	
Key Adaptation Options for Heaalth Sector	Strengthen healthcare infrastructure and improve disease surveillance; Training of Health Workers on Water-Borne Diseases: Educating local communities on effective water storage and safe water treatment options; Develop Mobile Healthcare Units;	
Key Adaptation Options for Transportation Sector	Improve Drainage Systems against flood; Elevate and Reinforce Roads and Bridges; Implement Early Warning Systems for Flood Risks; Promote Alternative Transport Modes eg rail; Adopt Soil Stabilization Techniques against erosion; Improve Rural Connectivity with alternative routes	

Table 3.7: Summary of Climate Risk Assessment and Adaptation Options for the Sudan Savannah Zone

3.5.4.5 Sahel

The Sahel Zone in Nigeria forms a transitional belt between the arid Sahara Desert and the humid Sudan Savannah. This agroecological zone is found mainly in the country's northernmost states: Sokoto, Zamfara, Yobe and Borno, with a significant presence in Katsina, Jigawa and Kebbi States. The Sahel agroecological zone is characterised by a semi-arid climate, sparse vegetation and highly variable rainfall, with an annual average of 500 mm (Nicholson, 2013) at its southern limits. The region's vegetation consists predominantly of drought-resistant species, including grasses, thorny shrubs, and scattered acacia and baobab trees (Table 3.8). Seasonal watercourses known as wadis provide temporary water resources during the brief rainy season but remain dry for most of the year. The landscape is largely composed of flat plains interspersed with sand dunes, reflecting the influence of wind erosion and desertification (Zhang et al., 2020).

Climate Hazards	Key Vulnerabilities	Key Impact
Rain induced Flood, High temperature, Sea-level Rise	Environmental factors: Land Degradation due to overgrazing, deforestation, and unsustainable farming practices; Water Scarcity due to declining water resources from rivers, lakes and ground water; Biodiversity Loss due to habitat destruction and climate stress Socio-economic factors: Poverty and low illiteracy; Insecurity and Conflicts; Agricultural production is subsistence-based and rainfed; Rapid population growth against limited resource; Limited access to healthcare services; Lack of Policy Implementation eg desertification	lower crop yields and Productivity because of desertification; drying up the water bodies; mass death of livestock; damage crops from excessive rainfall and flooding; forced migration of residents; desert encroachment; contamination and the spread of waterborne diseases, Higher incidence of vector-borne diseases like malaria and meningitis;forced migration of Populations
Key Adaptation Options for Agriculture	Implement Large-Scale Reforestation Programs against desertification; Promote Agroforestry and Sustainable Farming Practices; Expand Irrigation Infrastructure; Promote drought-Resilient Crop Varieties; Develop Early Warning Systems; Improve Rangeland Management against heat waves; Invest in Drought-Resistant Animal Breeds; Promote Alternative non-farming Livelihoods; Encourage Integrated Land Use Planning	
Key Adaptation Options for Water Resources	Promote efficient irrigation techniques such as drip irrigation; Expand Water Recycling and Reuse; Strengthen Groundwater Management Plans; Upgrade Water Infrastructure Resilience; Enhance Water Quality Monitoring; Restore Wetlands and Natural Buffer; Expand Access to Alternative Water Supplies;	

Key Adaptation Options for Energy Sector	Increase investment in off-grid solar, wind, and hybrid systems; Improve Energy Efficiency Measures; Improve Energy Storage Solutions; Implement Water Resource Management Plans; Develop heat-resistant photovoltaic materials and cooling mechanisms.
Key Adaptation Options for Health Sector	Enhance Healthcare Capacity for Heat Emergencies; Expand Access to Clean Drinking Water; Promote Hygiene and Sanitation Campaigns; Support Climate-Resilient Agriculture to address poor yield; Enhance Disease Surveillance and Early Warning System; Develop Flood-Resilient Healthcare Facilities
Key Adaptation Options for Transportation Sector	Invest in Climate-Resilient transport Infrastructure; Improve Drainage Systems; Develop Alternative Transport Routes; Support Local Transport Cooperatives to develop and maintain local transport systems

Table 3.8: Summary of Climate Risk Assessment and Adaptation Options for the Sahel Zone



Chapter Four

Adaptation Priorities

The Government of Nigeria recognises that achieving sustainable development in the country depends on prioritising and effectively addressing poverty, inequalities, insecurity, and climate change. In identifying adaptation priorities for Nigeria's critical sectors, due regard was given to current levels of risk and vulnerability, findings from the economic appraisal, and stakeholders' perceptions, which are crucial to achieving a shared understanding of the context. An economic appraisal conducted under the NAP process provides a rigorous economic basis for prioritisation and investment in climate change adaptation.

To ensure the effectiveness of NAPs, adaptation strategies were differentiated into short-term measures that address already-exposed climate vulnerabilities and long-term strategies determined through scientific modelling that monitors future uncertainties and vulnerabilities. The uncertainties associated with climate change are better addressed through an adaptation framework that enables estimation of the cost implications and potential benefits of implementing the identified adaptation measures. The Economic Appraisal for NAP Nigeria (FME, 2025) further streamlined the prioritisation undertaken under the National Climate Change Policy and Response Strategy (NCCP-RS) from eight to three sectors and additionally evaluated the cost-benefit analysis of the prioritised adaptation strategies across these sectors (Table 4.1).

	Sector	Option	Strategy
1	Floods management (Agriculture)	Floods	i. Flood management and control systems (flood resistant animal ranching (prevention), community radios for communications for flood management ii. Post-flood livestock relocation and recovery (proposal of flood resistant animal ranching (post-flood)

2	Heat impacts management (Agriculture)	Heat stress/ Heatwave	i. Comprehensive Livestock Cooling and Climate management coming from the proposal of shed and water reservoirs for animal husbandry) ii. Comprehensive Livestock Cooling and Climate management coming from the proposal of shed and water reservoirs for animal husbandry) iii. Irrigation Expansion (for crops production) iv. Climate-resilient crops (heat resistant crops) v. Comprehensive fish and aquaculture climate adaptation (coming from the proposal of heat action for fishery sector)
3	Cross-cutting (Agriculture)	Multi-risks	i. Smart agriculture production management ii. Agriculture insurance
4	Water resources management	Drought /flood adaptation strategies	i. Water Supply Augmentation (Rainwater harvesting, Groundwater development, Water treatment and recycling) ii. Flood Protection & Demand Management (Flood-resistant water infrastructure, Early warning systems, Emergency response protocols)
5	Energy	Multi-risks	i. Renewable Energy Expansion (solar, wind) ii. Grid Hardening and Modernization iii. Energy Storage & Backup Syste

Table 4.1: Prioritised Adaptation Strategy and Option for Agriculture, Water resources and Energy. (Source: CRA 2025)

It is noteworthy that almost all the current national development plans and climate change-related policies identify Agriculture, Water Resources and Energy as high-priority sectors because of their macroeconomic contributions, their importance

to Nigeria's development agenda, and their high vulnerability to climate change.

i. Agriculture: About 10 to 25% decline in agricultural yield has been projected by 2080 because of climate crisis under Business-as-Usual scenario (FMEnv, 2021b).

ii. Water Resources: Nigeria is already experiencing rainfall variability across all the 4 major ecological zones. The southern states are having increasing rainfall intensity with attendant flooding, but the northern states are facing a decline in precipitation, causing more drought and decreasing surface water. Current reports show that Lake Chad has lost over 93 percent of its 45,000 km^2 area as at 1960 which is largely attributed to climate change.

iii. Energy: Nigeria's power sector is fragile and climate-sensitive, with grid collapses linked to both floods and extreme heat. Floods inflicted ₦329 million in direct damage to power assets in 2012, while heatwaves caused two national grid collapses in one week as surging demand overloaded infrastructure in 2024 (CBA, 2025). Energy infrastructure remains concentrated in high-risk flood zones, further elevating baseline losses. Energy access is poor, with over 86 million people lacking electricity, and the situation is worse in rural areas, where only 26% have access.

The cost-benefit analysis (CBA) of the main adaptation options for the agriculture, water resources, and energy sectors, based on a 35-year evaluation period and a 10% social discount rate, used net present value (NPV), benefit-cost ratio (BCR), and internal rate of return (IRR) as key indicators and yielded highly favourable economic outcomes (CBA, 2025). At the aggregate level, the portfolio of adaptation options generates highly positive results. By 2050, the total capital costs amount to ₦18.3 trillion (USD 12.4 billion), increasing to ₦26.5 trillion (USD 18.0 billion) by 2080. These are substantial investments, yet the present value of benefits far exceeds the costs, with BCRs of 2.32 in 2050 and 2.41 in 2080, and Net Present Values (NPVs) of ₦15.3-16.6 trillion (USD 10.4-11.3 billion). This means that for every ₦1 (or dollar) invested in adaptation across all sectors, Nigeria gains at least ₦2.3 (or dollars) in avoided damages, productivity gains, and resilience co-benefits. This exceeds the standard threshold of 1, clearly confirming the economic viability of the combined climate adaptation portfolio.

Value in Naira	Floods management Agricultural Adaptation Strategy		Heat impacts management Agricultural Adaptation Strategy		Cross-cutting Agriculture Adaptation Strategy		Water supply augmentation		Flood Protection Systems		Energy Sector Adaptation Strategy	
Appraisal period (years)	2050	2080	2050	2080	2050	2080	2050	2080	2050	2080	2050	2080
Capital Costs	123,00	213,00	1 637,00	2 630,00	248,00	458,00	1 085,00	1 685,00	120,75	172,50	15 075,00	21 300,00
Present Value of Benefits	273,55	289,22	2 200,81	2 326,94	683,87	723,06	2 036,06	2 152,75	426,49	450,93	21 249,67	22 467,46
Present Value of Costs	64,05	66,18	933,50	957,09	117,84	122,82	641,75	656,00	77,37	78,60	9 752,63	9 900,49
Benefit Cost Ratio	4,27	4,37	2,36	2,43	5,80	5,89	3,17	3,28	5,51	5,74	2,18	2,27
Net Present Value	209,50	223,04	1 267,31	1 369,85	566,03	600,24	1 394,31	1 496,74	349,12	372,33	11 497,05	12 566,98
Value in USD (1 472.1475 per USD1)												
Appraisal period (years)	2050	2080	2050	2080	2050	2080	2050	2080	2050	2080	2050	2080
Capital Costs	83,55	144,69	1 111,98	1 786,51	168,46	311,11	737,02	1 144,59	82,02	117,18	10 240,14	14 468,66
Present Value of Benefits	185,82	196,46	1 494,97	1 580,64	464,54	491,16	1 383,06	1 462,32	289,70	306,31	14 434,47	15 261,69
Present Value of Costs	43,51	44,96	634,11	650,13	80,04	83,43	435,93	445,61	52,56	53,39	6 624,76	6 725,20
Benefit Cost Ratio	4,27	4,37	2,36	2,43	5,80	5,89	3,17	3,28	5,51	5,74	2,18	2,27
Net Present Value	142,31	151,51	860,86	930,51	384,50	407,73	947,13	1 016,71	237,15	252,91	7 809,71	8 536,49

Table 4.2: Results of the CBA of all prioritised adaptation strategies (Source: CBA, 2025)

When ranked by relative performance and attractiveness using BCR, Cross-cutting agriculture measures (BCR 5.80-5.89) and Flood Protection Systems (5.51-5.74) stand out as the most attractive options, delivering nearly six dollars in benefits for every dollar invested. They are "quick win" measures that can deliver resilience dividends at low cost. When ranked by absolute net benefits (NPV), Energy sector adaptation (USD 7.8-8.5 billion) stood out with a higher magnitude than all other strategies combined, followed by Water Supply Augmentation (USD 947 million-1.0 billion) and Heat Impacts Management in Agriculture (USD 861-931 million). This pattern reflects Nigeria's acute vulnerability to costly power outages and the very high economic value of avoided disruptions in the energy sector. Others are Cross-cutting agriculture (USD 385-408 million) and Flood Protection Systems (USD 237-253 million).

4.1 Identified Sectoral Needs to Support NAP Process

Government of Nigeria understands that climate change affects every sector of the economy, however the prioritization of the three sectors considered the level of vulnerability, national coverage, possible finance investment required to address them on both medium and long term. Under the NAP process sectoral gaps and needs assessment were undertaken through a robust consultative process which recognizes the elements that limit national ability to attain effective adaptive capacity and climate resilience.

4.2 Specific Sectoral Adaptation Strategies

4.2.1 Agriculture

Sectoral strategies:

- Floods management Agricultural Adaptation Strategy

- Heat impacts management Agricultural Adaptation Strategy
- Cross-cutting Agriculture Adaptation Strategy

Strategic Sectoral Need	Recommended Measures	Benefits of the Action	Benefit-Cost Ratio	Net Present Value
Floods management Agricultural Adaptation Strategy	Flood management and control systems using Physical infrastructure (dykes, drainage canals, retention basins); Resilient pastoral management practices (raised ranching); Community-based warning and information systems (local radio stations, contingency plans). Post-flood relocation and recovery using Rapid evacuation logistics mechanisms; Emergency funds for the rehabilitation of pastoral infrastructure; Recovery kits for affected households	Reduced animal and plant losses during floods; Protection of agricultural infrastructure; Reduced emergency expenses (food, rehousing); Improved resilience of rural communities to hydrometeorological shocks; Maintained economic activity in vulnerable areas.	4.27 in 2050 and 4.37 in 2080	USD 142.3 million in 2050 and USD 151.5 million in 2080
Heat impacts management Agricultural Adaptation Strategy	Smart irrigation systems including Improved water-use efficiency for crops; Solar-powered pumps and precision irrigation; Expansion of irrigated areas in heat-prone zones. Climate-resilient crops by introducing drought- and heat-tolerant varieties Investment in breeding and seed dissemination programs Technical support to farmers for adoption Post-harvest cooling and storage Climate-resilient fisheries using improved aquaculture management Deployment of solar cooling systems; Investment in fish feed and energy-efficient infrastructure	Increased crop yields due to irrigation and climate-resilient seeds Water use efficiency increase Reduced losses from drought and post-harvest spoilage Improved revenues for farmers and fishers through higher-quality production, Strengthened food systems and year-round supply Creation of jobs in irrigation, cooling services, and aquaculture.	From 2.36 in 2050 and 2.43 in 2080	USD 860.9 million in 2050 to USD 930.5 million in 2080

Cross-cutting Agriculture Adaptation Strategy	Smart agriculture production management (Agro-ecosystem services enhancement) Sustainable land management Improved seeds and farm inputs Agricultural insurance	Enhanced ecosystem services (soil fertility, water regulation, carbon sequestration) More social co-benefits (migration reduction, empowerment, stronger safety nets) Protection of farmers' incomes and productive capital through subsidized insurance	From 5.8 in 2050 to 6.1 in 2080	USD 384.5 million in 2050 and USD 407.7 million in 2080
	Reduce pre- and post-harvest food losses by 50% Develop and regularly update a risk and vulnerability atlas	Increased income to farmers Improved quality of farm products		
	Programme to promote digital agriculture including deployment of technologies and innovation such as AI, yield mapping and remote sensing; Early warning system; Drip Irrigation and meteorological forecasts to lower impact of climate extremes Irrigation; mini-dams; Promote mechanisms for weather and climate information sharing to farmers Enhance the adoption of improved livestock production systems Develop and implement a comprehensive National Livestock Information and Management System (NLIMS) to track, analyse and report on the livestock industry Increase access to adaptation finance, especially to empower women and youth, through economic incentives	Increased yield for farmers Increased income Better access to information		

Capacity Building	Programme for capacity strengthening of Institutions to plan for CSA, regenerative agriculture and e-agriculture to be formulated and implemented including practical demonstration of more resilient crop and livestock practices (including demonstration plots and other mechanisms) Promote knowledge transfer of research output farmers and other value-chain actors through an extension service system Undertake an Impact, Risk, and Vulnerability Assessment for the Food and Agriculture sector by 2030 Support genetic Improvement Program for Crops, Fish and Livestock to enhance agricultural sector productivity through breeding and upgrading of indigenous species	Enhances skills to increase agricultural yield and income Reduce negative impact of agriculture on environment Improve data gathering and reduce losses from extreme climate events		

Table 4.3: Activities in the Prioritized Area and the Benefits for promoting resilience in Agriculture sector

4.2.2 Water Resources

Sectoral strategies

Water Supply augmentation

- Flood Protection and Demand Management

Strategic Sectoral Need	Recommended Measures	Benefits of the Action	Benefit-Cost Ratio	Net Present Value
Water Resource Management Adaptation Strategy - Water supply augmentation	Increase water availability for households and communities through borehole development; solar-powered pumping and water storage systems Water treatment units and distribution networks Programmes to enhance resilience to prolonged droughts and seasonal water shortages. Programmes to improve the resilience of water supply infrastructure to extend and improve water supply infrastructure in urban and rural areas	Expanded access to water Time savings and reduced burden on women and children Increased public health through improved hygiene and sanitation; Stabilized agricultural and small-scale industrial productivity; Strengthened drought resilience and reduced migration pressures. Reduce time spent collecting water, especially for women and children	from 3.17 in 2050 to 3.28 in 2080	USD 947 million in 2050 and USD 1.02 billion in 2080

Water Resource Management Adaptation Strategy - Flood Protection and Demand Management	Programmes to reinforce water-related infrastructure to withstand floods (e.g., elevated pumping stations, protected treatment plants, reinforced pipelines) Establish early warning systems for flood anticipation and rapid response Erosion control systems and catchment protection to maintain water quality Promote contingency protocols for emergency water allocation, ensuring equitable and efficient distribution during crisis periods.	Avoided damage to critical infrastructure (pumping stations, reservoirs, pipelines) Reserved water quality due to reduced sedimentation and contamination during floods Reduced erosion losses in key watershed areas, leading to lower treatment costs Avoided land-use losses in flood-prone zones, preserving access to land and reducing resettlement needs.	5.51 in 2050 and 5.74 in 2080	USD 237.2 million in 2050 and increasing to USD 252.9 million in 2080,
	Integrated of water resources management (IWRM) approach for sustainable development into river basin development Protect and restore degraded watershed and wetlands to protect water resources and related ecosystems	Increase water access Prevent degradation of watersheds Provide water for irrigation		
	Water infrastructure improvement (dams, boreholes, surface water, irrigation schemes, and distribution Network) Development of a climate-smart hydropower dam to reduce the use of fossil fuels	Provide more water for irrigation Increase food production Reduce damage to energy infrastructure Increase reliability of power supply Reduce degradation of forests		
	Safe water (treatment, reduced contamination) Improve sanitation and hygiene facilities (facilities, education, treatment) with elevated structures in flood-prone areas	Improve portable water access; Reduce spread of communicable diseases; Reduce cost of health services		
	Programme to Improve the availability of water through the deployment of renewable energy source (solar), in various intervention schemes Building climate-resilient water infrastructure in the provision of potable water across the nation Programme to promote aquifer recharge, groundwater protection, and watershed management of boreholes and water supply schemes	Sustained water supply Improved sanitation Improved agricultural yield More time for women and children to engage in meaningful activities Reduce vulnerability of the people		

Table 4.4: Activities in Prioritized Area and the Benefits for promoting resilience in Water resources sector

4.2.3 Energy

Sectoral strategies

- Grid Hardening and Modernisation that include increased protective margins in the construction and placement of all new infrastructure (i.e. higher standards and specifications)
- Expand sustainable energy sources through Renewable Energy Expansion (solar and wind) and decentralise transmission to reduce the vulnerability of energy infrastructure to climate impacts
- Strengthen existing infrastructure using Energy Storage & Backup System, in part through early efforts to identify and implement all possible 'no regrets' actions
- Undertake risk assessment and risk reduction measures to increase the resilience of the energy sectors

Strategic Sectoral Need	Recommended Measures	Benefits of the Action
Energy Sector Adaptation Strategy	Wide deployment of solar power to replace fossil fuel-based generation and reduce vulnerability to supply shocks Promotion of wind power generation to diversify renewable sources and ensure continuity in dry seasons Grid hardening through upgraded infrastructure that is more resistant to climate-induced outages (heatwaves, storms) Establishment of energy storage systems to balance supply and demand fluctuations, especially during periods of extreme heat or drought.	Avoided outage costs as Grid modernization and storage reduce costly power interruptions; Renewable energy expansion which leads to significant fuel savings and enhanced energy security Job creation as renewable energy investments create hundreds of thousands of jobs and providing stable incomes in both rural and urban areas GHG emission reductions as: renewable energy, storage, and grid efficiency improvements prevent millions of tonnes of CO^2 emissions annually, delivering climate co-benefits aligned with Nigeria's NDC . Improve System resilience
Renewable Energy	Energy transitioning; opportunities affordable electricity for households and communities. Promote renewable energies, establishment of woodlots and activities related to green economy; Advance business solutions in renewable energy and clean cooking	Improve livelihoods Increase mitigation capacity; Reduce vulnerability; Improve well-being; Provide employment; Reduce pollution; Create jobs

	Promote programmes for the use of biofuels; and other renewable energy sources;	Improve livelihoods; Increase mitigation capacity Reduce vulnerability; Improve well-being; Provide employment; Reduce pollution; Create jobs
	Clean cooking Promote renewable energy e.g. solar; Digital transformation; Deepen rural electrification; Investment in infrastructure; Promote wide adoption of renewables for agricultural processing	Improve livelihoods; Increase mitigation capacity; Reduce vulnerability; Improve well-being; Provide employment; Reduce pollution of the environment

Table 4.5: Activities in the Prioritised Area and the Benefits for Promoting Resilience in the Energy Sector

4.2.4 Forestry

Sectoral strategies

- Implement a national Community-Based Natural Resources Management Programme.
- Support the implementation and review of the National Forest Policy
- Promote the adoption of nature-based solutions to support Forest and wetland conservation
- Support Regenerative Forestry and natural resources management
- Promote the private sector to help establish and restore community and private native forests, plantations and nurseries
- Improve the management of Forestry Reserves and control tree harvesting

Prioritised Area	Activities	Benefits
Strengthen Policy	Review and implement the National Forest Policy and related regulations at both federal and state levels	Promote sustainability; Improved coordination
Community based -Forest restoration	Restore degraded forest ecosystems Strengthen national Community-Based Natural Resources Management programme, consisting of sustainable forest management plans (SFMPs) and sustainable forest management practices and that recognize climate change Undertake an impact, risk, and vulnerability assessment on ecosystems Develop and regularly update a risk and vulnerability atlas	Reduce forest degradation Improved ecosystem services More benefit to host communities Increase participation and inclusiveness;
Nature based Restoration	Accelerate ecosystem-based adaptation using nature-based solution	Improved livelihood; generate employment; reduce vulnerability; Improved biodiversity protection Less forest degradation Improved co-benefits

Table 4.6 Activities in the Prioritised Area and the Benefits for promoting resilience in the Forestry sector

4.2.5 Health and Sanitation

Sectoral strategies

- Improve coordination and responsiveness of policies in the sector.
- Promote readiness of the health sector through disease surveillance.
- Promote sustainability through monitoring and capacity building of services and human resources.

Prioritised Area	Activities	Benefits
Strengthen Policy	Complete a national update of the climate and health vulnerability and adaptation assessment	Promote sustainability of the plan; Coordination of the sector improved
Improved Readiness of the Health sector	Develop a national integrated climate, health and environmental early warning system Promote training, education and awareness of health workforce and communities	Reduce incidence of disease outbreak Improved capacity in the sector Better readiness to deal with disease outbreaks Capacity of health workers is improved
Enhance sustainability of Services	Develop and agree on performance and accountability measures to track progress on delivery of climate adaptation and mitigation actions for climate and health on climate, its impact on health, and key protective actions Improve climate resilience and sustainability of health services across primary, secondary and tertiary care	Improved achievement of goals Increased output in the health sector Reduced vulnerability in the sector Funds are freed to tackle other challenges

Table 4.7: Activities in Prioritised Area and the Benefits for Promoting Resilience in the Health and Sanitation Sector

4.2.6 Transport

Sectoral strategies

- Improve coordination and make policies climate sensitive
- Increase the readiness of the transport Sector to adapt to climate change
- Include increased protective margins in the construction and placement of all new infrastructure (i.e. higher standards and specifications)
- Undertake risk assessment and risk-reduction measures to increase the resilience of the transport sector
- Strengthen existing infrastructure, in part through early efforts to identify and implement all possible 'no regrets' actions

Prioritised Area	Activities	Benefits
Strengthen Policy	Review of the policies in the transport sector to mainstream adaptation measures Strengthen the regulatory environment, the security architecture of Nigerian waterways, and the capacities of key government agencies to become enablers of blue economic growth	Improve coordination in the sector; Improve participation and decision-making process Improved enforcement of transport regulations
Increase readiness	Undertake an Impact, Risk, and Vulnerability Assessment of the infrastructure sector by programmes to boost implementation of the National Integrated Master Plan on Transport Infrastructure Develop and regularly update a risk and vulnerability atlas Transportation infrastructure: With respect to transportation infrastructure, develop and implement risk assessment and response plans that include infrastructure mapping and identification of high-risk infrastructure, and investment in reinforcing and/or relocating existing high-risk infrastructure for insurance purposes to undertake independent risk mapping and risk reviews to assess and provide recommendations to federal bodies in charge of transportations.	Reduced collapse of infrastructure in the sector Promote sustainability of the plan; Enhance participation; create employment; reduce vulnerability Improve the response capacity to emergencies Reduction in the incidence of infrastructure failure
Climate Proof Infrastructure	Identify and climate-proof existing vulnerable transport infrastructures Integrate climate change resilience in new infrastructures Digital transformation to promote business; promoting global market competitiveness; waste reduction; Activities related Green economy	Improve livelihoods; Create employment; Reduce vulnerability; Promote competitiveness and increase access to market
	Develop and execute measures to improve the resilience of the inland waterways' infrastructure	Improve livelihoods; Create employment; reduce vulnerability; Promote competitiveness and increase access to market

Table 4.8: Activities in Prioritised Area and the Benefits for Promoting Resilience in the Transport sector

4.2.7 Commerce and Industry

Strategies

- Protect the Nigerian industrial & commercial sectors from the adverse effects of climate change.
- Explore new markets and opportunities arising from climate change.

Prioritised Area	Activities	Benefits
Strengthen Policy	Integrate priority actions into policies	Promote sustainability of the plan; Improve equality and inclusiveness
	Risk assessments and mapping: to undertake risk assessments, including mapping and identifying high-risk areas for industry. These risk assessments and maps should be publicly available Incentives for re-location of industrial parks: should provide incentives to establish and promote industrial parks in less environmentally sensitive areas for commercial refugees. Promote awareness and capacity building programmes on climate change impacts and adaptation such as targeted training sessions for industry leaders and insurance companies on the impacts and opportunities of climate change incorporate climate change into annual and long term business plans, with particular emphasis on risk reduction, but additionally with a focus on new adaptation-related business opportunities (e.g. adapted agricultural inputs, increased demand for beverages and cooling equipment) Relocation: Industries should plan for relocation of facilities located in high risk, high impact areas	Promote sustainability of the plan; enhance participation; create employment; reduce vulnerability; promote competitiveness and and increase access to market
	Digital transformation to promote business; Promoting global market competitiveness; Waste reduction Activities related to Green economy	Improve livelihoods; Create employment; Reduce vulnerability; Promote competitiveness and increase access to market
	Investing and improving in telecommunications and electricity; Promote green economy in the sector, Practices that would reduce carbon emissions from industrial and commercial processes	Improve livelihoods; Create employment; Reduce vulnerability; Promote competitiveness and increase access to market

Table 4.9: Activities in the Priority Area and the Benefits for promoting resilience in the Commerce and Industry Sector

4.2.8 Livelihoods

Strategies

- To reduce the adverse effects of climate change on poverty eradication and livelihoods, in particular by promoting the use of adaptive social protection measures for all.

Prioritized Area	Activities	Benefits
Equitable Empowerment	Ensure equitable access to basic services, ownership and control over land and adequate, affordable, and sustainable housing, access to natural resources and appropriate new technologies, and financial services to preserve livelihoods	Increase the wellbeing of citizens Provide sustainable income Reduce vulnerability of the people
Safety net and capacity building	Incentives for livelihood activities: Access to finance, credit, and market linkages to ensure improved livelihoods and economic empowerment of rural communities, to promote empowerment of women, youth, Indigenous Peoples, and vulnerable communities to undertake economic activities Create a Disaster Risk Management Plan for frontline communities across the different ecological zones to enhance effective risk reduction responses and recovery Integrate a national insurance scheme with the country's broader climate adaptation strategy to protect livelihoods Livelihoods training and skills development: National Directorate of Employment (NDE) should expand and intensify training and skills development for people in high-risk areas, including in particular skills related to climate change adaptation (such as construction of water and grain storage and wind proofing buildings).	Provide sustainable income Reduce vulnerability of the people Create job opportunities

Table 4.10: Activities in the Priority Area and the Benefits for promoting resilience in the Livelihood sector

4.2.9 Cultural Heritage

Strategies

- Protecting cultural heritage from the impacts of climate-related risks by developing adaptive strategies to preserve cultural practices and heritage sites and by designing climate-resilient infrastructure, guided by traditional, Indigenous Peoples' and local knowledge systems.

Prioritised Area	Activities	Benefits
Conservation of Cultural heritage	Inventory and document indigenous knowledge and cultural practices relevant to building resilience of heritage sites to climate change	Increase knowledge of national heritage Protect cultural materials
Improving the readiness in the Cultural Heritage sector	Develop a master plan to preserve Indigenous Peoples' and local knowledge systems, and protect and conserve cultural practices and heritage sites	Reduce the vulnerability of cultural heritage to climate hazards Halting illegal export of cultural materials
Enhancing sustainability	Promote the adoption of indigenous culture and knowledge to build resilience to climate change	Creating job opportunities in the sector Improve living conditions in rural areas

Table 4.11 Activities in the Priority Area and the Benefits for Promoting Resilience in the Cultural Heritage sector

4.3 Nature-Based Solutions

Nigeria is adopting nature-based solutions (NbS) to address current and anticipated impacts of climate change and to realise low-carbon development. The approach was highlighted in both the NDC 2021 and the Climate Change Act. Nigeria is endowed with a wide diversity of ecosystems – from rainforests through dry savannahs to the dry Sahel, flat coastal zones, high-altitude plateaus, montane forests, and useful freshwater wetlands, all under various climatic conditions. Nigeria will use these natural ecosystems to significantly reduce the impact of extreme climate events such as floods, catastrophic storms, and severe droughts, and to offset vast amounts of the world's CO2 emissions (Nzegbule and Udoka, 2024). By implementing NbS actions, the country will address adaptation challenges with multiple additional benefits across sectors at relatively low cost, and, according to the estimate of Griscom et al. (2017), has the potential to meet 20% of its mitigation needs by 2050. Successful implementation of NbS actions will require eliminating policy barriers, such as improper agenda-setting and poor

strategic leadership, and improving coordination and long-term monitoring to tackle challenges. Institutions such as the NCCC, River Basin Development Authorities, the Great Green Wall Sahara Commission, and the National Parks Service will be strengthened to implement long-term adaptation solutions using NbS. Nigeria will improve the capacity of institutions to implement NbS actions.

4.4 Loss and Damage

4.4.1 Agrifood Systems and Related Livelihoods

4.4.1.1 Rationale and Scope

This subsection sets out the rationale for integrating Loss and Damage (L&D) into Nigeria's National Adaptation Plan (NAP). It places L&D within the continuum of climate action, alongside mitigation and adaptation, and clarifies its specific relevance in the Nigerian context. The subsection highlights the cross-sectoral nature of L&D, using agrifood systems and related livelihoods as the primary analytical lens.

Nigeria is already experiencing significant loss and damage associated with climate variability and climate-related extreme events, affecting natural systems, livelihoods, infrastructure, food systems and human well-being. Recent large-scale floods, recurrent droughts and rising temperature extremes have demonstrated the scale and frequency of climate-related disruptions across the country, particularly in agriculture-dependent regions (Federal Government of Nigeria, 2021a; Federal Government of Nigeria, 2025). Climate change is increasing the frequency and intensity of floods, droughts, heat stress and coastal erosion, resulting in escalating risks across key sectors of the economy, particularly agrifood systems.

Loss and damage refer to the adverse impacts of climate change that persist despite mitigation and adaptation efforts, including impacts that exceed existing adaptive capacities or where adaptation limits have been reached. These impacts include both economic losses, such as damage to crops, infrastructure and productive assets, and non-economic losses, including impacts on health, ecosystems, cultural heritage, social cohesion and human mobility. This definition aligns with the United Nations Framework Convention on Climate Change (UNFCCC), which recognises loss and damage as encompassing both sudden-onset events and slow-onset processes, particularly in vulnerable developing countries (UNEP & UNEP Copenhagen Climate Centre, 2026).

Agrifood systems are a critical entry point for addressing loss and damage in Nigeria, given their central role in livelihoods, food security and economic stability. Agriculture remains highly climate-sensitive and is characterised by strong dependence on rain-fed production systems, low productivity, weak infrastructure

and limited adaptive capacity (Federal Government of Nigeria, 2022; Federal Government of Nigeria, 2024). These structural constraints heighten exposure and sensitivity to climate hazards and amplify the scale and persistence of losses and damage. Given that a large proportion of Nigeria's rural population depends directly or indirectly on agrifood systems, climate-induced losses in this sector have significant multiplier effects on household welfare, the national food supply and broader economic stability.

Across Nigeria, communities have long relied on indigenous knowledge, traditional resource management practices, reciprocal social support networks and other locally developed coping mechanisms to manage climate variability and environmental change. However, the increasing frequency, intensity and compounding nature of climate-related hazards are progressively exceeding the capacity of many of these informal systems. Strengthening and scaling up effective locally led approaches through supportive policies, institutions and investment, therefore, represents an important opportunity to enhance national responses to climate-related Loss and Damage.

Consistent with emerging international guidance on strengthening national responses to loss and damage, this section adopts a comprehensive climate risk management approach that integrates risk assessment, risk reduction, anticipatory action, preparedness, response, recovery and rehabilitation (UNEP & UNEP Copenhagen Climate Centre, 2026). It also reflects the growing international emphasis on strengthening national systems to assess, address and finance loss and damage, including through enhanced access to global support mechanisms and technical assistance. The scope of this section covers both economic and non-economic losses and damages affecting crops, livestock, fisheries, forestry and associated value chains, arising from extreme weather events and slow-onset processes. It considers impacts at national and subnational levels, with particular attention to vulnerable populations, including smallholder farmers, pastoralists, fisherfolk, women, youth, older persons and persons with disabilities (Federal Government of Nigeria, 2021b). The section also highlights linkages with disaster risk management, food security and rural development frameworks to ensure policy coherence, while positioning loss and damage as a distinct but complementary pillar within Nigeria's broader climate resilience agenda.

4.4.2 Climate-Related Losses & Damages

This subsection synthesises the main climate hazards identified in the NAP, including both extreme weather events and slow-onset processes, and analyses how they translate into losses and damages across agrifood systems in Nigeria. It covers impacts on crops, livestock, fisheries, forestry, and broader value chains. While the current NAP already provides substantial evidence on hazards and vulnerabilities, this subsection reframes that information explicitly through a loss



A road connecting the two cities of Blantyre and Lilongwe in Malawi has been torn apart by Cyclone Freddy . Photo: Thoko Chikondi/AP (www.theguardian.com/environment/2023/mar/17/weather-tracker-cyclone-freddy-madagascar-mozambique-malawi)

and damage lens and incorporates updated country-specific insights to ensure continued relevance.

Climate-related hazards are already generating substantial and recurring losses across Nigeria's agrifood systems. Flooding events have caused widespread destruction of farmlands, rural infrastructure and productive assets, while recurrent droughts and rising temperatures are reducing crop productivity, degrading rangelands and affecting livestock systems (Federal Government of Nigeria, 2025; Federal Government of Nigeria, 2021a). Coastal erosion and sea-level rise are also increasingly affecting fisheries, coastal settlements and associated livelihoods, particularly in vulnerable coastal areas (Federal Government of Nigeria, 2025). For example, recurrent flooding in the Niger–Benue floodplains have resulted in widespread destruction of staple crops, displacement of farming communities and disruption of agricultural value chains and food supply systems. The 2022 floods further highlighted the systemic nature of climate-related losses across Nigeria's agrifood systems.

Box 1: 2022 Floods and Systemic Losses Across Nigeria's Agrifood Systems

The 2022 floods were among Nigeria's most severe climate-related disasters in recent decades, affecting over 4.4 million people across 34 States and causing widespread destruction of livelihoods and productive assets. Flooding damaged farmlands, irrigation systems, storage facilities, roads and market infrastructure, while also resulting in major crop and livestock losses in key agricultural areas. Staple crops such as rice, maize and sorghum were significantly affected, contributing to reduced food availability and rising food prices. Beyond direct economic losses, the floods disrupted agrifood value chains, displaced farming communities and increased food insecurity and humanitarian needs, particularly in already vulnerable regions. The event highlights how extreme climate hazards can generate cascading and systemic losses across interconnected agrifood systems, reinforcing the need for strengthened anticipatory action, resilient infrastructure and recovery systems in Nigeria (UN News, 2025; FAO Nigeria, 2025).

These impacts interact with pre-existing structural vulnerabilities in agrifood systems. Agriculture in Nigeria is predominantly rain-fed and heavily reliant on smallholder production systems, characterised by low technology adoption, weak extension services, limited irrigation coverage, inadequate storage and processing infrastructure, and constrained access to finance and markets (Federal Government of Nigeria, 2022; Federal Government of Nigeria, 2024). As a result, climate shocks frequently trigger cascading effects across production systems, value chains, household incomes, and food security systems.

High post-harvest losses, market disruptions and weak rural infrastructure further compound climate-related losses and reduce the capacity of communities and institutions to recover from shocks (Federal Government of Nigeria, 2024). Climate risks are increasingly interacting with insecurity, displacement and resource competition, particularly in fragile and conflict-affected regions, thereby contributing to compounded and systemic risks that affect livelihoods and rural economies (Federal Government of Nigeria, 2021a; Federal Government of Nigeria, 2022).

Climate-related impacts are therefore not isolated events but interconnected, cumulative risks that affect economic systems, livelihoods, ecosystems and social stability over time. In line with emerging international approaches to addressing loss and damage, there is growing recognition of the need to strengthen integrated, forward-looking responses that address both immediate climate impacts and the underlying vulnerabilities that amplify losses and damage (UNEP & UNEP Copenhagen Climate Centre, 2026). Overall, the evidence indicates that climate-related losses and damage in Nigeria's agrifood systems are not isolated or episodic but increasingly systemic, recurrent and compounding. This highlights the need for a structured national approach to assessing, monitoring and addressing loss and damage, including strengthening early warning and anticipatory systems, improving loss and damage data and analytics, and enhancing access to climate risk financing and recovery mechanisms.

Understanding how these climate hazards translate into both economic and non-economic losses provides the basis for identifying the sectors, livelihood systems and population groups that are most vulnerable to Loss and Damage.

4.4.3 Economic and Non-Economic Losses

This subsection categorises and describes both economic and non-economic losses and damages in agrifood systems, with particular emphasis on food security and livelihoods. Economic losses include reduced productivity, income losses, and damage to agricultural assets and infrastructure. Non-economic losses include impacts on health, cultural heritage, traditional knowledge, biodiversity,

ecosystem services, and social cohesion. As this dimension is largely absent from the current NAP draft, the subsection requires substantial development, including Nigeria-specific examples and validation by FAO Nigeria and the government. It also identifies key data gaps and methodological challenges to inform future assessments, monitoring and reporting.

Climate change is already causing significant economic and non-economic losses across Nigeria's agrifood systems, affecting livelihoods, food security, ecosystems and broader socio-economic stability. The increasing frequency and intensity of floods, droughts, heat stress and other climate-related hazards are disrupting agricultural production, damaging infrastructure and productive assets, reducing household incomes and weakening resilience among vulnerable populations. Recent climate-related shocks have shown that losses are not only recurrent but increasingly systemic, affecting interconnected agrifood value chains and rural economies.

These impacts are compounded by structural challenges, including dependence on rain-fed agriculture, weak rural infrastructure, limited access to finance and persistent socio-economic inequalities. Beyond measurable economic costs, climate change is also contributing to food insecurity, displacement, ecosystem degradation, loss of social cohesion and heightened resource-based conflicts, particularly in climate-sensitive and fragile regions. Collectively, these impacts demonstrate that loss and damage in Nigeria is increasingly systemic, with cascading effects across livelihoods, value chains and human well-being (Federal Government of Nigeria, 2021a; Federal Government of Nigeria, 2022; Federal Government of Nigeria, 2024). Recent food security assessments further demonstrate how climate-related shocks compound impacts on livelihoods, food systems and human well-being in Nigeria.

Box 2: Climate Shocks and Escalating Food Insecurity Risks in Nigeria

Climate-related shocks are increasingly contributing to food insecurity and livelihood instability across Nigeria, particularly in vulnerable agrifood systems already affected by poverty, conflict and economic pressures. According to recent FAO assessments, approximately 34.7 million people are projected to face acute food insecurity in 2025, driven by the combined impacts of flooding, drought, displacement, insecurity and rising food prices. Climate-related disruptions to crop production, livestock systems, fisheries and market access continue to reduce food availability and weaken household coping capacities, particularly among smallholder farming communities.

The growing scale of food insecurity highlights the interconnected nature of climate-related loss and damage across agrifood systems, livelihoods and human well-being. It also reinforces the need for integrated approaches that combine anticipatory action, resilient food systems, social protection and livelihood recovery measures to reduce future climate-related losses and vulnerabilities (FAO Nigeria, 2025).

4.4.3.1 Economic Losses

Economic losses associated with climate change are already significant across Nigeria's agrifood systems and are projected to increase under future climate scenarios. These losses are reflected in declining agricultural productivity, livestock mortality, damage to infrastructure and productive assets, disruptions to value chains, and reduced household incomes (Federal Government of Nigeria, 2021a).

Climate variability and extreme weather events continue to affect crop production, fisheries and livestock systems, particularly given the strong dependence on rain-fed agriculture and the limited adaptive capacity of many rural communities (Federal Government of Nigeria, 2022). Flooding has caused widespread destruction of farmland, storage facilities, roads and irrigation infrastructure, while droughts and heat stress are reducing yields, degrading rangelands and increasing production costs (Federal Government of Nigeria, 2025).

Economic losses are further exacerbated by structural constraints within agrifood systems, including weak rural infrastructure, limited irrigation coverage, inadequate storage and processing facilities, and high post-harvest losses across major value chains (Federal Government of Nigeria, 2024). Climate-related disruptions to transport, processing and market systems often extend losses beyond production, affecting food availability, affordability and supply chain stability.

At the macroeconomic level, climate change poses significant risks to national development and economic growth. Evidence from national assessments indicates that climate impacts could significantly reduce GDP and increase public expenditure requirements if loss and damage are not effectively addressed. (Federal Government of Nigeria, 2021a).

4.4.3.2 Non-Economic Losses

In addition to economic impacts, climate change is causing substantial non-economic losses that affect individuals, communities, ecosystems and institutions across Nigeria. These include the loss of ecosystem services, biodiversity decline, food and nutrition insecurity, deteriorating health outcomes, displacement, erosion of social cohesion and disruption of cultural and traditional livelihood systems (Federal Government of Nigeria, 2021a).

Non-economic losses are particularly pronounced among vulnerable populations whose livelihoods depend directly on climate-sensitive natural resources and agrifood systems. Women, youth, older persons, persons with disabilities, and other marginalised populations are disproportionately affected by structural inequalities

that limit access to land, finance, technology, information, and decision-making processes (Federal Government of Nigeria, 2021b).

Climate-related pressures on land, water and forest resources are also intensifying competition for resources and contributing to insecurity, farmer–herder conflicts and displacement in several regions of the country, particularly in fragile and conflict-affected areas such as the Lake Chad Basin (Federal Government of Nigeria, 2021a; Federal Government of Nigeria, 2022). These dynamics constitute critical non-economic losses, as they undermine social stability, human security and long-term development prospects.

Many non-economic losses are difficult to quantify and may be irreversible, particularly when climate impacts affect ecosystems, indigenous knowledge systems, cultural identity and social networks. Strengthening national systems to assess and monitor both economic and non-economic losses will therefore be critical to informing policy, planning and financing responses to loss and damage (UNEP & UNEP Copenhagen Climate Centre, 2026).

The unequal distribution of these impacts highlights the importance of identifying the communities, production systems and geographic areas most at risk, to inform targeted policy responses.

4.4.4 Data Gaps and Methodological Challenges

Significant data and methodological gaps remain in assessing loss and damage in Nigeria. These include:

- Limited systems for systematically tracking economic and non-economic losses across agrifood systems
- Insufficient disaggregation of climate impacts by sector, geography and population group
- Weak integration of loss and damage into national statistical, monitoring and reporting frameworks
- Limited methodologies for capturing non-economic losses, including ecosystem degradation, cultural loss and social impacts
- Challenges in attributing observed losses to climate change versus other drivers

Addressing these gaps will be critical to strengthening evidence-based decision-making, improving monitoring and reporting, and informing effective policy and financing responses.

4.4.5 Systemic Nature of Loss and Damage

Loss and damage in Nigeria is increasingly systemic, driven by the interaction of climate hazards with underlying socio-economic and institutional vulnerabilities. Structural weaknesses in agrifood systems, including low productivity, inadequate infrastructure, weak extension services and limited access to finance reduce the capacity of communities to absorb and recover from climate shocks. This, therefore, highlights the need for integrated approaches that address both climate risks and development constraints.

4.4.6 At-Risk Communities & Systems

This subsection identifies the populations, production systems, and geographic areas most vulnerable to loss and damage. It places particular emphasis on groups with heightened exposure and limited adaptive capacity, including women, youth, and Indigenous Peoples. The analysis links climate exposure to socioeconomic vulnerability and limited recovery capacity across agroecological zones and livelihood systems. While the current NAP includes relevant information on ecological zones and vulnerable groups, this subsection synthesises and reframes it from an L&D perspective, complemented by practical insights from FAO Nigeria and the government where available.

Vulnerability to climate-related loss and damage in Nigeria is shaped by the interplay of exposure to climate hazards, the sensitivity of livelihoods and limited adaptive capacity. Agrifood systems remain particularly at risk due to their central role in supporting livelihoods, food security and rural economies, as well as their high dependence on climate-sensitive natural resources and rain-fed production systems (Federal Government of Nigeria, 2022; Federal Government of Nigeria, 2024).

Smallholder farmers, pastoralists, fishing communities and other rural populations whose livelihoods depend directly on agriculture, livestock, fisheries and natural resource-based activities are among the most vulnerable groups. These populations often operate under conditions of limited access to irrigation, agricultural inputs, finance, markets, extension services and climate information, which reduces their capacity to anticipate, absorb and recover from climate shocks (Federal Government of Nigeria, 2022). Recent assessments among smallholder farming communities further highlight how recurrent flooding is affecting livelihoods, food security and recovery capacities in vulnerable rural areas.

Box 3: Flood Vulnerability Among Smallholder Farmers in Niger State

Assessments conducted among smallholder farming communities in Niger State indicate that recurrent flooding is increasingly undermining agricultural production, livelihoods and household resilience. Flood impacts include destruction of cropland, loss of inputs and productive assets, reduced yields, disruption of market access and damage to rural infrastructure. Smallholder farmers are particularly vulnerable due to high dependence on rain-fed agriculture, limited access to irrigation, weak financial capacity and inadequate access to climate information and early warning systems.

The assessment further highlights that repeated flood events are eroding household coping capacities and increasing food insecurity and livelihood instability among vulnerable rural communities. These findings demonstrate how localized climate hazards can generate cumulative and long-term loss and damage impacts across agrifood systems, reinforcing the importance of anticipatory action, resilient agricultural practices and targeted support for vulnerable farming communities (Assessment of Flood Vulnerability among Smallholder Farmers in Niger State, Nigeria, 2025).

Climate risks are particularly severe in regions already affected by poverty, environmental degradation and insecurity. Northern parts of the country face increasing exposure to drought, desertification and resource-based conflicts, while southern and coastal areas are increasingly affected by flooding, erosion and sea-level rise affecting farming systems, fisheries and coastal livelihoods (Federal Government of Nigeria, 2021a; Federal Government of Nigeria, 2025). For example, flood-prone areas within the Niger–Benue river basins are especially vulnerable to recurrent flooding, which continues to result in widespread crop destruction, damage to rural infrastructure, displacement of communities and disruption of agricultural value chains and market systems. In many cases, these climate pressures interact with insecurity and displacement, further disrupting agrifood systems and weakening local resilience capacities (Federal Government of Nigeria, 2021a; Federal Government of Nigeria, 2022).

Climate-related human mobility can also generate secondary impacts on host communities receiving displaced populations. Increased pressure on land, water resources, public services, rural infrastructure and livelihood opportunities may create new vulnerabilities and intensify competition for natural resources. Strengthening resilience therefore requires integrated support for both displaced populations and host communities.

Women, youth, older persons, persons with disabilities and other marginalised populations are disproportionately affected by climate-related loss and damage due to structural inequalities that limit access to productive resources, land, finance, technology, climate information, essential services and participation in decision-making processes (Federal Government of Nigeria, 2021b). Older persons and persons with disabilities may also face additional barriers to accessing early warning information, evacuation support and livelihood recovery opportunities

following climate-related disasters. Ensuring that Loss and Damage responses are inclusive and accessible is therefore essential to reducing disproportionate impacts on these population groups. Women play a critical role across agrifood systems, contributing significantly to agricultural labour and household food security, yet often face greater constraints in accessing climate-resilient technologies, financial services and livelihood opportunities (Federal Government of Nigeria, 2021b; Federal Government of Nigeria, 2022).

The concentration of vulnerability within climate-sensitive agrifood systems underscores the importance of integrated, inclusive approaches to addressing loss and damage, with particular attention to livelihoods, food-systems resilience and the needs of the most vulnerable populations (UNEP & UNEP Copenhagen Climate Centre, 2026). These patterns show that loss and damage is not evenly distributed but concentrated in specific livelihood systems, populations and geographic hotspots, requiring targeted, inclusive policy responses.

Addressing these vulnerabilities requires coordinated strategic interventions that not only reduce climate risks but also strengthen the capacity of institutions and communities to avert, minimise and address Loss and Damage.

4.4.7 Strategic Priority Solutions to Avert, Minimise, and Address Loss and Damage

This subsection outlines priority measures structured around a Comprehensive Risk Management (CRM) approach, as advanced by the UNFCCC WIM Executive Committee and FAO. It covers: risk assessment and monitoring; risk reduction and anticipatory action; response, recovery, and rehabilitation; and risk transfer and protection mechanisms, including insurance and social protection. While many relevant adaptation measures are already included in the NAP, this subsection reorganises and strengthens them within a coherent L&D framework.

Addressing loss and damage in Nigeria requires a comprehensive and integrated climate risk management approach that combines risk reduction, anticipatory action, preparedness, response, recovery and long-term resilience building across agrifood systems and related livelihoods (UNEP & UNEP Copenhagen Climate Centre, 2026). Strengthening the resilience of agrifood systems is particularly critical given their central role in livelihoods, food security, employment and economic stability (Federal Government of Nigeria, 2024).

4.5 Risk Assessment and Monitoring

Strengthening climate risk assessment, early warning and monitoring systems will be essential for improving preparedness and reducing climate-related losses. This includes enhancing hydro-meteorological services, climate information systems, vulnerability assessments and systems for tracking economic and non-economic

losses and damages across agrifood systems (Federal Government of Nigeria, 2025). Improved data integration and information-sharing across sectors and institutions will support evidence-based planning and decision-making (Federal Government of Nigeria, 2022).

4.6 Risk Reduction and Anticipatory Action

Reducing climate risks within agrifood systems will require scaling up climate-smart agriculture, sustainable land and water management, ecosystem restoration and resilient rural infrastructure (Federal Government of Nigeria, 2022; Federal Government of Nigeria, 2024). Expanding irrigation systems, promoting drought-tolerant crops, improving soil and water conservation measures and strengthening extension services can help reduce vulnerability to climate shocks and improve adaptive capacity among smallholder producers.

Strengthening community-led approaches, including indigenous knowledge, traditional resource management practices and locally developed climate risk management mechanisms, can complement scientific approaches while improving the effectiveness, ownership and sustainability of Loss and Damage interventions.

Strengthening anticipatory action and early response systems, including community-based early warning mechanisms and forecast-based action, will also be critical for reducing losses before climate hazards escalate into disasters (UNEP & UNEP Copenhagen Climate Centre, 2026). Investments in digital agriculture, climate information services and improved risk communication systems can further support preparedness and timely decision-making (Federal Government of Nigeria, 2022).

4.6.1 Response, Recovery and Rehabilitation

Strengthening disaster response, livelihood recovery and rehabilitation systems is necessary to support affected communities and restore productive capacities following climate-related shocks. This includes rehabilitation and climate-resilient reconstruction of climate-damaged rural infrastructure, restoration of degraded ecosystems, rebuilding disrupted agrifood value chains and supporting recovery processes that reduce future climate risks rather than recreating previous vulnerabilities (Federal Government of Nigeria, 2024).

Targeted support to vulnerable populations, particularly women, youth, displaced populations and conflict-affected communities, will be important to ensure equitable recovery and resilience-building processes (Federal Government of Nigeria, 2021b). Integrating loss and damage considerations into humanitarian response and recovery planning can further strengthen resilience in fragile and climate-vulnerable contexts (Federal Government of Nigeria, 2021a).

4.6.2 Risk Transfer and Social Protection

Strengthening climate risk financing, agricultural insurance and shock-responsive social protection systems will be critical for addressing residual risks and supporting vulnerable populations affected by climate-related losses and damages (Federal Government of Nigeria, 2022). Expanding access to agricultural insurance, contingency financing and financial services for smallholder producers can improve the capacity of households and enterprises to absorb and recover from climate shocks.

Strengthened social protection systems linked to climate and disaster risk management can also help reduce vulnerability and support livelihood stability in the face of recurrent climate impacts (UNEP & UNEP Copenhagen Climate Centre, 2026).

4.6.3 Strengthening Resilient Agrifood Systems

Many of the priority actions required to address loss and damage are already reflected within Nigeria's agricultural and climate policy frameworks, including climate-smart agriculture, digital agriculture, sustainable resource management, value chain strengthening and improved access to agricultural finance and insurance (Federal Government of Nigeria, 2022; Federal Government of Nigeria, 2024). However, reframing them through a loss and damage lens strengthens their focus on preventing irreversible impacts and addressing residual risks. Hence, strengthening implementation and coordination of these interventions provides an important opportunity to enhance resilience, reduce vulnerability and minimise future climate-related losses and damages across agrifood systems and related livelihoods.

4.6.4 Indicative Outcomes Framework

To operationalise these strategic priority solutions, an indicative outcomes framework is proposed (Table X) to guide coordinated action across key agrifood systems and related livelihood sectors. The framework translates the strategic directions outlined above into priority outcomes, outputs and corresponding actions to avert, minimise and address climate-related loss and damage. It emphasises integrated approaches that strengthen climate risk management, resilient agrifood systems, anticipatory action, social protection, recovery systems, inclusive governance and climate risk financing, while recognising the interconnected nature of climate risks, livelihoods and food systems in Nigeria. The framework also provides a basis for strengthening coordination among government institutions, development partners, communities and private sector actors in implementing loss and damage responses at national and sub-national levels (Federal Government of Nigeria, 2022; Federal Government of Nigeria, 2024; UNEP & UNEP Copenhagen Climate Centre, 2026).

Strategic outcome	Key outputs	Priority actions
Outcome 1: climate risk information, monitoring and early warning systems strengthened to reduce climate-related losses and damages across agrifood systems	Integrated climate risk and loss & damage monitoring systems established	• Establish national systems for tracking economic and non-economic loss and damage across agrifood systems •Strengthen hydro-meteorological and climate information systems • Improve vulnerability and risk mapping across agrifood systems • Integrate community-based observations and indigenous knowledge into climate risk assessment and early warning systems.
	Community-centred early warning and anticipatory action systems operationalised	• Strengthen community-based early warning systems • Scale up forecast-based financing and anticipatory action approaches to reduce pre-disaster losses • Improve climate risk communication and dissemination systems
Outcome 2: resilience of agrifood systems and climate-sensitive livelihoods strengthened to avert and minimize loss and damage	Climate-smart and resilient agricultural production systems scaled up	• Promote climate-smart agriculture and agroecological approaches • Expand irrigation and sustainable water management systems • Scale up drought-tolerant crops and resilient livestock systems
	Sustainable natural resource management and ecosystem restoration strengthened	• Restore degraded ecosystems and watersheds • Promote sustainable land and forest management • Promote & strengthen ecosystem-based adaptation approaches to reduce long-term non-economic losses • Scale up locally led climate resilience and community-based natural resource management practices.
	Resilient agricultural value chains and rural infrastructure strengthened	• Improve storage, processing and market infrastructure •Climate-proof and rehabilitate infrastructure damaged by recurrent climate events _ • Reduce post-harvest losses • Strengthen climate-resilient rural transport and market systems
Outcome 3: vulnerable communities and livelihood systems better protected from residual climate risks and shocks	Shock-responsive social protection and livelihood recovery systems strengthened	• Support participatory livelihood recovery planning for both climate-affected and host communities. • Expand adaptive social protection systems linked to climate and disaster risk triggers • Support livelihood diversification and recovery programmes • Strengthen food security and nutrition support mechanisms
	Climate risk financing and insurance systems expanded	• Scale up agricultural insurance and risk transfer mechanisms to address residual losses • Improve access to climate finance and contingency financing • Strengthen financial inclusion for vulnerable producers and enterprises

Outcome 4: Governance, coordination and institutional capacity for loss and damage strengthened	Institutional coordination mechanisms for loss and damage enhanced	• Strengthen cross-sectoral coordination frameworks specifically for loss and damage governance_ • Integrate loss and damage into national and sectoral planning frameworks
	Capacity, knowledge management and data systems strengthened	• Strengthen national capacity for l&d assessment, reporting and planning • Strengthen knowledge-sharing and research systems • Improve data integration and evidence-based decision-making
	Inclusive and gender-responsive approaches strengthened	• Promote meaningful participation of women, youth, older persons, persons with disabilities and other vulnerable populations in loss and damage planning, implementation and decision-making. • Strengthen access to resources, services and climate information • Mainstream gender and social inclusion across loss and damage interventions
Outcome 5: integrated recovery, rehabilitation and resilience-building systems strengthened in fragile and climate-affected regions	Recovery and rehabilitation systems for climate-affected agrifood systems strengthened	• Rehabilitate damaged agricultural infrastructure and productive assets following climate shocks • Restore disrupted livelihood and food systems • Support resilient reconstruction and ecosystem recovery
	Climate-security and displacement-sensitive approaches strengthened	• Integrate climate-security considerations into loss and damage responses • Support climate-resilient livelihoods in displacement-affected areas • Strengthen local peacebuilding and natural resource governance mechanisms

Table 4.12: Indicative Outcomes Framework for Addressing Loss and Damage in Agrifood Systems and Related Livelihoods in Nigeria

Implementation of the above strategic outcomes and priority actions will require strengthened coordination across national and sub-national institutions, sustained investment in resilient agrifood systems and livelihoods, and inclusive approaches that prioritise vulnerable populations and climate-sensitive regions. Strengthening integrated climate risk management systems, anticipatory action, recovery mechanisms and climate risk financing will be critical for reducing future losses and damages while supporting resilient, equitable and sustainable development pathways in Nigeria (Federal Government of Nigeria, 2021a; Federal Government of Nigeria, 2024; UNEP & UNEP Copenhagen Climate Centre, 2026).

4.7 Enabling Conditions

This subsection identifies the enabling conditions required to effectively avert, minimise, and address loss and damage at the national level. Key elements include institutional coordination and governance across sectors; data, information, and assessment systems; financing mechanisms and access to international climate finance; technical capacity and knowledge systems; and partnerships at national, regional, and global levels. While related elements are addressed elsewhere in the NAP (e.g., implementation, finance, and monitoring and evaluation), this subsection consolidates and tailors them specifically to L&D.

Effective implementation of measures to avert, minimise, and address loss and damage will require strengthened enabling conditions across institutional, financial, technical and governance systems. Given the systemic nature of climate-related risks affecting agrifood systems and livelihoods, coordinated and integrated approaches will be essential to strengthen resilience, reduce vulnerability and support long-term recovery and adaptation processes (UNEP & UNEP Copenhagen Climate Centre, 2026).

4.7.1 Institutional Coordination and Governance

Strengthening coordination across federal, state and local institutions will be critical to ensuring coherent and effective responses to loss and damage. This includes enhanced collaboration among institutions responsible for agriculture, the environment, water, finance, disaster risk management and social protection, as well as the strengthened integration of loss and damage considerations into national and sectoral planning frameworks (Federal Government of Nigeria, 2025).

Given the interconnected nature of climate risks, food systems, livelihoods and natural resource management, integrated governance approaches will also be required to support coordinated implementation across agrifood systems and climate-sensitive sectors (Federal Government of Nigeria, 2022).

4.7.2 Data, Knowledge and Technical Capacity

Strengthening climate information systems, risk assessments, monitoring systems and knowledge management platforms will be essential for improving evidence-based decision-making and supporting effective planning and implementation (Federal Government of Nigeria, 2025).

There is also a need to strengthen national capacities for assessing and tracking economic and non-economic losses and damages, including impacts on livelihoods, ecosystems, food security and vulnerable populations (UNEP & UNEP Copenhagen

Climate Centre, 2026). Enhancing extension services, climate information dissemination systems and research partnerships will further support adaptive capacity and resilience-building across agrifood systems (Federal Government of Nigeria, 2022).

4.7.3 Climate Finance and Risk Financing

Addressing loss and damage at scale will require strengthened access to climate finance, risk financing mechanisms and long-term investment in resilient agrifood systems and livelihoods. Existing financing gaps, particularly in vulnerable and climate-affected regions, continue to constrain preparedness, recovery and resilience-building efforts (Federal Government of Nigeria, 2021a).

Strengthening access to agricultural finance, insurance systems, contingency financing and international climate finance mechanisms, including emerging global loss and damage financing arrangements, will therefore be critical for supporting integrated and inclusive responses (Federal Government of Nigeria, 2024).

4.7.4 Inclusive and Participatory Approaches

Inclusive and gender-responsive approaches will be essential to ensure that interventions effectively address the differentiated impacts of climate-related loss and damage. Women, youth and marginalised groups often face disproportionate climate risks due to unequal access to resources, services, information and decision-making processes (Federal Government of Nigeria, 2021b).

Strengthening the meaningful participation of vulnerable populations in planning, implementation, monitoring and decision-making can improve locally grounded responses, reinforce existing community resilience mechanisms, strengthen social cohesion and support more equitable resilience outcomes across agrifood systems and climate-sensitive communities.

4.7.5 Partnerships and Multi-Stakeholder Engagement

Effective responses to loss and damage will require strengthened partnerships among government institutions, local communities, development partners, research institutions, civil society organisations, and the private sector. Multi-stakeholder collaboration will be particularly important for mobilising technical expertise, financing, innovation and locally led solutions that support resilient agrifood systems and sustainable livelihoods (Federal Government of Nigeria, 2024).

4.7.6 Key Messages

Climate-related loss and damage in Nigeria's agrifood systems is already significant and increasing, with major implications for livelihoods, food security and economic stability.

These impacts extend beyond direct economic losses to include non-economic and systemic effects that disproportionately affect vulnerable populations and high-risk regions.

Addressing loss and damage in Nigeria, therefore, requires integrated, inclusive and forward-looking approaches that strengthen resilience across agrifood systems while addressing the underlying vulnerabilities that amplify climate risks.

Strengthened coordination, sustainable financing, robust evidence systems, locally led action and inclusive governance will be essential for averting, minimising, and addressing climate-related Loss and Damage while safeguarding livelihoods, food security and long-term resilience across Nigeria's agrifood systems.

4.8 Conflict Resolution and Peace Building

In Nigeria, climate-induced hazards worsen poverty, creating economic instability and social unrest; they cause social and cultural disruptions and generate conflicts. Hazards such as drought intensify competition for land, water, and mineral resources, thereby worsening conflicts, creating humanitarian crises, overloading state institutions, and creating the environment for recruiting armed groups.

The increasing frequency of droughts, desertification, heatwaves and changing rainfall patterns leads to water and food scarcity, causing tensions and conflicts over access to these resources in different regions of Nigeria. One impact of climate change is the altered migration patterns of herders and their animals, creating conflicts over access to pasture and water. NAP implementation will integrate conflict-sensitive and peacebuilding approaches.

4.9 Climate Mobility

Effective adaptation to climate-related mobility in Nigeria requires moving beyond reactive, crisis-driven responses toward a proactive, rights-based, and gender-sensitive governance framework. Drawing on existing evidence and emerging policy commitments—including the domesticated Kampala Convention, the NDC 3.0, and sub-national initiatives from Lagos State—this section sets out seven interrelated goals and the strategies required to achieve them.

Objectives:

- **Protect the Right to Stay:** Strengthen climate risk awareness and adaptive capacities for in-situ resilience so that populations facing climate risks retain a genuine choice to remain in their communities with dignity and without deepening vulnerability.

- **Prevent Trapped Populations:** Ensure that no population is left stranded in high-risk environments due to poverty, marginalisation, or institutional barriers, by guaranteeing access to the resources and protection needed to adapt in place or move safely and in dignity.

- **Enable Mobility in Dignity:** Formally recognise mobility as a legitimate climate adaptation strategy and strengthen safe, regular and livelihood-enhancing mobility pathways within and beyond Nigeria.

- **Adopt Rights-based Mobility Responses:** Protect and uphold the rights of people on the move and climate-affected populations, including IDPs, pastoralists, seasonal labourers, and migrants.

- **Anticipate Disasters and Support Recovery:** Reduce climate-driven harm through anticipatory action, mobility-sensitive preparedness and post-hazard response, including post-flood relocation and recovery support, relocation zones, and temporary/permanent camps for affected communities where necessary to secure sustainable solutions.

- **Foster Social Cohesion and Economic Inclusion in Destination Communities:** Ensure that climate migrants and displaced populations are enabled to be active contributors to their new communities and have equitable access to services, livelihoods, and economic opportunities, reducing conflict risk and building shared foundations for coexistence and adaptive governance.

- **Institutionalize and Localize the Climate Mobility Response:** Institutionalize data, planning and finance for mobility-sensitive adaptation across national and subnational levels, including by strengthening capacities for urban planning.

4.9.1 Climate Mobility as a Cross-Cutting Adaptation Priority

Climate mobility — encompassing displacement, migration, pastoralist movement, and the inability to move when movement is necessary — does not sit within any single ministerial mandate. It is simultaneously a humanitarian concern, a land use and planning challenge, a labour market question, a fiscal and financing problem, and a social protection obligation. Its cross-cutting nature is precisely why climate mobility has historically fallen between institutional responsibilities and received insufficient policy attention.

This chapter makes a deliberate choice to surface climate mobility as a coherent policy domain in its own right, rather than treating it as a sub-component of existing sectoral plans. The seven objectives above define the normative foundation — what Nigeria is aiming to achieve for people on the move and those at risk of being left behind. They are organised around the lived experience of climate-affected populations: the right to stay, the right to move safely, the right to be protected wherever one is, and the right to recover and rebuild.

The table that follows translates these objectives into action through the lens of existing policy agendas, frameworks and institutional mandates to provide guidance for the relevant parts of government. It straddles the institutional responsibilities of:

- National Climate Change Council & Federal Ministry of Environment: national coordination, Taskforce convener, MRV linkage.
- Ministry of Humanitarian Affairs & National Emergency Management Authority: operational emergency response and post-hazard assistance for displaced persons.
- Ministry of Labour & Employment & Ministry of Interior: labour mobility governance and protection measures.
- Federal Ministry of Agriculture and Rural Development & State Agriculture/ Environment Ministries: pastoral corridor governance, livestock recovery.
- National Bureau of Statistics and relevant technical partners: data collection, analysis and reporting.
- Federal Ministry of Housing and Urban Development (FMHUD) in coordination State Planning Boards and local planning authorities: urban and land use planning
- Sub-national governments: participatory planning, governance of land tenure and water access rights, social services, enforcement of codes (labour, environment, building)
- Development partners and private sector: technical support and co-financing.

Realising the recommended measures for the climate mobility sector will require blended finance from public budgets, climate finance and partners, private sector and diaspora resources. Resource mobilisation for the NAP provides an opportunity to allocate a defined share to mobility-sensitive measures and to build a dedicated project pipeline for funding. As is foreseen in the NAP financing strategy, it is critical that adaptation funding reaches the local level to address the different levels of risk exposure, community vulnerability and displacement pressures across the country.

Area	Challenge	Goal	Key actions	Benefits
Coordination & Planning	Climate mobility cuts across many ministries. No single agency currently has the mandate to address it.	Give climate mobility a clear institutional home and back it with budget.	• Establish a national climate mobility taskforce led by the Ministry of Environment • Assign lead responsibilities across ministries and publish an annual progress note tracking action against the NAP, NDC3.0 and other relevant strategies • Create dedicated budget lines and a pipeline of proposals to international climate funds	Unlocks international climate finance, which requires clear governance.
Data & Monitoring	Nigeria does not systematically track who is moving – or unable to move despite being at risk – because of climate hazards, and where they go.	Build the evidence base on climate-related movement and immobility.	• Add mobility questions to the census and national household surveys • Map mobility preferences in climate risk and vulnerability hotspots such as grazing corridors, flood zones, coastal areas, and urban arrival areas • Include community knowledge and local monitoring in official risk monitoring and early warning systems	Ensures resources are allocated where most needed, facilitates community buy-in for local adaptation planning, and allows Nigeria to track the impact of funds.

Area	Challenge	Goal	Key actions	Benefits
Early Warning	Warnings about floods and droughts often do not reach the people most at risk or arrive too late to act on.	Turn forecasts into action before people are displaced.	• Link NiMET and NEMA forecasts to pre-agreed response plans and standby funding • Extend warnings to mobile and hard-to-reach populations • Cover flood, drought, coastal erosion and conflict risk in a single system	Saves lives, protects homes and livestock, and avoids a more expensive emergency response later.
Anticipatory Disaster Response & Recovery	Floods and droughts displace large numbers of Nigerians every year, and many remain displaced long after the emergency has passed.	Prepare before disasters strike and plan for recovery and displacement solutions.	• Agree evacuation, shelter and assistance plans ahead of flood and drought seasons • Strengthen NEMA and state agencies to lead recovery, not only emergency response • Make return, local settlement or resettlement an explicit part of every recovery plan	Early action costs far less than crisis response and prevents temporary displacement from becoming permanent.
Pastoralist Movement	As rainfall becomes less reliable, herders travel further for pasture and water, increasing clashes with farming communities — particularly across the Middle Belt.	Secure grazing routes and support local agreements over land and water.	• Protect and restore priority grazing corridors, crossing points and water points • Support dialogue platforms between farmers, herders and local authorities • Bring veterinary, health and documentation services to markets and transit points	Reduces Nigeria's most immediate climate-driven conflict risk, protecting food production and lowering security costs.

Area	Challenge	Goal	Key actions	Benefits
Rural-Urban Migration	Climate pressure on farming is pushing people into cities faster than housing, services and jobs can absorb them.	Reduce the pressure to leave, and prepare cities for those who arrive.	• Invest in rural livelihoods and farm productivity in the areas where people leave • Help (secondary) cities plan land use, housing, water and services for population growth • Connect rural workers to urban jobs through training and job matching	Turns unmanaged urban pressure into planned growth, easing strain on housing, jobs and public services.
Labour Mobility	Climate stressors affecting rural livelihoods increases pressure for people to move for work, often in unsafe conditions, without contracts, documentation or legal protection, leaving them exposed to exploitation.	Ensure adequate protections for migrant workers in the context of increased climate risk.	• Regulate recruitment agents and brokers to reduce exploitation and debt bondage • Provide pre-departure information and documentation support for workers moving within Nigeria and across borders and embed worker protections in skills training and placement initiatives • Ensure migrant workers, including seasonal and informal workers, can access health services, grievance mechanisms and legal remedy	Protects a growing share of the national workforce and reduces forced labour risk. Well-protected migration returns remittances and skills to origin communities. Also advances Nigeria's existing ECOWAS free movement and ILO commitments.

Area	Challenge	Goal	Key actions	Benefits
Coastal Resilience & Relocation	Coastal erosion and rising seas are making some communities unlivable. Relocation is already happening informally, without rules or safeguards.	Set the rules for relocation before large-scale moves are underway.	• Adopt a national policy setting out when and how relocation is decided • Assess which coastal communities can adapt in place and which cannot, working with the affected communities to design their adaptation journeys • Protect land rights, secure tenure, livelihoods and social ties before people move	Protects rights, avoids lasting grievances and legal liability and opens access to international loss and damage funding.
Climate Mobility Finance & Social Protection	Government budget alone cannot cover a comprehensive climate mobility response. People who move should not lose access to financial supports buffering them against climate shocks.	Raise money beyond the government budget and keep support with people when they move.	• Develop fundable project concepts for international climate funds • Expand insurance for farmers and climate-vulnerable households • Issue diaspora-targeted green and blue bonds • Make social benefits accessible across state boundaries	Multiplies every naira of government spending by drawing in international climate funds, diaspora savings and insurance markets.

4.9.2 Alignment with NAP Investment Priorities

Given the NAP's clear prioritization of sectors for adaptation investment and M&E, the following table illustrates how those priorities align with the above climate mobility objectives and what a complementary approach to integrating climate mobility into the NAP strategic investment and monitoring framework could look like.

NAP Investment Priority	Climate Mobility Objectives	Opportunities to integrate climate mobility into investment design & the NAP Strategic Monitoring Framework	Complementary climate mobility actions
Flood / Heat / Water / Energy Management in Agricultural Adaptation	**Obj 1:** "Right to Stay" **Obj 3:** Mobility with Dignity	• **Use climate mobility modelling and assess mobility aspirations and capabilities in communities** targeted for sector investment, so interventions work with — not against — people's own adaptation preferences.	• **Expand data systems to map mobility preferences and climate mobility hotspots** to inform adaptation policy and investments.
Flood Management in Agricultural Adaptation	**Obj 2:** Prevent Trapped Populations **Obj 5:** Post-Disaster Recovery **Obj 7:** Institutionalize the Response	• **Integrate displacement tracking into flood response MRV:** track whether flood-affected households return home, integrate locally, or remain displaced, so recovery investment reflects actual outcomes. • **Embed mobility criteria in flood-resilient ranching proposals:** ensure flood-resistant ranching designs account for pastoralist movement and don't block seasonal corridors or transhumance routes.	• **Develop standards for flood-related relocations:** establish rights-based criteria, consent protocols, and minimum service standards for flood-triggered relocations, consistent with the Kampala Convention. • **Embed durable solutions planning in flood recovery:** require flood recovery programmes to define a durable solutions pathway — return, local integration, or resettlement — with timelines, agencies, and financing set from the outset.

Heat Impacts Management in Agricultural Adaptation	**Obj 1:** Protect the Right to Stay **Obj 2:** Prevent Trapped Populations **Obj 6:** Social Cohesion at Destination	• **Monitor heat-driven farmer-herder conflict as a risk indicator:** integrate conflict and resource-competition data into heat impact monitoring — a leading driver of forced mobility in the Middle Belt. • **Extend heat adaptation benefits to mobile and seasonal workers:** design cooling, irrigation, and crop resilience interventions to reach seasonal labourers and pastoralists, not only settled households.	• **Develop urban heat resilience planning for receiving communities:** support state and LGA planning to integrate heat stress projections, with attention to informal settlements absorbing migrants from heat-stressed rural areas. • **Link heat adaptation investments to rural livelihood diversification:** pair heat resilience with income diversification, giving households facing declining agricultural viability alternatives to distress migration.
Cross-Cutting Agricultural Adaptation Strategy	**Obj 1:** Protect the Right to Stay **Obj 2:** Prevent Trapped Populations **Obj 3:** Enable Mobility in Dignity	• **Track rural-urban migration as an outcome indicator for agricultural adaptation:** monitor whether investments are reducing distress outmigration in targeted communities, linking investment performance to mobility outcomes. • **Design agriculture insurance to reduce distress migration triggers:** ensure agriculture insurance covers the climatic events most linked to forced displacement — drought, flooding, extreme heat — and prioritize enrolment for smallholders in high-risk areas.	• **Scale micro-risk insurance to support mobility decisions:** extend index-based insurance to non-agricultural assets and household goods, reducing the asset loss that forces distress migration and enabling planned, voluntary moves. • **Integrate climate mobility awareness into agricultural extension services:** train extension workers to identify households at high risk of distress migration and connect them to livelihood diversification, insurance, and safe migration information.

Water Resource Management Adaptation Strategy	**Obj 1:** Protect the Right to Stay **Obj 4:** Rights-Based Mobility Responses **Obj 6:** Social Cohesion at Destination	• **Include displaced persons in water infrastructure planning:** ensure flood protection and water supply investments cover IDP settlements and informal areas absorbing migrants, not only formally recognized communities. • **Embed mobility sensitivity in water early warning protocols:** ensure flood and drought early warning triggers account for population movement — evacuation routes, water access along displacement corridors, and provision at hosting sites.	• I**nclude mobile and displaced populations in local water governance:** develop participatory water governance that includes pastoralists, seasonal migrants, and displaced persons as rights-holders in allocation decisions, reducing conflict-driving resource competition. • **Use water infrastructure investments to incentivize conflict prevention:** design shared water points, reservoirs, and irrigation schemes in conflict hotspots as peace infrastructure — neutral resources under joint management that build cross-community cooperation.
Energy Sector Adaptation Strategy	**Obj 3:** Enable Mobility in Dignity **Obj 6:** Social Cohesion at Destination **Obj 7:** Institutionalize the Response	• **Include displaced persons in energy access planning:** ensure grid hardening, solar mini-grids, and backup systems cover IDP settlements and informal areas receiving migrants, rather than reinforcing existing inequalities. • **Use energy infrastructure investments to reduce inter-communal conflict risk:** in conflict hotspots, design solar-powered water pumps and community energy facilities as shared, jointly managed resources that support social cohesion alongside energy access.	• **Create green jobs and skills pathways for climate-vulnerable communities:** prioritise renewable energy training and jobs for young people in high-outmigration rural areas and informal settlements receiving migrants, turning investment into a managed labour mobility pathway. • **Establish community co-benefit frameworks for energy infrastructure maintenance:** structure energy investments so host communities and displaced persons share responsibility for — and benefit from — maintaining energy assets, building interdependence that reduces conflict risk.

Table 4.13: Aligning Nigeria NAP Investment Priorities and Climate Mobility Objectives

Chapter Five

Imperatives for NAP Implementation

5.1 Sustainable Financing

The economic appraisal of Nigeria's NAP Process revealed the crucial need to mobilise substantial financial resources to implement the NAP, taking into account the country's population size, landmass and ecological diversity. The following are potential sources of funding for the National Adaptation Plan:

- **Government Budget:** The Nigerian Government is showing greater political will to finance climate action, having recognised the significant costs of inaction. However, mobilising funds from domestic budgetary allocations has often proved challenging because of a lack of clear policies and regulatory frameworks, low national budgetary allocations, and limited awareness (NAPF, 2020). The Climate Change Act of 2021 and the creation of the Climate Fund demonstrate the Government's commitment to achieving global climate change targets. Already, Nigeria's Sovereign Green Bond, the Ecological Fund Office (EFO), and the Great Green Wall Commission are existing avenues to raise domestic funding for adaptation projects.

- **Bilateral and Multilateral Partners:** Several bilateral and multilateral agencies have supported Nigeria in implementing various climate change activities in the country. Multilateral climate funders such as the Green Climate Fund (GCF), the Global Environment Facility (GEF), the Adaptation Fund (AF), and the World Bank are the major sponsors of adaptation action in Nigeria.

- **Private Sector Engagement:** The Implementation of Nigeria's NAP encourages private sector participation through innovative investments that align with identified adaptation needs and support the management of associated risks. Government is creating the enabling environment through provision of policies, innovative incentive, access finance and building capacity and dismantling barriers to private sector engagement in NAPF, 2021. Both the national and sub-national governments will engage the private sector and financial institutions to participate in the mobilization of adaptation finance



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and form partnerships to access GCF funds and other international climate finance organizations. Nigeria is introducing green taxation, promoting green bond; supporting corporate social and environmental responsibilities for adaptation; and encouraging public-private partnerships as strategies for private sector engagement.

Nigeria's NAP process is consistent with the Green Climate Fund (GCF), which supports developing countries like Nigeria in responding to climate change through mitigation and adaptation actions. The key GCF goals in Nigeria include (i) helping communities and ecosystems adapt to climate impacts such as flood risk, desertification, changing rainfall, and coastal erosion; (ii) building climate-resilient infrastructure; and (iii) strengthening early warning systems, all of which are consistent with the strategic priorities of Nigeria's NAP. The GCF is already supporting the Conservation of Biodiversity and Sustainable Use of Lowland Forests in Ogun, Edo, Delta, and Ondo States, which aligns with NAP objectives. Another GCF project is the Nigerian Climate Investment Platform (NCIP), launched to mobilise up to US$500 million for infrastructure, resilience, and adaptation projects. The existing relationship will be further explored to facilitate additional financing for adaptation projects in Nigeria.

NAP also aligns with other climate actions implemented by international organisations such as the Adaptation Fund (AF) and the Special Climate Change Fund (SCCF) of the UNFCCC. AF finances concrete adaptation projects in Nigeria, especially for vulnerable people. The National Emergency Management Agency (NEMA), which coordinates disaster risk reduction activities in the country, receives support from AF. In 2025, AF approved a regional project involving Nigeria,

Cameroon, the Central African Republic, Chad, and Niger, to be implemented by the World Meteorological Organization, worth about US$11.6 million for the development of climate data and early warning systems. The Special Climate Change Fund (SCCF) not only prioritises adaptation activities but also undertakes activities covering technology transfer, energy, transport, industry, forestry, agriculture, and economic diversification. These align with the strategies of the NAP process. Similarly, the World Bank has funded the Nigeria Erosion and Watershed Management Project (NEWMAP) and is currently supporting Nigeria to implement the Agro-Climatic Resilience in Semi-Arid Landscapes (ACReSAL) Project. The goal of ACReSAL is to control land degradation and desertification, reduce climate risks, and improve agricultural productivity in the semi-arid and dryland areas of the 19 northern states. The goals of these two World Bank Projects align with the NAP process.

5.1.2 Enhanced Transformative Capacity and Technology Transfer

Achieving transformation in adaptive capacity and improved resilience against climate change impact requires strengthening capacity across sectors and communities. NAPF (2020) recognized the MDAs of government as key nodes in the adaptation efforts. Elaboration of NAP document offered an opportunity to achieve a transformational change in the institutional and implementation capacity of adaptation programs and actions on both medium- and long-term to reducing vulnerability to the impact of climate.

5.1.3 Capacity Building

Nigeria has always underscored the importance of capacity building in realising a climate-resilient economy by promoting awareness, establishing institutional arrangements and partnerships, empowering skills for assessments, and creating an enabling environment, etc. This is in accordance with the Paris Committee on Capacity Building goals. However, as there are uncertainties with climate change, so is the demand for continuous updating of sectoral capacity to meet current and future scenarios for national resilience.

Nigeria needs to continuously strengthen coordination and cooperation mechanisms that accommodate stakeholders' involvement and integrate defined roles into the planning and implementation of adaptation activities. Enhancing the capacity of various actors involved in coordination, mobilisation of finance and MRV is necessary to gather relevant data, track performance and understand current adaptation measures, ultimately supporting the attainment of the objectives of the NAP.

5.1.4 Stakeholder Engagement Strategy

A Stakeholders' Engagement Strategy was developed by Nigeria's NAP process to avoid incoherent implementation of adaptation strategies and to bridge any identified gaps, ensuring that NAP implementation is holistic and collective. The Stakeholders' Engagement Strategy helps to build consensus and trust among parties, ensures inclusiveness, enhances the credibility of decisions, and guarantees the sustainability of outcomes. The strategy supports Nigeria's NAP in gaining country-wide ownership and legitimacy through adequate representation. Therefore, deliberate efforts will be made to integrate viable Indigenous practices and knowledge into the implementation of the NAP. Similarly, the NAP Stakeholders' Engagement Strategy encourages leveraging partnerships and networks, empowering individuals, and integrating viable Indigenous practices and community knowledge, all of which are needed for success.

5.1.5 Effective Communication of NAP Process

Inadequate communication was identified as one of the limitations of previous adaptation initiatives in Nigeria (ADCOM, 2021, p. xviii). Nigeria's NAP Process developed a programme-specific communication strategy to enhance citizens' awareness and promote political and public participation in the implementation of NAP. The communication strategy recognises the importance of disseminating the appropriate type of information to the target audience; the roles of different actors in information sharing; clear and consistent communication messages; and the selection of effective media channels and messengers (Annex 6). Nigeria's NAP adheres to Article 6 of the UNFCCC, which emphasises education, training and public awareness on climate actions to promote climate communications at all levels in accordance with national laws and regulations. Thus, Nigeria's NAP also embodies awareness creation, behavioural change; information and knowledge sharing; capacity building; public participation and sustainable national development. The executive, legislative and judicial arms of Government, with their respective MDAs across the 3 tiers of government, are recognised as primary stakeholders in the NAP process. Effective channels for conveying adaptation messages include traditional and conventional means such as mobile phones, radio and social media, as well as roadshows. NAP implementation messages will be location-specific, gender-responsive and delivered in local languages. The DCC should coordinate the implementation of Nigeria's NAP communication strategy.

5.1.6 Inclusive Participation and Coordination

Although the impacts of climate change affect sectors differently and solutions need to be pursued simultaneously on some occasions, the interconnected and overlapping nature of these processes demands effective coordination. Different

adaptation actions are being carried out across sectors, with varying scopes at national and subnational levels, which requires harmonisation. The Federal Government of Nigeria established an Inter-Ministerial Committee on Climate Change (ICCC) in 2010, which oversees inter-agency coordination on climate change issues and works with the DCC on various aspects of climate change programming. The institutional arrangements in Nigeria are key to the realisation of the national mandate and strategic plan for the NAP.

The demand for a broader and participatory institutional arrangement led to the establishment of NCCC to coordinate and guide the Climate change process. Part of the purpose of Nigeria's NAP is to increase capacity and involvement of the sub-nationals and facilitate bringing the adaptation solutions closer to the grassroots A strong political commitment is required to enable uptake and practice of adaptation planning at subnational.

5.1.7 Private Sector Engagement

The private sector is pivotal to the implementation of the NAP, as it contributes significantly to the national GDP. The sector was recognised in the Stakeholders' Engagement strategy as being affected by climate change and as a vehicle for implementing actions to reduce vulnerabilities across key sectors. It is envisaged that the private sector will play a key role in the realisation of NAP goals, particularly in financing investments in adaptation solutions in key priority sectors such as agriculture, energy, and water resources. To realise full participation of the private sector in the implementation of the NAP, several bottlenecks identified in the NAPF must be addressed, including a lack of an enabling environment, poor coordination, and limited awareness and capacity. Government and the private sector may need to come together through different investment arrangements to de-risk some investments or improve credit access for the private sector, thereby facilitating investment in adaptation measures.

5.1.8 Gender and Social Inclusion

Climate change is a significant threat to Nigeria's development, as its impacts strongly affect all economic sectors, making the ambition to strengthen national gender programming imperative. Nigeria recognises the complexity of gender dynamics and incorporates them as an essential component of the framework to foster inclusive climate resilience, promote social justice, and ensure sustainable development (FME 2020; Jangbrand, 2022; Dazé, 2020). Whilst Nigeria has made efforts to achieve gender equality, significant gaps remain in addressing the strong influence of cultural, religious, and traditional beliefs on gender norms and roles, including dimensions such as the patriarchal nature of societies, education and employment, and changing gender dynamics in the country (FME-DCC 2023). The

Gender Analysis Report builds on the achievements of the NAPGCC to integrate lessons learned into the implementation of the NAP for gender-responsive climate actions in Nigeria.

The NAP Process incorporated measures to address the differential vulnerability to climate change and the adaptive capacities of women and other vulnerable groups, caused partly by inequalities rooted in biased gender norms and patriarchal social customs, and by disparities that deny women access to and ownership of productive resources, thereby limiting their ability to adapt to climate change (Aderiye, 2019; Onwutuebe, 2019). The NAP Process also aligns with the National Gender Policy (2021-2026) and the NAPGC, which set the minimum national standards expected to meet the country's mandate for gender equality, good governance, accountability, and social responsiveness to the needs of its vulnerable groups, including the implementation of national climate change initiatives, with a focus on the NDC to the Paris Agreement. As recommended in the Gender Analysis Report, Nigeria's NAP Process included opportunities to address gender imbalances in the conduct of climate scenario modelling and vulnerability assessment; review of existing policies; a comprehensive stakeholder mapping exercise; resource mobilisation and selection of monitoring and evaluation indicators; and the use of sex-disaggregated data in adaptation initiatives and recruitment (Annex 5).

Furthermore, the implementation of Nigeria's NAP shall adopt the following strategies to ensure that gender issues play a crucial role in achieving success in climate change adaptation in the country:

i. Enable reduction in gender inequality in adaptation needs, opportunities and capabilities;

ii. Improved access to productive resources, education and other economic benefits to avoid exacerbating pre-existing inequalities and vulnerabilities hampering women's ability to adapt to climate change

iii. Promotion of opportunities for equitable participation by women and men on climate change adaptation decision-making process on how adaptation measures are formulated and implemented in Nigeria;

iv. Facilitate equitable access to financial resources to procure modern technology and to invest in sustainable farming practices capable of building resilience of women and men, e.g., modern agricultural inputs, climate-resistant seeds, as well as financial resources, making them more susceptible.

v. Identify and incorporate gender-responsive strategies in priority sectors for the NAP process such as training and support programs for vulnerable groups in agriculture and catering for the unique needs and safety concerns of women and girls during extreme weather events in the water and sanitation sector.

Chapter Six

NAP Financing Strategy

6.1: Introduction

Given the enormous climate crisis Nigeria may face in the future and the consequences of continuing business-as-usual, it is inevitable that the country invest in relevant adaptation solutions. According to ND GAIN, Nigeria is the 53rd most vulnerable and the 6th least-ready country for climate change adaptation. The devastating 2012 floods, which displaced more than seven million people and caused losses estimated at Naira 2.6 trillion (USD 17.3 billion) (CBA, 2025), were a national wake-up call. Similarly, the 2022 flood displaced 1.4 million people, destroyed over 400,000 hectares of farmland, and led to significant public health emergencies, including a cholera outbreak. Nigeria developed NASPA-CCN (2011) and NAPF (2020); however, a lack of funds stalled the implementation of the prescribed adaptation measures, and the level of coordinated national adaptation action required was not achieved. Financing was identified as a critical enabler, with explicit recognition of the importance of mobilising both domestic resources and international climate finance. Nigeria's NAP was developed to address the deficiencies of past frameworks and to provide a more robust plan with the potential to attract investors.

6.2 NAP Financing

The Government of Nigeria has committed to achieving NDC 3.0 by developing adaptation policies and solutions but needs to demonstrate strong political will by devoting significant annual budgetary allocations to the implementation of climate actions. It is estimated that between 2013 and 2020, the Government of Nigeria spent USD 88 million to finance climate investments and actions, and between 2017 and 2022, green bond issuance raised 165 million USD as a financing tool to support national projects in the environment, agriculture, power, and energy-efficient transportation (Onyimadu and Uche, 2021). Whilst Nigeria estimates it will plough in USD 17.7 billion annually to meet its conditional NDC commitments between 2021 and 2030 (FMEnv, 2021), no share of this fund was specifically allocated to adaptation measures. Other reports project that the Government of Nigeria could conservatively spend USD 12 billion per annum on adaptation and



Three-tier coins representing investment growth (Vecteezy.com)

resilience actions, amounting to 120 billion USD by 2030; though this adaptation figure is considered underestimated, as it is not based on actual sectoral actions (NATCCC, 2023; FGN, 2021). Nigeria will accelerate the operationalisation of the Climate Change Fund, which consists of payments to the government from carbon taxes, emissions trading schemes, appropriations from the National Assembly, funding from international organisations, fines issued to private and public entities, subventions, grants, etc., as contained in the Climate Change Act, 2021. The Climate Change Fund is intended primarily for both mitigation and adaptation initiatives to meet the NDC targets. Other funding sources for adaptation and resilience initiatives include the Development of Natural Resources Fund (DNRF), the Ecological Fund Office (EFO), Great Green Wall funding, the Clean Technology Investment Fund (CTF), and the Forestry Trust Fund, Nigeria's Green Bond Market, the private sector, and development partners such as the World Bank Group, the French Development Agency (AFD), and the African Development Bank Group, etc.

6.3 Current Sources of Adaptation Funding

The major sources of climate finance are domestic and international (multilateral) development banks and bilateral donors. In 2019/2020, USD 1.9 billion of public and private capital was invested in climate-related activities in Nigeria. This is only 11% of the estimated USD 17.7 billion needed annually to meet the conditional NDC target of reducing emissions by 47% relative to the Business-As-Usual (BAU) scenario. Despite being the third-largest recipient of climate finance on the continent (behind Egypt and Morocco), the tracked USD 1.9 billion of climate finance flowing to and within Nigeria is meagre compared to the size of the country's economy and the opportunities for low-carbon development. Climate finance flowing into

Nigeria was largely from the public sector (USD 1.5 billion, 77%), while the private sector lagged behind (USD 0.4 billion, 23%). Multilateral development banks were the largest source of climate finance, contributing approximately 67% of public climate finance. In 2021/22, MDBs channelled about USD 1.2 billion in climate finance to Nigeria. Approximately 83% of adaptation funding is delivered through debt instruments. Bilateral donors primarily provide grants directed towards policy support, national budget support, and capacity building (CBA, 2025).

The issuance of sovereign green bonds is a domestic mechanism for financing climate adaptation and mitigation projects in Nigeria. Proceeds from green bonds were allocated to renewable energy, afforestation, and micro-utilities. Nigeria may have made progress in securing climate adaptation finance, but current flows remain grossly inadequate to meet the country's adaptation needs. Increased international support, diversified financial instruments, and enhanced domestic capacity are critical to effectively addressing climate change impacts.

6.4 Finance Gap for Adaptation Action

According to the CBA (2025), by 2050, Nigeria needs to spend ₦18.3 trillion (USD 12.4 billion) to implement all 5 priority strategies (flood management in agriculture; heat-impact management in agriculture; cross-cutting agriculture, strengthening ecosystems and insurance systems; water supply augmentation alongside flood protection systems; and energy sector adaptation). These investments are highly profitable, as every Naira invested generates at least ₦2.3 in benefits, with BCRs of 2.32-2.41 and NPVs of USD 10.4-11.3 billion (Annex 3).

In 2019/2020, Nigeria invested an average of USD 1.9 billion per year in climate-related activities from public and private capital, indicating that only 11% of the required finance was met, given that the 2021 NDC estimated that USD 17.7 billion was needed annually to meet the conditional NDC targets of reducing emissions by 47% below business-as-usual by 2030.

Therefore, Nigeria needs a stronger financial mobilisation from both domestic and international sources in order to meet the expected NDC adaptation targets.

6.4.1 Past Financing Pattern

Climate finance to Nigeria was largely provided by public actors (USD 1.5 billion, 77%), while the private sector lagged (USD 0.4 billion, 23%). Public climate finance in Nigeria was primarily committed via Multilateral Development Finance Institutions (DFIs) (55%), followed by Bilateral DFIs (20%) and governments (19%), with the majority channelled as debt. Private sector investment significantly lags behind public investment, accounting for 23% of total climate finance committed

in 2019/2020. Private finance was largely split between equity (USD 226 million, 52%) and debt (USD 174 million, 40%), with institutional investors providing only grant-based finance (USD 21 million) (CFF, 2025). Private actors in the Nigerian climate finance ecosystem are focusing most of their investments on Energy Systems (63%), largely using equity-based financing. Evidently, Nigeria has accessed a very meagre amount of international public climate finance to date and continues to face significant challenges in accessing international funds because of the complexity of the application and approval process and the need to mediate access to international climate finance through an MDB, with Nigeria previously having no accredited entity.

6.4.2 NAP Gap Funding Strategy

Nigeria offers a profitable investment environment for adaptation, with confirmed positive returns from all five adaptation strategies, as evidenced by Benefit-Cost Ratios (BCRs) above 2 and strongly positive Net Present Values (NPVs) across both the 2050 and 2080 horizons (CBA, 2025). For a clearer investment strategy, the elements of the adaptation strategies that can be privately funded are separated from those that should be funded from the public purse, given their public goods nature.

6.4.2.1 Flood Management in Agriculture

Flood management in agriculture investments which includes flood-resistant ranching, community radios, and drainage infrastructure etc. represents a low-cost, high-return adaptation strategy. The government of Nigeria is required to play a catalytic role by initiating investments and reducing risks in a portfolio of complementary financial instruments because of the market failure problem associated with this adaptation strategy. Flood management in Agriculture will require Public Investment (Federal & State Budgets) of ₦40 billion (~$27m, ~37%) to cover the high fixed costs of infrastructure and to create public goods; Investment from International Climate Finance & Donors (GCF, GEF, Adaptation Fund, MDBs) of ₦30 billion (~$20m, ~28%) to bridge the funding gap for large upfront investments; Private Sector & Insurance will make (Domestic Banks, Agribusiness, Insurers) of ₦25 billion (~$17m, ~23%) to complements infrastructure investment by covering residual risks; Blended Finance & Impact Investors of ₦13 billion (~$9m, ~12%) to enable allow risk-sharing between the state, donors, and private investors (Table 6.1). The total funding secured for the flood management strategy amounts to ₦108 billion (approximately $72 million)

Adaptation Options	Financing Approaches	Actors and Amounts (₦)
Flood management and control systems (flood resistant animal ranching (prevention), community radios for communications for flood management) Post-flood livestock relocation and recovery (proposal of flood resistant animal ranching (post-flood))	• Public Investment & Budgeting o Allocate dedicated budget lines for flood-control infrastructure (dikes, drainage, embankments). o Integrate agricultural flood resilience in national disaster risk reduction budgets.	40% – catalytic financing for flood-resistant ranching and community early warning - ₦43.2 –
	• Climate Finance & Donor Support o Tap Green Climate Fund (GCF), Adaptation Fund, and Global Environment Facility (GEF) for climate-proofing irrigation and rural roads.	30% – GCF, GEF, Adaptation Fund for infrastructure and capacity-building - ₦32.4
	• Private sector & Insurance & Risk Transfer o Expand index-based flood insurance for farmers (subsidized premiums via government + private sector co-financing). o Catastrophe bonds and sovereign risk pools (e.g., African Risk Capacity).	20% – PPPs for storage facilities and markets; impact investors for flood-tolerant crops- ₦21.6
	• Blended Finance & Private Capital o PPPs for flood-resilient storage facilities and market centers. o Impact investors in climate-smart agriculture targeting flood-resilient seed varieties.	10% – index-based flood insurance, catastrophe bonds - ₦10.8

Table 6.1: Financing Approaches for Flood Management in Agriculture Adapted from CFF (2025)

6.4.2.2 Heat Impacts Management in Agriculture

Heat impacts management in agriculture is a capital-intensive but transformative adaptation strategy with a strong pay-off, a BCR of 2.4 and a positive NPV of $861–931 million. Nigeria will adopt a layered financing approach with the participation of different actors because of the high upfront CAPEX and technology and knowledge gaps. This requires $900 million (₦1.35 trillion) in total financing (2026–2050), with the Public Sector investing $270 million (30%); International Climate Finance: $250 million (28%); Credit & Microfinance: $180 million (20%); CSR & Private Sector: $120 million (13%); Carbon Finance & Impact: $80 million (9%) (Table 6.2).

Adaptation Options	Financing Approaches	Actors and Amounts (~~NB~~)
1) Comprehensive Livestock Cooling and Climate management coming from the proposal of shed and water reservoirs for (animal husbandry) 2) Post-harvest Cooling and Storage Systems (for crops production) 3) Irrigation Expansion (for crops production 4) Climate-resilient crops (heat resistant crops) 5) Comprehensive fish and aquaculture climate adaptation (coming from the proposal of heat action for fishery sector))	• R&D and Innovation Funding o Establish national agricultural innovation funds for heat-tolerant crop/livestock breeds. o Use venture capital to finance agri-tech startups offering heat-resilient solutions.	20% - ~~₦~~286.5 R&D, extension support for climate-resilient crops/ livestock
	• Credit & Microfinance o Expand concessional agricultural credit for adoption of shade nets, cooling systems, and drip irrigation. o Target women and smallholder farmers with micro-adaptation loans.	35% -GCF, GEF, Adaptation Fund Grants and concessional finance for irrigation, aquaculture, post-harvest cooling -~~₦~~500.875
	• CSR & Private Sector Engagement o Agribusinesses finance farmer capacity-building and adaptation trials under sustainability commitments.	25% – agribusiness financing; venture capital in agri-tech startups - ~~₦~~358.125
	• Carbon Finance o Monetize avoided emissions from energy-efficient irrigation and soil conservation practices.	20% –monetization of avoided emissions from energy-efficient irrigation and cooling ~~₦~~286.5

Table 6.2: Financing Approaches for Heat Impacts Management in Agriculture

6.4.2.3 Cross-cutting Agriculture

Cross-cutting agriculture, as a multiple-hazard response strategy, stands out as the most efficient agricultural adaptation option for Nigeria, addressing systemic vulnerabilities in the agricultural sector and delivering direct economic returns (reduced losses, stabilised incomes) and social co-benefits (reduced migration, gender empowerment, improved ecosystems). Nigeria will adopt a layered, complementary approach to this strategy, engaging the private sector alongside state leadership and donor support, given the systemic and collective nature of the interventions. Between 2026 and 2050, of the USD 1.1–1.2 billion required, the state will provide 35%, anchored through budget tagging and insurance co-financing; donors will fund systemic, multi-year programmes via a pooled trust fund (40%); the private sector will focus on delivery, insurance expansion, and telco partnerships (15%); and innovative finance will build long-term sustainability through blockchain traceability and carbon credits (10%) (Table 6.3).

Adaptation Options	Financing Approaches	Actors and Amounts (~~NB~~)
1) Smart agriculture production management 2) Agriculture insurance	• **Mainstreaming in National Budgets** o Introduce "climate budget tagging" in agricultural ministries to ensure allocation to adaptation projects.	10% (~~₦~~21.3)– budget tagging, guarantees, catalytic subsidies for insuranc
	• **Programmatic Financing** o Pool resources through multi-donor trust funds for climate-resilient agriculture.	40% (~~₦~~85.2)– multi-donor trust fund for ecosystem management, insurance, early warning, gender program IFAD (rural resilience)-
	• **Equity & Gender Financing** o Use gender-responsive climate finance to ensure women, youth, and vulnerable groups benefit equitably.	30% (~~₦~~63.9)– PPPs and concessional loans for smart agriculture, digital farmer registries
	• **Digital Finance & Data Systems** o Finance early warning systems and climate information services via blended donor and telco partnerships.	5% (~~₦~~10.65)– revolving funds for O&M and local stewardship
	• **Innovative Financing** o Use blockchain-based traceability for climate-smart food supply chains that attract green premiums.	5% (~~₦~~10.65) – blockchain traceability, carbon credits

Table 6.3: Financing Approaches for Cross-cutting Agriculture Strategy

6.4.2.4 Water supply augmentation associated with Flood protection systems

Nigeria faces a dual water challenge of recurring floods that destroy infrastructure, farmland, and other assets, and chronic water scarcity that threatens health, agriculture, and the energy system. The Federal and Local governments, acting as catalysts and anchors, will invest in water supply augmentation, which has strong social dividends (BCR ~3.2; NPV ~USD 1 billion), and in flood protection systems, which are a quick-win strategy with very high efficiency (BCR ~5.5–5.7; NPV USD 237–253 million), to tackle these challenges. Because the 2 strategies are largely public goods with diffuse benefits and markets alone are unlikely to mobilise the necessary investment, the State, with international support, will invest in them and later be assisted by private and community participation. Until 2050, a total investment of USD 1.2–1.3 billion is required for the 2 strategies, with annual operation and maintenance costs of around USD 100–120 million (Table 6.4).

Adaptation Options	Financing Approaches	Actors and Amounts (~~NB~~)
1) Water Supply Augmentation (Rainwater harvesting, Groundwater development, Water treatment and recycling) 2) Flood Protection & Demand Management (Flood-resistant water infrastructure, Early warning systems, Emergency response protocols)	**Public Infrastructure Finance** o Sovereign green bonds for irrigation modernization, small dams, and catchment restoration. o Local government climate funds for watershed protection.	20% (₦219.425)– green bonds, local government climate funds, anchor public infrastructure
	Payments for Ecosystem Services (PES) o Farmers paid for watershed management, financed by utilities and beverage companies.	10% (₦109.7) – PES for watershed management
	Private Sector Participation o PPPs in rural water distribution, desalination, and wastewater reuse projects.	25% (₦274.3) – rural water distribution, desalination, wastewater reuse
	International Climate Funds o Leverage GCF/GEF projects for integrated water resource management (IWRM)	40% (₦438.85)– concessional finance for IWRM, flood early warning, emergency response
	Community Finance o Water user associations contribute revolving funds for maintenance of local water systems.	5% (₦54.85)– revolving funds for maintenance and ownership

Table 6.4: Financing Approaches for Water Supply Augmentation Associated with Flood Protection Systems

6.4.2.5 Energy Sector Adaptation

Nigeria's energy sector needs investment to climate-proof the grids, expand rural electrification, enhance agricultural productivity, build cold storage, and promote industrial growth as measures to reduce vulnerability to climate extremes. Strategic adaptation investments in the energy sector are **capital-intensive but offer high absolute returns (NPV of USD 7.8–8.5 billion), with a proportion of them profitable for the private sector. This requires that, between 2026 and 2050, a total of USD** 10–11 billion, including O&M, be mobilised using blended finance to maximise the mobilisation of both private capital and concessional funding (Table 6.5).

Adaptation Options	Financing Approaches/ Instruments	Actors and Amounts (~~NB~~)
1) Renewable Energy Expansion (solar, wind) 2) Grid Hardening 3) Energy Storage & Backup System and Modernization	**Renewable Energy Investments** o Issue sovereign green bonds and sukuk for solar-powered irrigation and cold storage. o Crowd-funding for mini-grids and off-grid solar for rural agribusinesses.	State 10–15% (~~₦~~1,403.75) – green bonds/sukuk; regulatory incentives; catalytic anchor
	Blended Finance o Use concessional donor funds to de-risk private investment in rural energy infrastructure.	30–35% (~~₦~~4,211.25) – concessional funding, first-loss protection, guarantees
	Corporate Investment o Energy companies commit to adaptation as part of CSR/ESG frameworks (e.g., financing resilient energy access in rural areas).	20–25% (~~₦~~2,807.5) – IPPs, energy utilities, CSR/ESG financing
	Carbon Markets o Register solar irrigation and biogas digesters under Article 6 carbon trading mechanisms.	30–35% (~~₦~~4,211.25) – blended finance (concessional + equity; guarantee-backed debt; equity co-investment);
	Insurance & Risk Hedging o Energy utilities hedge against climate-induced supply disruptions via climate risk insurance.	5% carbon & insurance

Table 6.5: Financing Approaches for Energy Sector Adaptation Strategies

6.4.3 Potential Financing Sources and Structure for Adaptation Actions

Nigeria's NAP goals can only be realised by aligning adaptation aspirations with the vigorous mobilisation of funds from specific sources. This is achievable, given that the priority strategies have strong evidence of economic viability and offer resilience dividends, as shown in the CBA (2025). Climate finance sources can be classified into 3 funding sources.

6.4.3.1 Public Finance

Public finance comprises financial resources raised through taxes and other government revenue streams for climate change projects. International climate finance comprises funds provided by international multilateral or bilateral entities, including multilateral development banks (MDBs) such as the GCF, GEF, Adaptation Fund, AfDB, and the World Bank. Private finance comprises funds raised from

the sector of the economy not controlled by the state, including individuals, co-operatives, corporations, investors, philanthropy, and financial institutions. Blended finance primarily refers to different patterns of mixing state, private, and international climate finance.

With the overall ₦18.3 trillion (≈ USD 12.4 b) required to finance the 5 highlighted adaptation strategies by 2050, the following sources and proportions of the financing arrangement are effective in advancing the NDC 3.0 targets within a diversified and layered architecture:

Public finance (25–35 %) — national and state budgets, climate-budget tagging, sovereign green bonds, and dedicated adaptation lines.

International climate finance (30–40 %) — GCF, GEF, Adaptation Fund, AfDB, World Bank, and bilateral donors, focusing on de-risking and capacity building.

Private & blended finance (20–30 %) — PPPs, concessional credit, impact investment, insurance pools, and agribusiness engagement.

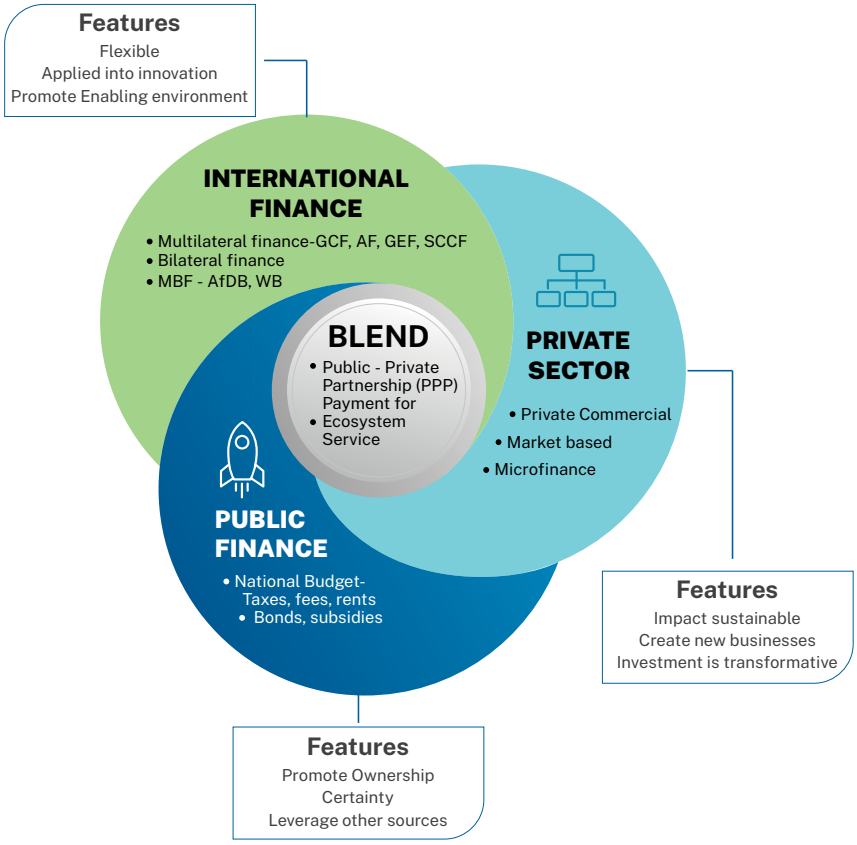


Figure 6.1: Potential Sources of NAP Finance and their Benefits

Innovative mechanisms (5–10 %) — carbon markets, blockchain traceability, PES, and community revolving funds — to ensure sustainability and inclusion. Nigeria's adaptation finance landscape is moving toward a strategic investment framework that combines public leadership, donor partnerships, and private-sector innovation.

6.4.4 Multilateral Climate Finance Sources Open to Nigeria NAP and their Benefits

Nigeria has not benefited commensurately from multilateral climate financing entities, given the available opportunities and the country's compelling level of vulnerability. The UNFCCC and a number of multilateral organisations have financial mechanisms that the country can access to fund investments related to NAP priority strategies. Such financial entities include:

The Global Climate Facility (GCF) promotes a paradigm shift towards low-carbon and climate-resilient development pathways in developing countries. It provides finance in the form of grants, loans, concessional loans, equity, guarantees, and results-based finance of up to USD 250 million to Nigeria through accredited access entities such as the Development Bank of Nigeria (DBN), UNDP, and UNEP. Proposals to the GCF require the endorsement of the Nigeria Designated National Authority (DNA), and importantly, projects must have a significant co-financing component.

The Adaptation Fund (AF) finances adaptation projects and programmes in developing countries to reduce the adverse impacts of climate change. Nigeria can access AF financing through the national implementing entity (NIE), the Federal Ministry of Environment. The AF may finance adaptation projects worth 20 million USD in Nigeria.

The Global Environment Facility (GEF) supports the implementation of multilateral environmental agreements in developing countries. Nigeria can access finance from GEF to fund its large- and small-grant projects addressing challenges in areas such as biodiversity, land degradation and climate change through the System for Transparent Allocation of Resources (STAR). GEF also offers non-grant instruments such as debt, equity and de-risking. The GEF Operational Focal Point is in the Federal Ministry of Environment.

Loss and Damage Fund: Established in 2022 under the Paris Agreement it is for financing project that address Loss and Damage associated with adverse impacts of climate change.

Africa Adaptation Accelerator Programme (AAAP) is an programme financed by both the African Development Bank AfDB) and the Global Centre on Adaptation (GCA).. It has upstream financing opportunity executed by GCA and the downstream financing executed by AfDB. The goal is to mobilise USD 25 billion by 2026 to accelerate and upscale climate adaptation action across the

Nigeria can transform its adaptation agenda into a catalyst for economic resilience, green jobs, and sustainable growth by institutionalising climate frameworks and capacity development, leveraging available finance mechanisms, scaling blended models, and taking advantage of strong cost–benefit actions.



Chapter Seven

Monitoring and Evaluation

7.1 Purpose of Monitoring & Evaluation

The purpose of monitoring and evaluation is to strengthen the capacity of the implementing agency, track progress in implementing adaptation interventions, and evaluate their effectiveness. The specific objectives, as described by FAO (2018), include: improving learning and understanding of adaptation practices across all sectors; enhancing adaptive management of actions in the face of climate uncertainty; ensuring that adaptation efforts are aligned with intended goals and objectives; enhancing accountability to national authorities and funding partners; meeting both national and international reporting obligations; and providing evidence to support adaptation financing and communicating adaptation priorities effectively to policymakers, stakeholders, and decision-makers.

7.2 Adaptation Monitoring and Evaluation Framework

The specific purpose of NAP's monitoring and evaluation (M&E) is to track iteratively the progress of NAP interventions and to determine how the plan's goals and activities are reducing vulnerability, improving adaptive capacity, and supporting the overall well-being of the people affected by the impacts of climate change (FMEnv, 2020). NAPF (2020) allowed the Ministry of Budget and National Planning (MBNP) and the National Bureau for Statistics (NBS) to undertake the monitoring and evaluation of NAP progress, as they have the constitutional mandate. Although MBNP and NBS have constitutional mandates to track and deliver on the M&E for government policies and programmes in Nigeria, total reliance on the two institutions for monitoring adaptation actions and processes may yield limited results because of their overloaded responsibilities. Fortunately, the Climate Change Act of 2021 strengthened this weak provision in NAPF by mandating, in Section 20(1), that "the Secretariat shall formulate a National Climate Change Action Plan every five-year cycle and subsequently direct the Director-General of the Council to submit to the National Assembly Committees on Climate Change a detailed report on the state of the nation with regard to climate change". Again, in Section 21(3), the Director-General is required, "within three months after the end of every financial year, to publish and submit to the National Assembly an evaluation report on the performance of climate change duties by private and public entities".

The provisions of the Climate Change Act 2021 mark a transformation in Nigeria's climate change governance, demanding greater accountability and transparency. The reporting expected from the DG of the Council to the National Assembly Committee on Climate Change represents an improvement that targets the timely achievement of monitoring and evaluation outcomes. The Act stipulates concrete steps for realising the significance of the M&E framework by introducing flexible management of adaptation actions under climate uncertainty; ensuring validation of adaptation processes and outcomes, as well as tracking the achievement of stated goals; measuring changes in the levels of vulnerability to climate change; and measuring adaptation processes and tracking changes in human and institutional capacity.

7.3: Monitoring and Evaluation Approach

A specific M&E framework was developed to strengthen tracking and assessment and improve the effectiveness of Nigeria's NAP strategies across the key thematic areas. Nigeria's NAP adopts the Results-Based Monitoring (RBM) with the Logical Framework approach to track progress in its implementation. The M&E framework covers all interventions under the core adaptation strategies identified in the NAP Readiness Project in Nigeria and the CRA framework (2024/25), which include: (i) Flooding Adaptation Strategies, (ii) Heat Stress/Heatwaves Adaptation Strategies, and (iii) Multi-Risk/Cross-Cutting Adaptation Strategies. With the NAP Readiness Project in Nigeria and the Climate Risk Assessment (CRA) already in place, the M&E sets out a Strategic Results Framework that describes goals, expected impact, outcomes, and outputs, as well as indicators and targets.

7.4: The National Adaptation Plan Strategic Monitoring Framework

Table 7.1 shows the Logical Framework Analysis for monitoring and evaluating the strategic priorities identified in the Economic Appraisal 2025.

Strategy	Strategic Options	Outcome	Strategic Objectives	targets
	Flood Management and Control Systems 1. Flood Management and Control Systems	Strengthened local capacity for flood prevention, preparedness, and response through flood-resistant animal ranching and community radio systems	To strengthened local capacity for flood prevention, preparedness, and response through flood-resistant animal ranching and community radio systems	100 flood-prone communities have infrastructure installed or rehabilitated 5,000+ livestock farms/ranches retrofitted or relocated to safe zones 0+ LGAs equipped with solar-powered community radios and trained broadcasters 70% of targeted communities trained and engaged in flood response
	Post-Flood Livestock Relocation and Recovery	Improved post-flood response for livestock keepers through safe relocation, restocking, and resilient ranching infrastructure	To Improve post-flood response for livestock keepers through safe relocation, restocking, and resilient ranching infrastructure	100+ relocation zones across flood-affected LGAs 5,000+ improved flood-resistant shelters built ≥500,000 animals supported annually after ≥50,000 herders trained in 6 geopolitical zones

Heat impacts management Agricultural Adaptation Strategy	Comprehensive Livestock Cooling and Climate Management	Increased availability and use of livestock cooling shelters and water reservoirs and heat-resilient livestock housing	To Increase availability and use of livestock cooling shelters and water reservoirs and heat-resilient livestock housing	≥5,000 units installed across 6 zones≥50,000 livestock shelters climate-adapted 10,000+ livestock units with improved water access systems 100,000 farmers and vet staff trained
	Post-harvest Cooling and Storage Systems (for Crops Production)	Reduce climate-related post-harvest crop losses and enhance food security and farmer incomes through climate-resilient cold storage systems	To reduce climate-related post-harvest crop losses and enhance food security and farmer incomes	≥5,000 units with 100,000 MT capacity installed in rural LGAs ≥50% of systems accessible to women/youth groups 100,000 farmers trained in storage & handling ≥60% of stored produce sold fresh or processed
	Irrigation Expansion (for Crops Production)	Expand, modernize, and promote equitable access to sustainable irrigation technologies among smallholders and cooperatives	To expand, modernize, and promote equitable access to sustainable irrigation technologies among smallholders and cooperatives	≥2,000 community-based schemes established or rehabilitated 100,000+ efficient units installed (drip, solar, sprinkler) ≥500,000 farmers trained across 6 geopolitical zones ≥40% of beneficiaries are women/youth
	Climate-Resilient Crops (Heat-Resistant Varieties)	Promote research, development, and adoption of heat-resistant crop varieties suitable for Nigeria's agroecological zones.	To promote research, development, and adoption of heat-resistant crop varieties suitable for Nigeria's agroecological zones.	≥15 new varieties (e.g., maize, rice, millet, cowpea, sorghum, tomato) released ≥10,000 agro-dealers/coops stocked with certified resilient seeds ≥2 million farmers trained (including 40% women/youth) ≥70% of target communities reached through radio, SMS, demo plots
	Comprehensive Fish & Aquaculture Climate Adaptation (Heat Action)	Develop and implement heat-resilient aquaculture technologies, water management practices, and capacity-building for fish farmers	To develop and implement heat-resilient aquaculture technologies, water management practices,	≥3,000 ponds upgraded with cooling & aeration systems ≥1 million heat-resilient fingerlings distributed ≥1,500 farms regularly monitoring and managing water quality ≥20,000 fish farmers trained across zones ≥50 communities reached with early warnings

Cross-cutting Agriculture Adaptation Strategy	Smart Agriculture Production Management	Adopt and scale smart agriculture production management systems that leverage climate information, precision farming, and digital tools.	To adopt and scale smart agriculture production management systems that leverage climate information, precision farming, and digital tools.	≥500,000 farmers regularly use tailored climate advisories ≥200,000 hectares under precision management ≥500,000 active users on digital agri platforms ≥1 million farmers trained ≥40% women and youth actively engaged
	Agriculture Insurance (for Climate Adaptation)	Promote affordable, accessible, and climate-responsive agricultural insurance schemes for smallholders and vulnerable farming groups.	To promote affordable, accessible, and climate-responsive agricultural insurance schemes for smallholders and vulnerable farming groups.	≥5 new climate-risk insurance products piloted PPP frameworks operational in ≥20 states ≥3 million farmers reached through awareness campaigns ≥1 million farmers access bundled input-insurance packages Subsidy schemes in place in ≥20 states ≥60% subsidy coverage for smallholders
Water Resource Management Adaptation Strategy	Water Supply Augmentation: Rainwater harvesting, Groundwater development, Water treatment & Recycling	Implement water supply augmentation systems including rainwater harvesting, sustainable groundwater development, treatment and reuse to support climate adaptation in high-risk areas	To implement water supply augmentation systems including rainwater harvesting, sustainable groundwater development, treatment and reuse to support climate adaptation in high-risk areas	≥500,000 rainwater systems operational (urban & rural) ≥3,000 new boreholes with solar pumps ≥500 recharge trenches/pits built ≥3,000 new boreholes with solar pumps ≥500 recharge trenches/pits built ≥1,000 small-scale treatment/ recycling units functional ≥1 million people reached through awareness campaigns ≥30 states with water augmentation plans Inter-agency coordination platform operational
	Flood Protection & Demand Management (Flood-resistant water infrastructure, Early warning systems, Emergency response protocols)	Strengthen flood protection infrastructure, early warning, and emergency management systems in flood-prone regions	To strengthen flood protection infrastructure, early warning, and emergency management systems in flood-prone regions	≥1,500 km of drainage channels upgraded ≥300 flood defenses constructed ≥5,000 communities with real-time alerts ≥80% response compliance ≥500 LGAs with trained emergency responders Response time reduced by 50% ≥774 LGAs with GIS-based flood risk maps Updated water demand plans ≥10 million people reached ≥1 million participate in drills
Energy Sector Adaptation Strategy	Renewable Energy Expansion for Climate-Resilient Development (Solar and Wind System	Scale up decentralized solar and wind energy systems for households, agriculture, health centers, and climate-sensitive sectors.	To reduce national dependence on GHG-generating fossil fuels	≥2,000 mini-grids / 10,000 standalone systems deployed ≥100 wind turbines installed; ≥100 MW generated ≥5,000 farms use RE-based systems ≥30% reduction in post-harvest loss ≥1,500 health and education facilities powered by RE ≥5,000 technicians trained ≥1,000 RE MSMEs supported

	Grid Hardening and Modernization	Modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs	To modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs	≥1,000 km of lines upgraded ≥100 substations climate-proofed ≥60% grid equipped with real-time diagnostics and automated control ≥300 microgrids installed in flood/drought-prone areas ≥5,000 personnel trained Resilience protocols adopted by all major DisCos and GenCos New grid resilience standards adopted Dedicated finance lines established
	Energy Storage & Backup Systems (for Climate-Resilient Energy Access and Emergency Preparedness)	Deploy decentralized energy storage (batteries, inverters) and clean backup systems (solar/gas) for critical infrastructure and climate-vulnerable communities	To deploy decentralized energy storage (batteries, inverters) and clean backup systems (solar/gas) for critical infrastructure and climate-vulnerable communities	≥15,000 storage systems installed in critical institution ≥2,000 communities with backup for flood, heat, and disaster response ≥500,000 households use solar + battery storage systems ≥80% of mini-grids integrated with storage (e.g., Li-ion, lead-acid) ≥5,000 technicians trained across all geopolitical zones

Table 7.1: Logical Framework for the Implementation of Adaptation Priority Options

7.5 Sector Based

The logical frameworks for the implementation of adaptation priorities in various sectors are given in Tables 7.2 to 7.14.

Sector	Outcome	Output	Indicators
Agriculture	Strengthened local capacity for flood prevention, preparedness, and response through flood-resistant animal ranching and community radio systems	1. Flood management infrastructure (e.g., levees, drainage channels) constructed or rehabilitated	# of LGAs with at least one functional flood control system
		2. Flood-resistant animal ranching systems implemented	# of ranches relocated or retrofitted # of animals protected
		3. Community-based flood early warning & radio systems installed	# of LGAs with functional community radio stations for flood alerts
		4. Stakeholder awareness and preparedness improved	% of target population demonstrating flood preparedness (e.g., drills, livestock evacuation plans)

Means of Verification	Responsibility	Assumptions and Risks
LG project files, satellite images, engineering verification	MoLD, FMEnv; State Govt; LGAs; Development Patners;	Infrastructure is maintained post-project
Livestock ministry reports, veterinary services, farmer surveys	MoLD, FMEnv; NEMA; Organised Private sector (OPS) ;	Land availability and willingness of ranchers to relocate
NIMET, community media orgs, broadcast logs	MoLD, FMCo, NCC; NiMET; FMEnv; Organised Private sector (OPS) ; NGOs, CBOs Development Patners; research Institutes/ centres, Radio Statio	Local staff retained, radios maintained, power access ensured
Training reports, drills evaluation, community feedback	MoLD, NEMA, NGOs, CBOs Development Patners; OPS; Radio Stations	Consistent engagement by local government/NGOs

Table 7.2 Logical framework for implementation of Adaptation priorities in Agriculture - Strengthened local capacity for flood prevention, preparedness, and response

Sector	Outcome	Output	Indicators
Agriculture	Improved post-flood response for livestock keepers through safe relocation, restocking, and resilient ranching infrastructure	1. Temporary and permanent relocation zones established for displaced livestock and herders	# of temporary relocation camps and permanent safe zones established
		2. Flood-resistant animal shelters constructed or rehabilitated	# of shelters built or upgraded (elevated pens, drainage systems)
		3. Veterinary and fodder support provided post-flood	# of animals vaccinated, treated, or provided feed
		4. Training and sensitization on climate-resilient animal husbandry	# of herders trained on flood risk, shelter design, disease control

Means of Verification	Responsibility	Assumptions and Risks
Site inspection, GIS mapping, LGA reports	MoLD, State Govt; LGAs; Development Patners;	Availability of safe, secure land for relocation
Engineering reports, photographic evidence, local monitoring	MoLD, FMEnv; State Govt; LGAs; Development Patners;	Materials and technical capacity available

Vet reports, feed distribution logs, FMARD data	MoLD, and the Line Agencies Development Patners;	Timely procurement and vet workforce availability
Training attendance, KAP surveys	MoLD, and the Line Agencies; NEMA; Development Patners;	Willingness of herders to participate in trainings

Table 7.3 Logical framework for implementation of Adaptation priorities in Agriculture - Improved post-flood response for livestock keepers

Sector	Outcome	Output	Indicators
Agriculture	Increased availability and use of livestock cooling shelters and water reservoirs and heat-resilient livestock housing	1. Solar-powered fans, misting and ventilation systems installed in livestock farms (esp. poultry, dairy)	# of systems installed Total animal units covered
		2. Heat-resilient shelters (e.g., reflective roofing, thatch sheds, airflow-friendly designs) constructed or upgraded	# of new/upgraded shelters % of shelters with passive cooling design
		3. Water access and hydration stations improved for livestock in heat-prone zones	# of hydration stations and rainwater systems % increase in water access
		4. Farmer and veterinary training in livestock heat stress management and early response	# of stakeholders trained Awareness score pre/post training
		1. Solar-powered fans, misting and ventilation systems installed in livestock farms (esp. poultry, dairy)	% of target population demonstrating flood preparedness (e.g., drills, livestock evacuation plans)

Means of Verification	Responsibility	Assumptions and Risks
Installer reports, solar vendor data, farm logs	MoLD, and the Line Agencies Development Patners;	Technical systems are well maintained
Site verification, photo records, LGA reports	MoLD, and the Line Agencies Development Patners;	Local builders and materials are available
Field assessment, livestock owner feedback	MoLD, and the Line Agencies Development Patners;	Water supply infrastructure remains functional
Attendance logs, test scores, training evaluations	MoLD, and the Line Agencies; NEMA Development Patners;	Training delivery not interrupted by security or logistical issues

Table 7.4 Logical framework for implementation of Adaptation priorities in Agriculture - Increased availability and use of livestock cooling shelters and water reservoirs and heat-resilient livestock housing

Sector	Outcome	Output	Indicators
Agriculture	Reduce climate-related post-harvest crop losses and enhance food security and farmer incomes through climate-resilient cold storage systems	1. Solar-powered and hybrid cold rooms and coolers installed in priority agricultural zones	# of systems installed, total cooling capacity (MT)
		2. Increased access to storage for women/ youth farmers and cooperatives	% of users from vulnerable groups (women, youth)
		3. Capacity of farmers and cooperatives built on storage best practices	# of farmers trained
	Reduce climate-related post-harvest crop losses and enhance food security and farmer incomes through climate-resilient cold storage systems	4. Market linkages improved through better storage infrastructure	% increase in produce reaching market in good condition

Means of Verification	Responsibility	Assumptions and Risks
Contractor reports, user feedback, project audits	MoLD, and the Line Agencies Development Patners;	Local manufacturers/ partners available
Farmer registries, cooperative records	MoLD, and the Line Agencies Development Patners; NGOs & CSOs	Equity-based targeting enforced
Training reports, attendance, knowledge assessments	MoLD, and the Line Agencies; CSOs; Development Patners;	Extension systems remain active
Market surveys, trader feedback, price tracking	MoLD, and the Line Agencies LGAs	Market demand and access remain stable

Table 7.5 :Logical framework for implementation of Adaptation priorities in Agriculture - Reduce climate-related post-harvest crop losses and enhance food security and farmer incomes

Sector	Outcome	Output	Indicators
Agriculture	Expand, modernize, and promote equitable access to sustainable irrigation technologies among smallholders and cooperatives	1. Small- and medium-scale irrigation infrastructure developed or rehabilitated	# of irrigation schemes installed/upgraded Hectares covered
		2. Water-efficient and climate-smart irrigation technologies promoted (drip, solar pumps)	# of solar/efficient systems deployed % reduction in water waste
		3. Institutional capacity and farmer training on irrigation use and maintenance strengthened	# of farmers trained % practicing proper irrigation scheduling
		4. Inclusive access ensured for women, youth and vulnerable farmers	% of irrigation beneficiaries who are women/youth

Means of Verification	Responsibility	Assumptions and Risks
Construction records, GIS, satellite verification	FMARD, FMWR; Development partners	O&M systems in place
Installer logs, tech provider records	FMARD, FMWR; REA; Development partners	Solar tech supply chain stable
Training records, extension reports	FMARD, FMWR; Development partners CSOs	Training is culturally appropriate and accessible
Farmer registry disaggregated by gender/age	FMARD, Development partners; CSOs FMW-Affairs	Gender-sensitive policy implementation sustained

Table 7.6: Logical framework for implementing adaptation priorities in agriculture - Expand, modernise, and promote equitable access to sustainable irrigation technologies among smallholders and cooperatives

Sector	Outcome	Output	Indicators
Agriculture	Promote research, development, and adoption of heat-resistant crop varieties suitable for Nigeria's agroecological zones.	1. Climate-resilient (heat- and drought-tolerant) crop varieties developed, tested and approved	# of new varieties developed & registered Time to market approval
		2. Certified seed multiplication and distribution systems strengthened	# of agro-dealers and cooperatives accessing improved seeds
		3. Farmers trained in climate-smart seed and agronomic practices	# of farmers trained % adopting recommended practices
		4. Public awareness campaigns on heat-tolerant crops and their benefits conducted	% increase in awareness among target communities

Means of Verification	Responsibility	Assumptions and Risks
IAR, IITA, NACGRAB, NASC, FMARD reports	FMARD, National Seed Service; Universities; Research Institutes	Research infrastructure functional; regulatory support consistent
NASC, seed company records, extension surveys	FMARD, National Seed Service; CSOs, Private sector	Private sector and seed companies involved
Training attendance logs, pre/post assessments	FMARD, National Seed Service; ADPs, CSOs	Extension systems are functional and inclusive
Communication campaign records, survey feedback	FMARD, National Seed Service; ADPs, CSOs; Radio Stations	Communication tailored to local languages and contexts

table 7.7: Logical framework for implementing adaptation priorities in agriculture - Promote research, development, and adoption of heat-resistant crop varieties suitable for Nigeria's agroecological zone

Sector	Outcome	Output	Indicators
Agriculture	Develop and implement heat-resilient aquaculture technologies, water management practices, and capacity-building for fish farmers.	1. Climate-resilient pond designs and water aeration/circulation systems installed	# of ponds upgraded with aerators, shading, cooling Reduction in pond water temperature fluctuations
		2. Heat-tolerant fish species and fingerlings developed and distributed	# of heat-tolerant fingerlings produced and distributed
		3. Water quality monitoring and management systems introduced (e.g., sensors, regular testing)	# of farms with monitoring systems % improvement in water quality parameters
		4. Training and awareness programs on climate-smart aquaculture and heat stress management conducted	# of ponds upgraded with aerators, shading, cooling Reduction in pond water temperature fluctuations
		5. Early warning and communication systems for heat stress and water crises established	# of communities/farmers receiving alerts Response rate

Means of Verification	Responsibility	Assumptions and Risks
Installation logs, field inspections, water temperature monitoring	FMARD, State Governments;	Technical systems maintained properly
Hatchery records, sales reports	FMARD, State Government	Breeding programs supported and funded
Lab test reports, farm logs	FMARD, State Government	Availability of affordable monitoring tech
Installation logs, field inspections, water temperature monitoring	FMARD, State Government	Training materials culturally appropriate
Communication logs, surveys	FMARD, State Government, CSOs, NiMET	Reliable communication infrastructure

Table 7.8: Logical framework for implementation of Adaptation priorities in Agriculture - Develop and implement heat-resilient aquaculture technologies, water management practices, and capacity-building for fish farmers.

Sector	Outcome	Output	Indicators
Agriculture	Adopt and scale smart agriculture production management systems that leverage climate information, precision farming, and digital tools.	1. Climate information services integrated into agricultural planning and decision-making	# of farmers accessing climate advisories Accuracy and timeliness of advisories
		2. Precision agriculture technologies (soil sensors, drones, satellite imagery) deployed and utilized	# of farms with precision tools % improvement in input efficiency
		3. Digital platforms and mobile apps for market info, pest alerts, and input supply established	# of active users Frequency of platform updates
		4. Farmer training and capacity building on smart agri-tech and climate-resilient practices	# of farmers trained % applying learned practices
		5. Mechanisms for gender- and youth-inclusive participation in smart agriculture developed	% of women and youth users of platforms and tech

Means of Verification	Responsibility	Assumptions and Risks
Platform user data, extension reports	FMARD, State Government	Meteorological services maintain accurate data
Tech deployment records, farm monitoring reports	FMARD, State Government	Technology affordable and reliable
Platform analytics, user surveys	FMARD, State Government NCC FMCo; Telcos	Mobile network coverage adequate
Training records, pre/post assessments	FMARD, State Government, CSOs;	Trainings are accessible and relevant
User demographics, platform analytics	FMARD, State Government NCC FMC; Telcos; CSOs	Gender-sensitive outreach maintained

Table 7.9 Logical framework for implementation of Adaptation priorities in Agriculture - Adopt and scale smart agriculture production management systems

Sector	Outcome	Output	Indicators
Agriculture	Promote affordable, accessible, and climate-responsive agricultural insurance schemes for smallholders and vulnerable farming groups	1. Climate-smart agriculture insurance products (index-based, weather-triggered, parametric) designed and piloted	# of new insurance products introduced % relevance to local risks
		2. Partnership frameworks established among insurers, government, and farmer groups	# of public-private partnerships formed Number of states engaged
		3. Farmer awareness and capacity building on climate insurance and risk transfer options	# of farmers sensitized % with increased knowledge on benefits
		4. Insurance bundled with climate-smart inputs (e.g. seeds, fertilizer, credit) and extension services	# of bundled services distributed % of uptake
		5. Policy incentives and premium subsidies provided to enhance uptake	% of farmers benefiting from premium subsidies States with enabling policy frameworks

Means of Verification	Responsibility	Assumptions and Risks
Insurance company reports, product documentation	FMARD; FMEnv; NAICO; CSOs; Private sector	Product development informed by local risk data
MoUs, policy documents, implementation reports	FMARD; FMEnv; NAICO; CSOs; Private sector	Political and institutional buy-in remains strong
Pre/post surveys, campaign reports, training logs	FMARD; FMEnv; NAICO; CSOs; ADPs	Outreach materials culturally adapted and accessible

Extension and partner distribution reports	FMARD; FMEnv; State Government; CSOs; Private sector	Coordination among service providers
State and federal budget reports, subsidy logs	FMARD; NAICO; CSOs; Private sector; State Governments	Budget allocations remain stable

Table 7.10: Logical framework for implementation of Adaptation priorities in Agriculture - Promote affordable, accessible, and climate-responsive agricultural insurance schemes

Sector	Outcome	Output	Indicators
Water Resources	Water Supply Augmentation: Rainwater harvesting, Groundwater development, Water treatment & Recycling	1. Rainwater harvesting systems established in households, schools, farms, and public facilities	# of rainwater harvesting systems installed Volume of water harvested
		2. Groundwater recharge and borehole development schemes implemented sustainably	# of boreholes developed # of aquifer recharge structures built
		3. Water treatment and small-scale recycling systems installed (urban & agro-industrial use)	# of water treatment and greywater reuse units installed Water quality improvement rate
		4. Awareness and capacity-building on water conservation and reuse promoted	# of communities trained % practicing conservation
		5. Institutional coordination and policy frameworks on decentralized water augmentation strengthened	# of LGAs and states with integrated water augmentation plans # of partnerships formed

Means of Verification	Responsibility	Assumptions and Risks
Field inspections, installation records	FMEnv; FMWR; Developmet Partners	Adequate rainfall and rooftop catchment areas
NIWRMC, WRAs, hydrogeological data	FMWR; FMEnv; NHSA NESREA	Aquifers not contaminated; groundwater sustainably managed
Water quality tests, installation records	FMWR; FMEnv; NHSA; CSOs; FMPower	Communities accept water reuse; power supply for treatment
Workshop logs, pre/post assessments, media reports	FMWR; FMEnv; NHSA	Behavioural change efforts are culturally tailored
State policy documents, stakeholder meeting minutes	FMWR; FMEnv; NHSA; State Governments	Policy reforms are implemented effectively

Sector	Outcome	Output	Indicators
Water Resources	Strengthen flood protection infrastructure, early warning, and emergency management systems in flood-prone regions	1. Flood-resistant water and drainage infrastructure constructed/upgraded (levees, culverts, embankments, storm drains)	# and km of protective structures built or upgraded % reduction in urban/rural flood inundation
		2. Community-based early warning systems (EWS) for floods operational	# of communities receiving timely flood warnings % of warnings acted upon
		3. Emergency preparedness and response protocols strengthened at local/state levels	# of LGAs with trained emergency teams and SOPs Time to emergency response
		4. Flood risk mapping, demand planning and zoning integrated into water and land use planning	# of LGAs with updated flood risk maps and plans
		5. Public awareness campaigns and drills on flood preparedness and water use efficiency	# of campaigns conducted Community participation rate

Means of Verification	Responsibility	Assumptions and Risks
FMWR reports, LG engineering records, satellite imagery	FMWH, FMWR, FMEnv; NEMA	Infrastructure designs climate-smart and maintained
NiMet, NEMA reports, SMS alert data	FMEnv, NIMET; Telcos; CSOs; Radio Stations	Mobile and radio networks functional during disasters
LGA emergency plans, NEMA performance reports	NEMA; LGAs, CSOs, NOA	Training sustained; contingency funding available
GIS records, land use plans, WRAs documentation	NEMA; FMSc&Tech;	Inter-agency data sharing and enforcement of zoning
Campaign logs, media coverage, NGO reports	NEMA; CSOs; State Governments	Communication tailored to language and literacy levels

Table 7.12: Logical framework for implementation of Adaptation priorities in Water resources - Strengthen flood protection infrastructure, early warning, and emergency management systems

Sector	Outcome	Output	Indicators
Energy	Scale up decentralized solar and wind energy systems for households, agriculture, health centers, and climate-sensitive sectors	Scale up decentralized solar and wind energy systems for households, agriculture, health centers, and climate-sensitive sectors	# of mini-grids and solar systems installed % reduction in fossil fuel use in communities
		Scale up decentralized solar and wind energy systems for households, agriculture, health centers, and climate-sensitive sectors	# of wind turbines operational Installed wind capacity (MW)
		Scale up decentralized solar and wind energy systems for households, agriculture, health centers, and climate-sensitive sectors	# of farms using solar for production/storage Post-harvest loss reduction %
		Scale up decentralized solar and wind energy systems for households, agriculture, health centers, and climate-sensitive sectors	# of health centers/schools powered % service uptime during climate shocks
		Scale up decentralized solar and wind energy systems for households, agriculture, health centers, and climate-sensitive sectors	# of technicians trained # of RE-based MSMEs supported

Means of Verification	Responsibility	Assumptions and Risks
Installation logs, energy meters, fuel consumption surveys	FMPower; REA; FMH; Private Sector	Affordable procurement and storage of components
Energy generation reports, satellite wind data	FMPower; REA; FMH; Private Sector	Wind mapping and siting studies completed
Farm records, FMARD data, cold storage performance	FMPower; REA; Private Sector	Adoption by farmers and cooperatives sustained
Facility audits, uptime reports, REA monitoring	FMPower; REA; FMH; Private Secto	Operation and maintenance budgets in place
Training reports, SME registries, employment surveys	FMPower; REA; FMH; Private Sector; CSO	Training programs are gender-inclusive and accessible

Table 7.13: Logical framework for implementation of Adaptation priorities in Energy Sector - Scale up decentralised solar and wind energy systems for households, agriculture, health centres, and climate-sensitive sectors

Sector	Outcome	Output	Indicators
Energy	Modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs.	Modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs.	# of km of climate-resilient lines upgraded # of elevated/ relocated substations in flood zones
		Modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs.	% of grid with smart controls and load balancing # of remote sensors operational
		Modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs.	# of community microgrids established Service continuity during climate events
		Modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs.	# of engineers trained in climate-resilient grid management # of utilities adopting resilience protocols
		Modernize and climate-proof national and regional electricity transmission and distribution systems through grid hardening, smart technologies, and decentralized resilience hubs.	# of resilience-focused investment policies and tariff mechanisms adopted

Means of Verification	Responsibility	Assumptions and Risks
Field reports, technical audits	FMPower; REA; Private Secto	Civil works not delayed by land issues or security
SCADA system logs, grid analytics reports	FMPower; REA; Private Sector	Digital infrastructure supported by reliable telecoms
REA reports, mini-grid operators' logs	FMPower; REA; Private Sector; CSOs	Local ownership and training in place
Training reports, utility audits, policy documents	FMPower; REA; Private Sector	Utilities invest in HR development and apply new skills
NERC directives, FMP, donor reports	FMPower; REA; Banks; Insurance co. Private Sector	Sector reforms are implemented and politically supported

Table 7.14: Logical framework for implementation of Adaptation priorities in Energy sector - Modernize and climate-proof national and regional electricity transmission and distribution systems

Sector	Outcome	Output	Indicators
Energy	Deploy decentralized energy storage (batteries, inverters) and clean backup systems (solar/gas) for critical infrastructure and climate-vulnerable communities.	1. Battery storage systems installed in health, education, and agro facilities	# of systems deployed % of uptime during power outages
		2. Backup power systems (solar+inverter, clean gensets) provided for climate emergencies	# of communities/institutions with functional backup Response time during emergencies
		3. Household-level energy storage supported in off-grid and vulnerable zones	# of households with functional battery/inverter systems % reduction in reliance on diesel
		4. Energy storage integrated into mini-grids and RE systems	% of RE mini-grids with storage Load balancing performance
		5. Technicians trained for storage installation, safety, and maintenance	# of trained technicians % of systems functioning after 12 months

Means of Verification	Responsibility	Assumptions and Risks
≥15,000 storage systems installed in critical institutions	FMPower, FMH; FMED; REA; Private Sector;	Facility energy reports, uptime logs
≥2,000 communities with backup for flood, heat, and disaster response	REA; Private Sector; NEMA	NEMA/LEMA/REA data, emergency performance reviews
≥500,000 households use solar + battery storage systems	Private Sector;	Survey data, REA installations, DisCo collaboration
≥80% of mini-grids integrated with storage (e.g., Li-ion, lead-acid)	REA; Private Sector	REA and private mini-grid reports, SCADA data
≥5,000 technicians trained across all geopolitical zones	REA; Private Sector; CSOs	Training records, post-installation system audits

7.6: Reporting on Progress on NAP under the UNFCCC

Nigeria's NAP integrates a reporting system during its implementation to facilitate continuous learning and adaptive management, including monthly reports from local implementation teams and quarterly progress reports submitted to the National Coordinator. The NAP implementation will be monitored annually and reviewed biennially, with stakeholders conducting annual and biennial review workshops. As a reporting tool, public dashboards and GIS-based platforms are introduced for real-time data sharing and transparency.

The National Adaptation Plan M&E system will be monitored by project-executing entities and supervised by the Federal Ministry of Environment (the national implementing entities (NIEs)). Findings from continuous learning will be integrated into planning and budgetary processes, disseminated through policy briefs and meetings, and shared across states and among other stakeholders, including NGOs, communities, and the private sector.

Capacity development for M & E: To ensure effective implementation of the NAP monitoring and evaluation framework, the plan includes a process of continuous training for relevant stakeholders on the various facets of M & E, with progress to be reported. It is an opportunity to identify gaps along the chains of M&E actors across sectors to support the realisation of the M&E objectives.

7.7: Revising and updating the NAP

As a living document, Nigeria's NAP will be revised and updated every five-year cycle, taking into account the uncertainty associated with climate change and the country's diversity. The review and updates will take into account progress achieved and previous adaptation barriers to address them, as well as social and environmental trends, new adaptation issues and policies, and national and international government pledges. The five-year review and updating process will incorporate lessons learned during the previous five years to increase the effectiveness of the NAP, tackle uncertainties posed by climate change, and maximise the country's resilience and sustainable development gains. Other features that will be embodied in the revision include:

- Identified gaps in the past five-year cycle of implementing the plan
- Emerging risks and potential mitigation measures
- Progress and obstacles in achieving goals and strategies
- Suggestions on ways of mobilising climate finance and implementing adaptation measures
- Mechanism of addressing identified capacity and communication needs to enhance implementation

The Federal Ministry of Environment, as the national implementing entity (NIE), shall initiate the revision process one year before each five-year cycle to enable the integration of lessons into the national development planning process. The revised NAP, which addresses gaps and integrates new findings and risks related to climate change in Nigeria, will be considered by the Inter-ministerial Committee and endorsed by the Federal Government before submission to the UNFCCC.

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